Gene Essentiality from the Cell Graph

The backbone against flux balance analysis and Flux Cone Learning,

and why the solve helps the prediction where the metabolic module does not

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Abstract

Gene essentiality is the one endpoint on which a flux model has a published, non-trivial score: on the released Merzbacher 2025 yeast split, flux balance analysis reads an AUROC of 0.69 over all 1,121 labeled genes and Flux Cone Learning 0.73 on the 223 held-out genes. This document scores the cell graph transformer on the same held-out genes, puts FBA on every medium the ontology defines and a classical ladder of models on the same split, and asks two questions. The first is how far the held-out AUROC goes: a width-12 transformer reading the chromosome-pathway indicator and ProtT5, with FBA's four growth ratios on its logit, scores 0.893 [0.84, 0.94] under Flux Cone Learning's own fold protocol, 0.15 above it on the same genes, and every part of the gain is an input the ontology computes for every gene at no training cost: a two-layer MLP on the same three inputs reads the transformer's per-cell number in six seconds and 0.886 as the same fold ensemble. The second is why the solve helps and the metabolic module, which is built on the same genome-scale model, does not: routed through the module's growth column the prediction never beats the backbone alone, and the rounds locate the reason in the layer's constraint set, whose enzyme-capacity ceilings as parameterized contain no growing flux, so its alternating projections never reach the polytope FBA optimizes over and a linear program under its bounds is infeasible. On the genome-scale model's own bounds the module carries the solve, whether handed it as a warm start or solving for it in the forward pass with the gradient through the dual, and the gated module cell reaches the level of the solve appended as an input. Every number here is at a validation-selected epoch on a pinned test set, with a gene bootstrap and the across-seed range, and every table is generated from the result files.

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1 The endpoint, its labels and its splits ✗

1.1 Vocabulary ✗

AUROC Area under the receiver operating characteristic, computed as the Mann-Whitney rank statistic with mid-ranks on ties: the probability that a randomly drawn essential gene scores above a randomly drawn nonessential one.

backbone The cell graph transformer over the nine-graph perturbed gene graph with `prot_T5_all` node features unless another encoding is named, read at the pooled perturbed tokens. The *pooled* arm is this backbone with a scalar head and no metabolic module.

chromosome-pathway indicator The 197-dimensional per-gene encoding `normalized_chrom_pathways` (`sgd_gene_graph_hot.py`): five SGD numeric gene features scaled to $[0, 1]$ (protein length, molecular weight, isoelectric point, the median and the median absolute deviation of the gene's expression), a one-hot of the gene's chromosome, and a multi-hot over the SGD pathways it is annotated to. *chrom-pathway* in figure labels; the name understates it, since five of its columns are continuous gene properties and two of those are expression.

FBA affine The readout that adds $w^\top r + b$ to the backbone's logit, where r holds the deleted gene's knockout-to-wild-type growth ratio on each of four media from one FBA solve per gene; the *fba* arm.

fold ensemble Five models trained on the five cross-validation files (each a different fifth of the 787 labeled non-test genes held for validation), their logits averaged on the held-out genes; Flux Cone Learning's own test protocol.

gene bootstrap Resampling the scored genes with replacement (2,000 draws) and reporting the 2.5 and 97.5 percentiles of the statistic.

GO multi-hot The per-gene encoding over Gene Ontology terms (`go_term_hot.py`): one column per term, a gene's row marking the terms SGD annotates it to, propagated to every ancestor through *is_a* and *part_of* (the true-path rule), the three namespace roots dropped, and a term kept only if ten or more genes carry it: 2,362 columns over 5,501 annotated genes. *GO direct* keeps the annotated terms alone (2,887 columns, two or more genes); *GO BP*, *MF* and *CC* keep one namespace (1,580, 430, 352). The SGD file carries no evidence codes, so annotations inferred from mutant phenotypes are in the vector.

held-out genes The 223 genes Merzbacher 2025 held out of its own training (`test = 1` in the released split), of which 195 resolve as single-deletion genotypes in the build. Every model number in this document is on those 195 unless the gene set is named.

layer that solves The metabolic module with its flux replaced by the growth optimum of the linear program under the layer's own bounds, one solve per genotype in the forward pass, the bound duals as its gradient. Its *elastic* form lets the model's one forced flux (ATP maintenance) go unmet at a price, so the program is feasible for every genotype by construction.

message passing The update of a node's representation from an aggregate of its neighbors' representations, $h_i' = \gamma(h_i, \bigoplus_{j \in \mathcal{N}(i)} \phi(h_i, h_j))$; graph convolution (GCN), graph attention (GAT, GATv2), GraphSAGE and the graph isomorphism network (GIN) are choices of ϕ, $\bigoplus$ and γ. *Relational* message passing (RGCN, RGAT) keeps one weight or one attention per edge type over a multigraph.

metabolic module The enzyme-constrained thermodynamic flux layer over yeast-GEM 9.0.2 that maps the perturbed representation to a flux vector by K alternating projections onto the mass balance $Sv = 0$ and the box of bounds; here it is read at the objective column `r_2111` (growth). The *anchored* layer tightens the GEM's box by the enzyme-capacity ceilings and the second-law directions; the *stoichiometry-only* layer keeps the GEM's box.

parsimonious FBA Two linear programs: the growth optimum over the polytope $\{Sv = 0,\ \ell \le v \le u\}$ with the deleted gene's reactions closed through the gene-protein-reaction rules, then the flux of least total absolute value at that optimum. *pFBA*; *the solve* when the object is the optimum rather than the method.

released labels The 1,121 essentiality calls in the released split. The column named `essential` is a viability flag (1.0 marks a nonessential gene); it is inverted once and the four class counts are asserted against the paper's Methods.

selected epoch The argmax of a centered five-epoch moving average of the validation AUROC. The held-out genes are scored once, at that epoch.

solve-derived encodings Per-gene vectors computed from pFBA rather than from sequence: the deletion's own flux vector, its difference from the wild-type vector, and eight statistics of the gene's reactions under the wild-type solve and its flux variability (`ess_reference_flux_table.py`).

warm start The flux layer's proposal written as a reference flux plus a null-space residual of relative size α, $v = v_{\text{ref}} + \alpha N z$, before the alternations; with a per-genotype table the reference is the deletion's own pFBA flux.

Weisfeiler-Leman test Color refinement: every node starts with a color, and each iteration replaces it by a hash of its own color and the multiset of its neighbors' colors; the multiset of colors after k iterations is the graph's 1-WL signature. Different signatures prove two graphs non-isomorphic; equal signatures prove nothing. A message-passing network is at most as expressive as the test.

1.2 Where the labels come from ✗

The build is a union of the SGD essential-gene list (1,140 genotypes, all essential) and the single-mutant fitness screens (Costanzo 2016, Kuzmin 2018; 4,540 genotypes), 5,656 single-deletion genotypes in all. No fitness value enters any objective: the head is one logit scored by binary cross entropy against a 0/1 label, and every genotype whose gene carries no label arrives masked. Two label sources are used and named throughout:

Table 1: The split files, one per label source: genes per split with the essential count in parentheses. The test block is the released held-out genes under either source; validation is a class-stratified fifth of the remaining labeled genes at seed 0. Under the genome source the released label wins on every disagreement (12 genes).

split file	labels	train (ess)	val (ess)	test (ess)
`ess_fcl_val0.2_s0`	released (1,121)	629 (100)	158 (25)	195 (31)
`ess_genome_heldout_val0.2_s0`	released + SGD essential + viable fitness (5,572)	629 (100)	158 (25)	4674 (974)
`ess_genome_val0.2_s0`	released + SGD essential + viable fitness (5,795)	4,368 (879)	1,093 (220)	195 (31)

1.3 The protocol inherited from 027, and what it got wrong ✗

The wave-9 protocol pinned the held-out genes and let the data module draw train and validation at 80/10/10 over the remaining 5,461 records. Of those, 926 carry a label, so the validation set held 75 labeled genes with about 12 essential, on which a single AUROC read has a Hanley-McNeil standard error near 0.07; 10 percent of the labeled genes sat in a random test block nothing reports; and 4,535 unlabeled records were forwarded every epoch under a False mask. A split file replaces the draw: it names every record's split, is class-stratified, carries the labels it was stratified on, and the trainer drops every record it does not name.

2 The comparators, computed rather than quoted ✗

2.1 Flux balance analysis on four media ✗

The published FBA comparator is cobrapy `single_gene_deletion` with glucose as the carbon source, scored on all 1,121 labeled genes. The phenotype it is compared against was called on rich medium, and the model here is scored on the 195 held-out genes, so the comparison as published is neither on the same medium nor on the same genes. Table 2 recomputes it on yeast-GEM 9.0.2 under the exchange bounds the model ships with and under the three media the ontology defines, with the essentiality score being the negative knockout-to-wild-type growth ratio, rounded to 10^{-3} so that the roughly 900 deletions that grow at 0.99 to 1.00 of the wild type rank as the tie they are (without the rounding two identical runs differed by 0.03 AUROC on the held-out genes).

Table 2: FBA single-gene deletion on yeast-GEM 9.0.2 (cobrapy 0.30.0), scored as the negative knockout-to-wild-type growth ratio rounded to 10^{-3}. AUROC on all 1,121 labeled genes, on the 223 held-out genes, and on the 195 held-out genes the model scores with a 2,000-draw gene bootstrap interval. 1121 of the labeled genes are in the GEM; the rest keep wild-type growth. Bold marks the highest value in a column.

medium	WT growth (1/h)	lethal	all 1,121	held-out 223	our 195 [95% CI]
shipped bounds (glucose minimal, aerobic)	0.086	186	0.707	0.633	0.654 [0.56, 0.75]
SM, synthetic minimal	0.314	160	0.707	0.642	0.666 [0.57, 0.76]
SC, synthetic complete	0.543	91	**0.719**	**0.653**	**0.666** [0.57, 0.76]
YPD approximation (no peptone)	0.535	104	0.715	**0.653**	0.664 [0.57, 0.76]

The medium moves the number of lethal deletions by a factor of two and the AUROC by about 0.01. The published 0.69 on all labeled genes is reproduced to within 0.02.

2.2 yeast-GEM's own benchmark, and accuracy against balanced accuracy ✗

yeast-GEM ships its own essentiality benchmark and prints it in its README: on 1,107 genes, an accuracy of 0.902 with 933 true non-essential, 65 true essential, 94 false non-essential and 15 false essential calls. Its release file (`results/external/yeast-GEM-9.1.1/`, version 9.1.1 at commit `2d594ae`, 2026-08-30) says how: the Kennedy synthetic complete medium is written onto the exchanges (every exchange closed, 28 uptakes at −0.5, glucose at −20, fifteen inorganic uptakes open), each gene is deleted through the gene-protein-reaction rules, and a deletion is inviable when its growth ratio is below 10^{-6}; the genes are the SGD verified ORFs in the model, and the labels are the Stanford deletion project's inviable list of 2011, scored against the model at that one rule, a protocol the repository attributes to Heavner and Price (2015). The released split's labels are the same truth: on the 1,104 genes both name, the two sources agree on 1,102.

Table 3 puts every scorer in this document at that rule. Two things are visible at once. The first is that accuracy is the wrong statistic for this endpoint: 86 percent of the genes are nonessential, so a scorer that calls nothing essential reads 0.856 on yeast-GEM's set and 0.841 on the 195 held-out genes, and yeast-GEM's 0.902 is that floor plus four and a half points; its recall on the essential class is 0.41 and its balanced accuracy 0.70. FBA on the previous release, 9.0.2, under the ontology's own synthetic complete recipe reads 0.907 on the released labels with a recall of 0.46, the same rule on the same truth with a different medium composition, so the 9.1.1 number is reproduced to within half a point. The second is where the transformer sits at a threshold: at its loss's decision boundary it reads 0.856, the majority floor, with a recall of 0.68 and a balanced accuracy of 0.78, and at the threshold that calls 31 genes essential it reads 0.867 with precision and recall both at 0.58. The released Flux Cone Learning predictions at $P(\text{viable}) < 0.5$ call eight genes essential on the 195, all eight correctly, for a recall of 0.26. A ranking statistic is what separates the scorers, and the balanced accuracies, 0.63 to 0.72 for the solvers and 0.78 for the transformer, order them the way the AUROCs do.

Table 3: Every scorer at the rule yeast-GEM reports its benchmark with: a deletion is called essential when the criterion in the rule column holds, and the call is scored against the label as a confusion matrix with essential as the positive class (TP/FN/FP/TN). Accuracy is followed by the majority class's share of the gene set in parentheses; precision and recall are for the essential class; balanced accuracy is the mean of recall and specificity. yeast-GEM 9.1.1's row is its own release file on its own labels (the Stanford deletion project's inviable list over the model's verified ORFs); those labels and the released split agree on 1102 of the 1,104 genes both name. Every other row is on the released labels. The transformer is the round-12 fold ensemble at the loss's decision boundary and at the threshold that calls as many genes essential as the labels hold. Bold marks the highest value in a column.

scorer	rule	gene set	ess/n	TP/FN/FP/TN	acc (maj)	prec	rec	spec	bal
yeast-GEM 9.1.1 (its own benchmark)	growth ratio < 1e-6, Kennedy SC medium	1,107 verified ORFs, Stanford inviable list	159/1,107	65/94/15/933	0.902 (0.856)	0.812	0.409	0.984	0.696
FBA, yeast-GEM 9.0.2, shipped	growth ratio < 1e-6	1,121 labeled (released labels)	157/1,121	82/75/104/860	0.840 (0.860)	0.441	0.522	0.892	0.707
FBA, yeast-GEM 9.0.2, shipped	growth ratio < 1e-6	195 held-out (released labels)	31/195	11/20/11/153	0.841 (0.841)	0.500	0.355	0.933	0.644
FBA, yeast-GEM 9.0.2, SM	growth ratio < 1e-6	1,121 labeled (released labels)	157/1,121	78/79/82/882	0.856 (0.860)	0.487	0.497	0.915	0.706
FBA, yeast-GEM 9.0.2, SM	growth ratio < 1e-6	195 held-out (released labels)	31/195	11/20/7/157	0.862 (0.841)	0.611	0.355	0.957	0.656
FBA, yeast-GEM 9.0.2, SC	growth ratio < 1e-6	1,121 labeled (released labels)	157/1,121	72/85/19/945	**0.907** (0.860)	0.791	0.459	0.980	0.719
FBA, yeast-GEM 9.0.2, SC	growth ratio < 1e-6	195 held-out (released labels)	31/195	11/20/2/162	0.887 (0.841)	0.846	0.355	0.988	0.671
FBA, yeast-GEM 9.0.2, YPD	growth ratio < 1e-6	1,121 labeled (released labels)	157/1,121	73/84/31/933	0.897 (0.860)	0.702	0.465	0.968	0.716
FBA, yeast-GEM 9.0.2, YPD	growth ratio < 1e-6	195 held-out (released labels)	31/195	11/20/2/162	0.887 (0.841)	0.846	0.355	0.988	0.671
Flux Cone Learning (released predictions)	P(viable) < 0.5	223 released held-out (released labels)	31/223	8/23/0/192	0.897 (0.861)	**1.000**	0.258	**1.000**	0.629
Flux Cone Learning (released predictions)	P(viable) < 0.5	195 held-out (released labels)	31/195	8/23/0/164	0.882 (0.841)	**1.000**	0.258	**1.000**	0.629
FBA (released predictions)	growth < 1e-6	195 held-out (released labels)	31/195	11/20/11/153	0.841 (0.841)	0.500	0.355	0.933	0.644
transformer, round-12 fold ensemble	logit > 0	195 held-out (released labels)	31/195	21/10/18/146	0.856 (0.841)	0.538	**0.677**	0.890	**0.784**
transformer, round-12 fold ensemble	top 31 logits (label prevalence)	195 held-out (released labels)	31/195	18/13/13/151	0.867 (0.841)	0.581	0.581	0.921	0.751

2.3 Flux Cone Learning on the same genes ✗

Recomputed from the authors' released per-gene predictions: 0.7308 on the 223 held-out genes (published 0.73) and 0.742 on the 195 genes the model scores. On those same 195 genes the released FBA predictions score 0.640, so the published pair overstates FBA by about 0.06 relative to a head-to-head on one gene set. The paper prints a third number, 0.78 ± 0.065 in the legend of its Figure 3a, which is the mean over five cross-validation folds drawn inside its 898 training genes; a model scored on the held-out genes is compared with the 0.73 line, never with that one.

2.4 The classical ladder on the same split ✗

The fitness and expression lines each carry a ladder of models that need no graph and no training beyond a solve, and essentiality gets the classification form of it: FBA's growth ratio with no fitting, then four model classes (L_2 logistic regression, a random forest, a radial-basis support vector classifier, cosine nearest neighbors) on each of twenty-four gene encodings, every pair fit twice, with and without the four FBA growth ratios appended. The encodings are the seventeen the fitness ladder carries (codon frequency, the chromosome and pathway indicator, six nucleotide windows, the two FUDT embeddings, the species language model, CaLM, ProtT5 and ESM2 with and without dubious genes, and their concatenation, `mixed`) plus seven controls: a one-hot gene identity, and standard normal vectors of dimension 1, 10, 100, 1,000, 6,607 (the one-hot's width) and 8,192, drawn once at seed 0. A one-hot rung must score chance on held-out genes, since every held-out column is zero in training, and a random vector must score chance at every width; either scoring above chance would mean the split leaks. Every fitted rung is fit on the split file's 629 training genes, its one hyperparameter chosen by five-fold stratified cross-validation inside those genes, and scored once on the 158 validation and 195 held-out genes, on which nothing was chosen. The transformer cells in Sections 3 and 4 train on the same 629 genes and score on the same 195. Three further encodings come from the solve itself rather than from the sequence, added after round 8 (`ess_reference_flux_table.py`): the deletion's own parsimonious FBA flux vector on YPD (4,131 reactions), that vector's difference from the wild-type vector, and eight numbers about the gene's own reactions under the wild-type solve and its flux variability at 90 percent of the optimum (reaction count, reactions carrying flux, the largest and the summed absolute flux, the largest forced flux, the smallest and largest variability range, and the deletion's growth ratio). Figure 1 shows all 216 fitted rungs; Table 4 keeps, per encoding, the model class whose cross-validation AUROC is highest.

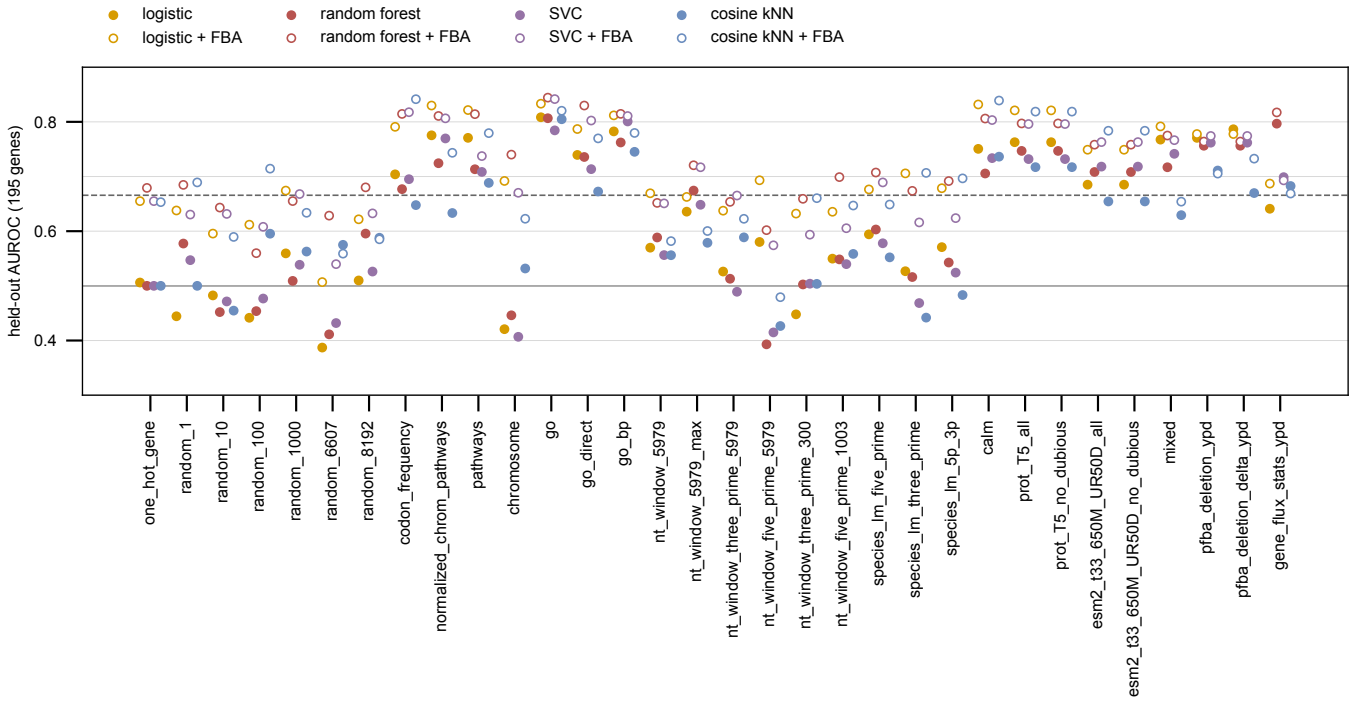


Figure 1: The classical ladder on the released-label split. Held-out AUROC on the 195 genes for every model class (color) on every encoding (column), filled without and open with the four FBA growth ratios appended; the dashed line is FBA alone on the shipped medium (0.654), the solid line chance. The last three columns are the solve-derived encodings. The hyperparameter of each mark was chosen by five-fold cross-validation inside the 629 training genes. Written by `experiments/028-gene-essentiality/scripts/logistic_baselines.py`.

Table 4: The classical ladder on the released-label split (`ess_fcl_val0.2_s0.json.gz`). Per encoding, the model class whose five-fold cross-validation AUROC inside the 629 training genes is highest, with and without FBA's four growth ratios appended; the validation genes and the 195 held-out genes (31 essential) are scored once, on genes no fit touched, the held-out number with a 2,000-draw gene bootstrap. FBA alone is not fitted. Every model and encoding is in Figure 1. Bold marks the highest value in a column.

encoding	best model (chosen)	dim	CV	val	held-out [95% CI]
FBA growth	medium=shipped	–	–	0.683	0.654 [0.56, 0.75]
FBA growth	medium=SM	–	–	0.710	0.666 [0.57, 0.76]
FBA growth	medium=SC	–	–	0.726	0.666 [0.57, 0.76]
FBA growth	medium=YPD	–	–	0.714	0.664 [0.57, 0.76]
one_hot_gene	logistic (C=10)	6,579	0.504	0.508	0.506 [0.50, 0.52]

encoding	best model (chosen)	dim	CV	val	held-out [95% CI]
one_hot_gene + FBA	random forest (max_depth=8)	6,583	0.775	0.730	0.679 [0.59, 0.77]
random_1	random forest (max_depth=4)	1	0.598	0.560	0.577 [0.46, 0.69]
random_1 + FBA	cosine kNN (k=10)	5	0.806	0.699	0.689 [0.56, 0.80]
random_10	cosine kNN (k=50)	10	0.549	0.481	0.455 [0.35, 0.56]
random_10 + FBA	logistic (C=0.001)	14	0.786	0.684	0.596 [0.46, 0.72]
random_100	random forest (max_depth=None)	100	0.522	0.376	0.454 [0.34, 0.57]
random_100 + FBA	random forest (max_depth=4)	104	0.773	0.662	0.560 [0.42, 0.70]
random_1000	random forest (max_depth=4)	1,000	0.475	0.490	0.509 [0.40, 0.62]
random_1000 + FBA	random forest (max_depth=None)	1,004	0.778	0.749	0.655 [0.53, 0.78]
random_6607	cosine kNN (k=1)	6,607	0.549	0.477	0.575 [0.49, 0.66]
random_6607 + FBA	random forest (max_depth=None)	6,611	0.734	0.713	0.628 [0.50, 0.75]
random_8192	cosine kNN (k=50)	8,192	0.523	0.515	0.588 [0.49, 0.69]
random_8192 + FBA	random forest (max_depth=8)	8,196	0.768	0.748	0.680 [0.57, 0.79]
codon_frequency	SVC (C=1)	64	0.700	0.663	0.695 [0.59, 0.79]
codon_frequency + FBA	SVC (C=1)	68	0.849	0.772	0.818 [0.73, 0.90]
normalized_chrom_pathways	logistic (C=10)	197	0.705	0.722	0.775 [0.68, 0.86]
normalized_chrom_pathways + FBA	logistic (C=0.01)	201	0.802	0.791	0.830 [0.75, 0.90]
nt_window_5979	logistic (C=0.01)	2,560	0.579	0.532	0.570 [0.46, 0.67]
nt_window_5979 + FBA	logistic (C=0.01)	2,564	0.743	0.705	0.669 [0.55, 0.79]
nt_window_5979_max	logistic (C=0.001)	2,560	0.626	0.608	0.636 [0.54, 0.73]
nt_window_5979_max + FBA	logistic (C=0.01)	2,564	0.777	0.786	0.663 [0.54, 0.78]
nt_window_three_prime_5979	logistic (C=0.001)	2,560	0.555	0.580	0.526 [0.41, 0.64]
nt_window_three_prime_5979 + FBA	logistic (C=0.001)	2,564	0.732	0.720	0.637 [0.51, 0.76]
nt_window_five_prime_5979	cosine kNN (k=25)	2,560	0.594	0.518	0.427 [0.32, 0.54]
nt_window_five_prime_5979 + FBA	logistic (C=0.01)	2,564	0.733	0.657	0.693 [0.59, 0.79]
nt_window_three_prime_300	random forest (max_depth=4)	2,560	0.564	0.537	0.503 [0.39, 0.62]
nt_window_three_prime_300 + FBA	random forest (max_depth=None)	2,564	0.752	0.757	0.659 [0.53, 0.78]
nt_window_five_prime_1003	logistic (C=0.001)	2,560	0.541	0.485	0.550 [0.43, 0.68]
nt_window_five_prime_1003 + FBA	logistic (C=0.01)	2,564	0.727	0.665	0.636 [0.52, 0.75]
fudt_upstream	SVC (C=1)	768	0.611	0.583	0.578 [0.47, 0.69]
fudt_upstream + FBA	random forest (max_depth=None)	772	0.775	0.747	0.707 [0.60, 0.81]
fudt_downstream	SVC (C=10)	768	0.589	0.553	0.469 [0.36, 0.58]
fudt_downstream + FBA	logistic (C=0.1)	772	0.770	0.735	0.706 [0.59, 0.81]
species_lm_5p_3p	logistic (C=0.001)	1,536	0.634	0.631	0.571 [0.47, 0.68]
species_lm_5p_3p + FBA	logistic (C=0.01)	1,540	0.763	0.733	0.679 [0.56, 0.79]
calm	SVC (C=1)	768	0.755	0.777	0.734 [0.64, 0.82]
calm + FBA	random forest (max_depth=8)	772	0.850	0.833	0.806 [0.72, 0.89]
prot_T5_all	SVC (C=10)	1,024	0.781	0.767	0.732 [0.62, 0.83]
prot_T5_all + FBA	logistic (C=0.01)	1,028	0.842	0.826	0.821 [0.74, 0.90]
prot_T5_no_dubious	SVC (C=10)	1,024	0.781	0.767	0.732 [0.62, 0.83]
prot_T5_no_dubious + FBA	logistic (C=0.01)	1,028	0.842	0.826	0.821 [0.74, 0.90]
esm2_t33_650M_UR50D_all	random forest (max_depth=8)	1,280	0.731	0.694	0.708 [0.61, 0.80]
esm2_t33_650M_UR50D_all + FBA	random forest (max_depth=8)	1,284	0.815	0.794	0.758 [0.66, 0.86]
esm2_t33_650M_UR50D_no_dubious	random forest (max_depth=8)	1,280	0.731	0.694	0.708 [0.61, 0.80]
esm2_t33_650M_UR50D_no_dubious + FBA	random forest (max_depth=8)	1,284	0.815	0.794	0.758 [0.66, 0.86]
mixed	logistic (C=10)	6,149	0.788	0.794	0.768 [0.66, 0.87]
mixed + FBA	logistic (C=10)	6,153	0.833	0.837	0.792 [0.69, 0.89]
pfba_deletion_ypd	random forest (max_depth=None)	4,131	0.770	0.723	0.756 [0.65, 0.86]
pfba_deletion_ypd + FBA	random forest (max_depth=None)	4,135	0.776	0.736	0.764 [0.66, 0.86]
pfba_deletion_delta_ypd	random forest (max_depth=None)	4,131	0.769	0.723	0.756 [0.65, 0.86]
pfba_deletion_delta_ypd + FBA	random forest (max_depth=None)	4,135	0.776	0.736	0.764 [0.66, 0.86]
gene_flux_stats_ypd	random forest (max_depth=None)	8	0.837	0.760	0.797 [0.70, 0.88]
gene_flux_stats_ypd + FBA	random forest (max_depth=None)	12	0.847	0.777	0.817 [0.73, 0.90]
pathways	logistic (C=0.01)	173	0.673	0.708	0.771 [0.68, 0.85]
pathways + FBA	random forest (max_depth=8)	177	0.808	0.776	0.814 [0.73, 0.89]
chromosome	cosine kNN (k=10)	17	0.523	0.537	0.532 [0.44, 0.63]
chromosome + FBA	logistic (C=0.1)	21	0.758	0.768	0.692 [0.58, 0.80]
go	random forest (max_depth=None)	2,362	0.852	0.895	0.807 [0.72, 0.88]
go + FBA	random forest (max_depth=None)	2,366	0.877	**0.915**	**0.844** [0.77, 0.91]
go_direct	logistic (C=0.01)	2,887	0.827	0.770	0.739 [0.63, 0.84]
go_direct + FBA	logistic (C=0.01)	2,891	**0.877**	0.805	0.787 [0.68, 0.88]
go_bp	random forest (max_depth=8)	1,580	0.828	0.862	0.762 [0.66, 0.86]
go_bp + FBA	random forest (max_depth=8)	1,584	0.856	0.884	0.815 [0.71, 0.90]

The controls behave. The one-hot rung scores 0.506 and the six random widths score 0.45 to 0.59, chance within the 195-gene noise, and with FBA appended they score 0.56 to 0.69, FBA's own number, so no rung reads anything but its features. Without FBA four encodings are clear of 0.70: ProtT5 0.732, CaLM 0.734, the concatenation 0.768, and the chromosome-pathway indicator 0.775 [0.68, 0.86], each level with Flux Cone Learning's 0.742 on the same genes. FBA lifts every one of them: logistic regression on ProtT5 with the ratios scores 0.821 [0.74, 0.90] from 1,029 parameters,

the support vector classifier on 64 codon frequencies 0.818 [0.73, 0.90], the random forest on CaLM 0.806 [0.72, 0.89], and logistic regression on the 197-dimensional chromosome-pathway indicator 0.830 [0.75, 0.90]. The rung with the highest cross-validation AUROC, the random forest on CaLM with FBA at 0.850, scores 0.806 on the held-out genes; the highest held-out number in the figure, 0.841 for cosine nearest neighbors on codon frequency with FBA, is the maximum over 216 rungs and is not a selected model. Neither an embedding nor the solver reaches 0.80 alone, and their sum does on four encodings. FBA's ratio is a feature the ontology computes for every gene at no training cost, so the transformer is measured against 0.80 to 0.83 from here on, not against the published 0.73.

The annotation vectors on their own. Logistic regression on the 173-column pathway multi-hot alone scores 0.771 [0.68, 0.85] with a cross-validation AUROC of 0.673, and with the four ratios 0.822, the level ProtT5 with the ratios reaches from 1,029 parameters; the chromosome one-hot alone is below chance (0.421) and 0.692 with the ratios. The GO multi-hot is the strongest single encoding on the ladder: logistic regression on the propagated vector scores 0.808 [0.72, 0.89] with no solve at all, above every other encoding without FBA, and 0.833 with the ratios; the random forest on it with the ratios scores 0.844 [0.77, 0.91] at a cross-validation AUROC of 0.877, the highest of any rung. The propagated form beats the direct annotations under every model (0.808 against 0.739 linear without the ratios), and the biological-process namespace alone reaches 0.812 with them.

The solve carries more than its growth ratio. The eight reaction statistics under a random forest score 0.797 [0.70, 0.88] with no sequence information, and 0.817 [0.73, 0.90] with the four ratios appended, at the highest cross-validation AUROC in the ladder (0.847); the growth ratio alone, which is one of the eight, scores 0.645 on the same genes. The full deletion flux vector scores 0.756 under its selected model and 0.771 under logistic regression, level with ProtT5 and with the pooled transformer, so a gene's position in the wild-type flux solution and how much the solve moves when it is deleted rank essential genes about as well as its protein sequence does, and the two sources are different: no other encoding reaches 0.80 without the solve, and the eight statistics come within 0.003 of it with nothing else.

2.5 Graph-only baselines: message passing over the gene graph ✗

The Gene_Graph project (2021 to 2023) classified essential genes by two-layer graph convolution and graph attention over YeastMine's physical, genetic and regulatory interaction graphs with all-ones node inputs, and saved no number. The same design is rerun here on the released split (`graph_baselines.py`): transductive node classification on one gene graph, where every gene's logit is a function of its neighborhood and its own features and no genotype is perturbed, on the physical interaction graph and on the union of the nine graphs the transformer reads, with all ones, with log degree and the clustering coefficient, or with ProtT5 as inputs, beside an MLP on ProtT5 with no graph (Table 5).

Table 5: Graph-only baselines on the released split: transductive node classification on one gene graph with no genotype perturbed, two layers of width 64, class-weighted binary cross entropy, 200 epochs, the epoch selected by a five-epoch moving average of the validation AUROC, scored once on the 195 held-out genes at seeds 2700 to 2702. The graph is the physical interaction graph or the union of the nine gene graphs the transformer reads; the inputs are all ones (the Gene_Graph design), log degree with the clustering coefficient, ProtT5, or ProtT5 with the chromosome-pathway indicator and the four FBA growth ratios as node features; two arms run 400 epochs. The MLP rows have no graph. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
MLP	none	ProtT5	1,024	69,825	0.754	0.003	0.756, 0.755, 0.750	62, 58, 56
GCN	physical	ones	1	4,353	0.579	0.003	0.577, 0.577, 0.583	88, 100, 49
GCN	physical	degree + clustering	2	4,417	0.582	0.009	0.592, 0.571, 0.583	29, 26, 3
GCN	physical	ProtT5	1,024	69,825	0.769	0.002	0.770, 0.766, 0.771	194, 193, 197
GATv2	physical	ones	1	8,897	0.495	0.034	0.455, 0.493, 0.538	137, 148, 50
GATv2	physical	degree + clustering	2	9,025	0.592	0.003	0.592, 0.588, 0.596	160, 192, 116
GATv2	physical	ProtT5	1,024	139,841	0.760	0.018	0.747, 0.747, 0.786	101, 93, 79
GCN	union of nine	ones	1	4,353	0.613	0.000	0.613, 0.613, 0.613	8, 5, 3
GCN	union of nine	degree + clustering	2	4,417	0.638	0.021	0.661, 0.644, 0.610	197, 197, 2
GCN	union of nine	ProtT5	1,024	69,825	0.834	0.003	0.836, 0.829, 0.836	197, 196, 197
GATv2	union of nine	ones	1	8,897	0.560	0.014	0.578, 0.556, 0.545	38, 107, 125
GATv2	union of nine	degree + clustering	2	9,025	0.702	0.025	0.699, 0.672, 0.733	197, 191, 197
GATv2	union of nine	ProtT5	1,024	139,841	0.796	0.020	0.809, 0.768, 0.811	118, 197, 119
MLP	none	ProtT5 + chrom-pathway	1,221	82,433	0.817	0.000	0.817, 0.817, 0.818	70, 61, 60
GCN	union of nine	ProtT5 + chrom-pathway	1,221	82,433	0.850	0.003	0.849, 0.854, 0.847	197, 196, 197
MLP	none	ProtT5 + FBA ratios	1,028	70,081	0.803	0.013	0.814, 0.785, 0.810	71, 197, 76
GCN	union of nine	ProtT5 + FBA ratios	1,028	70,081	0.832	0.002	0.829, 0.833, 0.834	195, 197, 197
MLP	none	ProtT5 + chrom-pathway + FBA ratios	1,225	82,689	0.870	0.001	0.870, 0.872, 0.869	55, 58, 55
GCN	union of nine	ProtT5 + chrom-pathway + FBA ratios	1,225	82,689	0.847	0.005	0.843, 0.844, 0.855	195, 197, 196
GCN	union of nine	ProtT5 (400 epochs)	1,024	69,825	0.857	0.001	0.859, 0.856, 0.857	392, 397, 394
GCN	union of nine	ProtT5 + chrom-pathway + FBA ratios (400 epochs)	1,225	82,689	**0.877**	0.004	0.874, 0.874, 0.883	381, 396, 392

Structure alone carries little and structure with sequence carries a lot. With all-ones inputs the models read 0.50 to

0.61, chance to the ladder's random-vector level, and with degree and clustering 0.58 to 0.70. With ProtT5 the MLP reads 0.754, the ladder's logistic rung plus 0.02, and graph convolution over the union of the nine graphs reads 0.836, 0.829 and 0.836, a seed mean of 0.834 with a spread of 0.003 from 69,825 parameters: the edges add 0.08 to the protein embedding, the physical graph alone adds 0.015, and attention adds nothing over convolution on either graph. That cell is above the pooled transformer (0.774) and level with the transformer with FBA's four ratios (0.821) with no solve at all, and its three seeds select epochs 196 to 197 of 200 with the validation curve still rising; at 400 epochs the same cell reads 0.857 and selects epochs 392 to 397, so the budget is what bounds it.

The transformer's inputs move these cells the way they move the transformer, and the comparison says where the transformer's number comes from. Given the chromosome-pathway indicator and the four FBA ratios as node features beside ProtT5, the MLP with no graph reads 0.870, 0.872 and 0.869, a seed mean of 0.870 with a spread of 0.001 in six seconds of CPU, which is the round-9 transformer's 0.871 from the same three inputs; graph convolution on the union reads 0.847 at 200 epochs and 0.877 at 400, still rising. The indicator alone beside ProtT5 lifts the MLP to 0.817 and the convolution to 0.850, the ratios alone to 0.803 and 0.832. On this endpoint, then, the three inputs carry the number and the architectures on top of them are level: a two-layer MLP, graph convolution over the nine graphs, and the perturbed-graph transformer read 0.87 to 0.88 per cell from the same inputs. Given the transformer's own averaging, five fold models on the cross-validation files with their logits averaged on the held-out genes, the MLP reads 0.886 [0.82, 0.94] from fold models of 0.866 to 0.892, and the convolution at 400 epochs 0.888 [0.83, 0.93] from fold models of 0.807 to 0.897 (`graph_baselines.py -arms folds`), against the transformer's 0.893 [0.84, 0.94]. Three ensembles within 0.007 of one another, each inside the others' intervals, are one number at the resolution of 195 genes, and the MLP's costs thirty seconds.

2.6 The message-passing architectures, as equations and as baselines ✗

Every graph model in the comparator set is an instance of message passing: a node's representation is updated from an aggregate of its neighbors' representations,

$$h_i^{(k)} = \gamma^{(k)}\Big(h_i^{(k-1)}, \bigoplus_{j \in \mathcal{N}(i)} \phi^{(k)}\big(h_i^{(k-1)}, h_j^{(k-1)}, e_{ji}\big)\Big), \tag{1}$$

with ϕ the message, $\bigoplus$ a permutation-invariant aggregation and γ the update, which is the form the MATH-490 project set out from and the form the Gene_Graph project's models take. The architectures differ in the three functions, and the differences are what the 1-WL result of the next subsection bears on.

GCN the symmetric-normalized mean with self-loops, $\hat{A} = A + I$, $\hat{d}_i = \sum_j \hat{A}_{ij}$: $h'_i = \Theta^\top \sum_{j \in \mathcal{N}(i) \cup \{i\}} \hat{A}_{ji}\,(\hat{d}_j \hat{d}_i)^{-1/2}\, h_j$.

GAT and GATv2 attention over the neighborhood, $h'_i = \alpha_{ii}\Theta h_i + \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \Theta h_j$, with

$$\alpha_{ij} = \mathrm{softmax}_j\big(a^\top \mathrm{LeakyReLU}(\Theta[h_i \| h_j])\big)$$

in the original form and the nonlinearity applied before the scoring vector in the v2 form, which is what makes the attention a function of both endpoints rather than a ranking of the neighbors alone; four heads here.

GraphSAGE the mean of the neighbors concatenated with the node's own representation, $h'_i = \Theta_1 h_i + \Theta_2 \,\mathrm{mean}_{j \in \mathcal{N}(i)} h_j$.

GIN the injective sum with a learned self-weight and a two-layer MLP, $h'_i = \mathrm{MLP}\big((1+\epsilon)\, h_i + \sum_{j \in \mathcal{N}(i)} h_j\big)$, the aggregator Xu et al. (2019) show to be as expressive as the 1-WL test, which the mean and the attention are not.

RGCN one weight per relation over a multigraph, $\mathcal{R}$ the relation set, $h'_i = \Theta_0 h_i + \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_r(i)} \frac{1}{c_{i,r}} \Theta_r h_j$, with Θ_r a combination of B shared bases so the parameter count does not grow with the relation count ($B = 4$ here).

RGAT the same relation-typed neighborhoods with attention within each relation, $h'_i = \sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_r(i)} \alpha_{ij}^{(r)} \Theta_r h_j$, the relational form Gene_Graph ran on its two-relation graph; here one GATv2 attention per relation on its own edge subset, summed, with no basis sharing, since the basis-shared implementation materializes a weight matrix per edge and 4.87 million typed edges times 64×64 is 80 GB.

The nine gene graphs the transformer reads enter the relational models as nine edge types on one node set, which is the multigraph the cell graph holds; the union of the nine loses the type. Gene_Graph's dual-primal convolution, which passes messages on the line graph as well as the graph, is described there and not rerun, since its attention on the primal graph reduces to GAT with self-loops. Table 6 runs GIN, GraphSAGE, RGCN and RGAT under the protocol of Table 5, with ProtT5 alone and with the transformer's three inputs, and Table 7 runs the MLP and GCN with the

chromosome-pathway indicator split into its blocks, the cheap half of the attribution that Section 3 completes on the transformer.

Table 6: The message-passing architectures of the MATH-490 and Gene_Graph projects on the released split, the protocol of Table 5: GIN and GraphSAGE on the union of the nine graphs, and RGCN and RGAT with the nine graphs as nine edge types (RGCN with a basis decomposition of four bases; RGAT as one GATv2 attention per relation, summed, since the basis-shared form holds a weight matrix per edge), with ProtT5 alone and with the transformer's three inputs as node features, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
GIN	union of nine	ProtT5	1,024	78,147	0.665	0.059	0.696, 0.583, 0.718	186, 171, 195
GraphSAGE	union of nine	ProtT5	1,024	139,457	0.783	0.002	0.784, 0.785, 0.780	87, 77, 82
RGCN	nine typed	ProtT5	1,024	348,425	0.836	0.007	0.844, 0.828, 0.836	119, 87, 81
RGAT	nine typed	ProtT5	1,024	1,258,049	0.852	0.011	0.853, 0.865, 0.839	196, 111, 113
GIN	union of nine	ProtT5 + chrom-pathway + FBA ratios	1,225	91,011	0.705	0.033	0.664, 0.744, 0.706	195, 182, 177
GraphSAGE	union of nine	ProtT5 + chrom-pathway + FBA ratios	1,225	165,185	0.887	0.002	0.886, 0.890, 0.885	98, 93, 92
RGCN	nine typed	ProtT5 + chrom-pathway + FBA ratios	1,225	412,745	0.860	0.009	0.847, 0.868, 0.866	181, 139, 134
RGAT	nine typed	ProtT5 + chrom-pathway + FBA ratios	1,225	1,489,601	**0.889**	0.004	0.887, 0.885, 0.895	153, 84, 118

Keeping the edge type is worth more than any choice of aggregator. On ProtT5 alone the relational models read 0.852 (RGAT) and 0.836 (RGCN) against 0.834 for graph convolution over the untyped union at the same 200 epochs, 0.783 for GraphSAGE and 0.665 for GIN; on the transformer's three inputs RGAT reads 0.887, 0.885 and 0.895 (mean 0.889), GraphSAGE 0.887, RGCN 0.860, GIN 0.705, against 0.847 for graph convolution. RGAT on the three inputs is the strongest of the cheap models in this document, level with the transformer's best single cell on the pathway input (Section 3.13) and 0.019 above the MLP on the same three inputs. GIN, the aggregator as expressive as the 1-WL test, is the weakest on both inputs and selects epochs 171 to 195 on five seeds of six, the sign of a model still moving at the end of its budget rather than of a converged one; what the next subsection measures is that 1-WL expressiveness is not the axis this endpoint turns on, so the ranking of the aggregators here says nothing about it. GraphSAGE, whose only difference from graph convolution is the separate weight on the node's own representation, gains 0.04 on the three inputs and loses 0.05 on ProtT5 alone.

Table 7: The chromosome-pathway indicator split into its three blocks (the 173-column pathway multi-hot, the 17-column chromosome one-hot, the seven scaled gene properties), each beside ProtT5 and the four FBA ratios, and the pathway block with the ratios and alone, on the MLP with no graph and on graph convolution over the union of the nine graphs, the protocol of Table 5, seeds 2700 to 2702. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
MLP	none	ProtT5 + pathways + FBA ratios	1,201	81,153	**0.880**	0.004	0.876, 0.885, 0.878	62, 56, 59
GCN	union of nine	ProtT5 + pathways + FBA ratios	1,201	81,153	0.846	0.001	0.845, 0.844, 0.847	197, 196, 197
MLP	none	ProtT5 + chromosome + FBA ratios	1,045	71,169	0.809	0.001	0.810, 0.809, 0.808	66, 64, 66
GCN	union of nine	ProtT5 + chromosome + FBA ratios	1,045	71,169	0.834	0.005	0.839, 0.837, 0.827	197, 197, 196
MLP	none	ProtT5 + properties + FBA ratios	1,035	70,529	0.798	0.012	0.788, 0.814, 0.791	197, 74, 196
GCN	union of nine	ProtT5 + properties + FBA ratios	1,035	70,529	0.834	0.003	0.834, 0.829, 0.838	196, 197, 197
MLP	none	pathways + FBA ratios	177	15,617	0.828	0.001	0.827, 0.828, 0.829	81, 72, 84
GCN	union of nine	pathways + FBA ratios	177	15,617	0.834	0.011	0.832, 0.847, 0.821	197, 197, 197
MLP	none	pathways	173	15,361	0.749	0.022	0.719, 0.761, 0.769	89, 81, 84
GCN	union of nine	pathways	173	15,361	0.871	0.000	0.871, 0.872, 0.871	197, 197, 197

The pathway block is the indicator. On the MLP, ProtT5 with the FBA ratios reads 0.803 (Table 5); adding the 173-column pathway multi-hot lifts it to 0.880, adding the chromosome one-hot leaves it at 0.809 and adding the seven gene properties at 0.798, both within one seed spread of the base, and the full 197-column indicator reads 0.870, below the pathway block alone. The pathway block with the four ratios and no ProtT5 reads 0.828, above ProtT5 with the ratios, so 177 columns carry more of the label than the 1,024-dimensional sequence embedding does. On graph convolution at 200 epochs the ordering is the same at a smaller spread (0.846 with the pathway block against 0.834 with either other block), with the selected epoch at 196 or 197 on every seed, the sign that the 200-epoch GCN is still climbing, as the 400-epoch rows of Table 5 showed. The one row where the graph matters is the pathway multi-hot alone: 0.749 on the MLP and 0.871 with convolution over the union of the nine graphs, the largest gain the graph gives any input in this document. The Gene Ontology multi-hot, the larger annotation of the same kind, runs under the same protocol (Table 8); its transformer round is Section 3.14.

Table 8: The Gene Ontology multi-hot on the MLP with no graph and on graph convolution over the union of the nine graphs, the protocol of Table 5: the propagated vector (2,362 columns), the biological-process namespace (1,580) and the direct annotations (2,887), each beside ProtT5 and the four FBA ratios; the propagated vector with the ratios and alone; the propagated vector with the pathway multi-hot beside ProtT5 and the ratios; and, as the control for the pathway block, one column that marks a gene annotated to any pathway (517 of 6,607 genes) beside ProtT5 and the ratios. Seeds 2700 to 2702. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
MLP	none	ProtT5 + pathway any + FBA ratios	1,029	70,145	0.774	0.019	0.761, 0.761, 0.801	197, 194, 75
GCN	union of nine	ProtT5 + pathway any + FBA ratios	1,029	70,145	0.832	0.003	0.831, 0.836, 0.829	197, 197, 197
MLP	none	ProtT5 + GO + FBA ratios	3,390	221,249	0.855	0.001	0.854, 0.854, 0.856	28, 27, 27
GCN	union of nine	ProtT5 + GO + FBA ratios	3,390	221,249	0.887	0.002	0.889, 0.888, 0.884	193, 195, 197
MLP	none	ProtT5 + GO BP + FBA ratios	2,608	171,201	0.819	0.003	0.822, 0.815, 0.820	191, 197, 178
GCN	union of nine	ProtT5 + GO BP + FBA ratios	2,608	171,201	0.885	0.003	0.882, 0.886, 0.887	196, 195, 197
MLP	none	ProtT5 + GO direct + FBA ratios	3,915	254,849	0.848	0.004	0.842, 0.851, 0.849	44, 16, 44
GCN	union of nine	ProtT5 + GO direct + FBA ratios	3,915	254,849	0.877	0.003	0.873, 0.881, 0.878	195, 197, 196
MLP	none	GO + FBA ratios	2,366	155,713	0.818	0.012	0.806, 0.834, 0.813	2, 29, 2
GCN	union of nine	GO + FBA ratios	2,366	155,713	**0.891**	0.003	0.895, 0.888, 0.890	195, 197, 197
MLP	none	GO	2,362	155,457	0.806	0.008	0.795, 0.810, 0.812	2, 13, 10
GCN	union of nine	GO	2,362	155,457	0.888	0.001	0.887, 0.889, 0.887	194, 196, 183
MLP	none	ProtT5 + GO + pathways + FBA ratios	3,563	232,321	0.859	0.002	0.862, 0.857, 0.859	28, 24, 30
GCN	union of nine	ProtT5 + GO + pathways + FBA ratios	3,563	232,321	0.884	0.003	0.885, 0.887, 0.880	196, 196, 193

Graph convolution on the GO multi-hot alone reads 0.888, with no sequence embedding and no solve, and 0.891 with the four ratios: level with the transformer's best single cell and above the MLP on all three of its inputs (0.870). On the MLP the same vector beside ProtT5 and the ratios reads 0.855, 0.025 under the pathway block, so without the graph GO carries less than the pathways and with it more; the propagated form beats the direct annotations on both models (0.887 against 0.877 on convolution), and the biological-process namespace alone loses nothing on convolution (0.885) and 0.036 on the MLP. Every convolution cell on GO selects its epoch between 183 and 197, still rising at the end of the budget. The one-column control that marks a gene as annotated to any pathway reads 0.774 on the MLP and 0.832 on convolution, the level of ProtT5 with the ratios, so the pathway block's gain is which pathway rather than whether there is one.

Given its budget, graph convolution on GO alone is the highest single model in this document. At 400 epochs it reads 0.912, 0.910 and 0.904 (mean 0.909, sd 0.003), with the four ratios 0.908, and beside ProtT5 with the ratios 0.901 (Table 9): two layers of convolution over the union of the nine gene graphs on 2,362 annotation columns, no sequence embedding and no solve, 43 seconds per seed on one card, above the transformer's best single cell (0.895) and above the reported fold ensemble's point (0.893) inside its interval. The three seeds still select epochs 382 to 397, but at 800 epochs the number does not move: 0.905, 0.911 and 0.901 (mean 0.906), 0.907 with the ratios, 0.903 beside ProtT5 (Table 10), the selected epochs again near the end of the budget and the held-out number the same, so the curve is flat past 400 and the selection rule is reading noise on its plateau. The sequence embedding and the solve add nothing to it at either budget, where on every other input they were the gain.

Table 9: Graph convolution over the union of the nine graphs on the propagated GO multi-hot at a 400-epoch budget, since every 200-epoch cell on it selected its epoch between 183 and 197: GO alone, GO with the four FBA ratios, and GO beside ProtT5 with the ratios. The protocol of Table 5 otherwise; seeds 2700 to 2702. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
GCN	union of nine	GO (400 epochs)	2,362	155,457	0.909	0.004	0.913, 0.910, 0.903	397, 395, 379
GCN	union of nine	GO + FBA ratios (400 epochs)	2,366	155,713	**0.909**	0.005	0.905, 0.906, 0.915	396, 392, 393
GCN	union of nine	ProtT5 + GO + FBA ratios (400 epochs)	3,390	221,249	0.901	0.003	0.900, 0.898, 0.905	397, 368, 391

Table 10: The three rows of Table 9 at an 800-epoch budget, since at 400 the three seeds still selected epochs 382 to 397. Seeds 2700 to 2702. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
GCN	union of nine	GO (800 epochs)	2,362	155,457	0.906	0.004	0.905, 0.911, 0.901	777, 784, 788
GCN	union of nine	GO + FBA ratios (800 epochs)	2,366	155,713	**0.908**	0.002	0.905, 0.911, 0.908	768, 790, 769
GCN	union of nine	ProtT5 + GO + FBA ratios (800 epochs)	3,390	221,249	0.903	0.001	0.905, 0.902, 0.903	769, 751, 792

On six seeds the number holds. Rerun on seeds 2700 to 2705 so its count matches the six-seed module arms (Table 11), the 400-epoch convolution on GO alone reads 0.912, 0.909, 0.902, 0.898, 0.901 and 0.905 (mean 0.905, sd 0.005), with the ratios 0.899 to 0.913 (mean 0.907) and beside ProtT5 with the ratios 0.895 to 0.905 (mean 0.900); the first three seeds differ from their earlier runs in the third decimal, and every one of the eighteen cells is above the transformer's seed mean on any input. The spread over seeds is 0.005, against 0.055 for the module's scalable form on the same six seeds (Section 4.22), whose one unfit run is the difference.

Table 11: The three rows of Table 9 on seeds 2700 to 2705, so the graph convolution's count matches the six-seed module arms of Section 4.22. The first three seeds are reruns of the Table 9 cells and differ from them in the third decimal. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
GCN	union of nine	GO (400 epochs)	2,362	155,457	0.905	0.005	0.912, 0.909, 0.902, 0.898, 0.901, 0.905	397, 397, 379, 387, 397, 396
GCN	union of nine	GO + FBA ratios (400 epochs)	2,366	155,713	**0.907**	0.004	0.904, 0.906, 0.913, 0.908, 0.899, 0.909	396, 392, 397, 397, 391, 396
GCN	union of nine	ProtT5 + GO + FBA ratios (400 epochs)	3,390	221,249	0.900	0.003	0.902, 0.898, 0.905, 0.898, 0.900, 0.895	397, 368, 391, 391, 386, 397

Under the reported protocol it is the highest number in this document. Five fold models of the 400-epoch convolution on GO alone score 0.910, 0.893, 0.918, 0.916 and 0.907 on the held-out genes, a fold mean of 0.909, and their averaged logits score 0.917 [0.87, 0.96]; at 800 epochs 0.915 [0.87, 0.95]. Against the transformer's reported ensemble (folds 0.860, 0.899, 0.889, 0.876, 0.884; 0.893 [0.84, 0.94]) the folds are above on four of five, the fold mean by 0.028 and the ensemble by 0.024, with the two intervals overlapping (Table 55). The five folds take two minutes on one card. With three seeds per fold, each fold's logit the mean of its seeds before the folds are averaged, the ensemble reads 0.917 [0.87, 0.95] at 400 epochs and 0.915 [0.87, 0.95] at 800, the same numbers as one seed per fold (folds 0.899 to 0.922, mean 0.910), so the ensemble is not sitting on a lucky seed. So the three architectures on the three inputs that were one number (0.886, 0.888, 0.893) now have a fourth point above them, and what moved it is the input and the graph together, not the architecture: the same convolution on the three inputs reads 0.888 as an ensemble, and the GO vector without the graph reads 0.855 on the MLP.

It is the convolution, not message passing in general. On GO alone at 400 epochs relational convolution reads 0.843, relational attention 0.844 and GraphSAGE 0.823 (Table 12), 0.07 to 0.09 under the symmetric-normalized mean, where on the three inputs those same forms were level with or above it. The pathway multi-hot added to GO reads 0.905, nothing on top of GO alone. What the normalized mean does that the others do not, on this input, is not measured; hypothesis, untested: the signal is the neighborhood's annotations smoothed over the graph, and a form with a separate path for the node's own 2,362 columns fits those instead.

Table 12: Relational convolution and relational attention over the nine graphs as nine edge types on the propagated GO multi-hot alone, GraphSAGE over the union on it, and graph convolution over the union on GO with the pathway multi-hot, all at 400 epochs, the protocol of Table 5 otherwise; seeds 2700 to 2702. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
RGCN	nine typed	GO (400 epochs)	2,362	776,585	0.843	0.019	0.831, 0.829, 0.869	101, 149, 91
RGAT	nine typed	GO (400 epochs)	2,362	2,799,425	0.845	0.009	0.836, 0.843, 0.858	34, 56, 25
GCN	union of nine	GO + pathways (400 epochs)	2,535	166,529	**0.904**	0.002	0.906, 0.901, 0.905	388, 396, 397
GraphSAGE	union of nine	GO (400 epochs)	2,362	310,721	0.823	0.001	0.822, 0.824, 0.824	307, 380, 394

2.7 The Weisfeiler-Leman test on the deletion graphs ✗

A message-passing model over a gene graph is bounded by what color refinement can separate: two graphs that the 1-WL test assigns the same signature are the same graph to it, whatever the model's weights (Xu et al. 2019). The transformer's perturbed graph is the gene graph with the deleted gene's token removed, so the first question is whether single deletions are even distinguishable by structure, and the second is whether structure alone ranks them. `wl_knockout_test.py` answers both on the physical interaction graph and on the union of the nine graphs the transformer reads (Table 13): for every labeled gene it hashes the graph with that gene's node removed by color refinement at one, two and three iterations with degree as the initial color, counts collisions, and fits logistic regression on the WL subtree kernel's feature map of each gene's 2-hop ego graph. Figure 2 illustrates the refinement and the kernel on a toy pair of six-node graphs, not the yeast graphs.

Table 13: The 1-WL test on the single-deletion graphs. For every labeled gene the gene graph with that gene's node removed is hashed by Weisfeiler-Leman color refinement with degree as the initial color; a collision is a set of deletions with one hash, which no 1-WL-bounded message-passing model can tell apart by structure. The WL kernel rung is logistic regression on the bag of colors of each gene's 2-hop ego graph refined for three iterations, hashed to 1,024 bins, with C chosen by five-fold cross-validation inside the training genes: physical: C = 0.001, CV 0.578, validation 0.626, held-out 0.490; union of nine: C = 0.01, CV 0.443, validation 0.478, held-out 0.564.

graph	edges	k	deletions	distinct	in collision	largest	essential	min
physical	139,463	1	1,121	1,118	6	2	0	11
physical	139,463	2	1,121	1,118	6	2	0	10
physical	139,463	3	1,121	1,118	6	2	0	18
union of nine	1,516,034	1	1,121	1,121	0	1	0	130
union of nine	1,516,034	2	1,121	1,121	0	1	0	91
union of nine	1,516,034	3	1,121	1,121	0	1	0	89

The test separates the deletions and the kernel does not rank them. On the physical graph, one iteration already gives 1,118 distinct signatures among 1,121 deletions, three pairs collide, none of the six genes in them is essential, and two more iterations change nothing; on the union every deletion has its own signature at one iteration. A single deletion changes the degree of every neighbor, and on graphs this dense (139,463 and 1,516,034 edges) that is enough for the test, so 1-WL expressivity is not what bounds a message-passing model on this endpoint. The kernel rung reads 0.490 on the physical graph and 0.564 on the union, chance on 195 genes, which is the same reading the graph-only baselines gave with all-ones and degree inputs: a gene's neighborhood structure, however finely refined, does not say whether the gene is essential, and every number above chance in this document comes from what is written on the nodes.

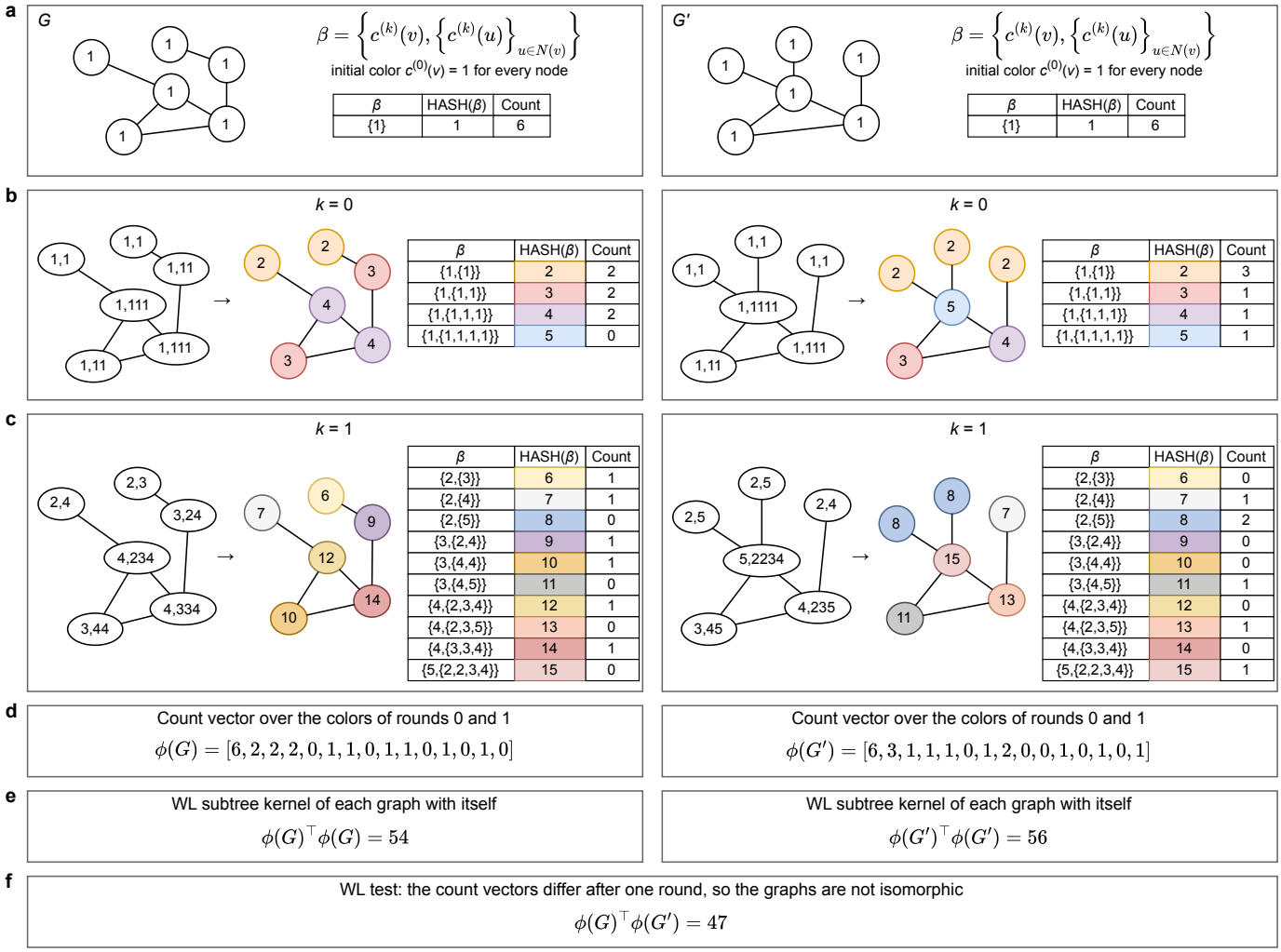


Figure 2: The 1-WL test and the WL subtree kernel on a toy pair of graphs. **a**, Two six-node graphs G and G' with six edges each, every node given the initial color 1; each round of color refinement replaces a node's color by the hash of the multiset β of its own color and its neighbors' colors, and the table counts the nodes of each color. **b**, Round $k = 0$: the multiset at each node (left), the hashed colors (right) and the count of each color; a node's new color is its degree in disguise, and G' alone holds a degree-4 node. **c**, Round $k = 1$: the same refinement on the round-0 colors, where the two graphs share one of the ten colors. **d**, The count vector ϕ over the colors of rounds 0 and 1, the WL subtree kernel's feature map. **e**, The kernel of each graph with itself, $\phi^\top \phi$. **f**, The kernel across the pair; the count vectors differ after one round, so the test declares the graphs non-isomorphic. Drawn in draw.io from `notes/assets/drawio/028-wl-test.drawio`; the yeast deletion graphs are hashed the same way by `wl_knockout_test.py`.

3 The backbone: splits, labels, weight decay and the shape of the curve ✗

Every cell in this section is the pooled backbone at width 12 with one transformer block and a one-layer preprocessor (16,874 parameters), trained for 80 epochs at three initialization seeds with the split pinned. Round 1 (jobs 2149 to 2181) moves one lever at a time away from the cell that trains on the released-label split file (`ess_fcl_val0.2_s0`, 629 training genes, 158 validation genes): the old 75-gene validation draw, the genome-wide labels, weight decay at 0, 10^{-2} and 10^{-1}, batch 32, learning rate at 3×10^{-4} and 3×10^{-3}, the class weight 772/126, and graph regularization off. Table 14 carries every number; Figure 3 the curves.

Table 14: Round 1, the backbone. Width 12, one block, one-layer preprocessor, 80 epochs, seeds 2700 to 2702. Held-out AUROC on the 195 genes: the seed mean with its standard deviation, the three seeds, the gene bootstrap of the pooled per-gene logits, the epochs selected on validation, and the median absolute epoch-to-epoch step of the validation curve. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
wave-11 protocol (75-gene val)	0.759	0.020	0.781, 0.755, 0.741	0.760 [0.67, 0.84]	18, 38, 14	0.0056
released labels, 158-gene val	0.725	0.019	0.745, 0.722, 0.708 ↓	0.738 [0.65, 0.83]	66, 77, 77	0.0026
weight decay 0	0.725	0.019	0.745, 0.720, 0.709 ↓	0.735 [0.64, 0.82]	66, 77, 77	0.0027
weight decay 1e-2	0.726	0.017	0.745, 0.722, 0.711 ↓	0.735 [0.64, 0.83]	65, 77, 77	0.0029
weight decay 1e-1	0.725	0.018	0.746, 0.721, 0.709 ↓	0.735 [0.64, 0.82]	66, 77, 77	0.0025
batch 32	**0.765**	0.005	0.766, 0.770, 0.759	0.764 [0.67, 0.85]	74, 73, 77	0.0032
genome labels (5,795 genes)	0.704	0.024	0.706, 0.726, 0.679 ↓	0.705 [0.61, 0.80]	77, 77, 69	0.0020
lr 3e-4	0.658	0.093	0.737, 0.682, 0.556 ↓	0.732 [0.64, 0.82]	77, 77, 74	0.0023
lr 3e-3	0.686	0.043	0.718, 0.704, 0.637 ↓	0.713 [0.62, 0.81]	77, 77, 50	0.0024
class weight 772/126	0.759	0.016	0.764, 0.772, 0.741	**0.769** [0.68, 0.86]	77, 75, 53	0.0034
graph regularization off	0.725	0.019	0.745, 0.722, 0.709 ↓	0.738 [0.65, 0.83]	66, 77, 77	0.0026

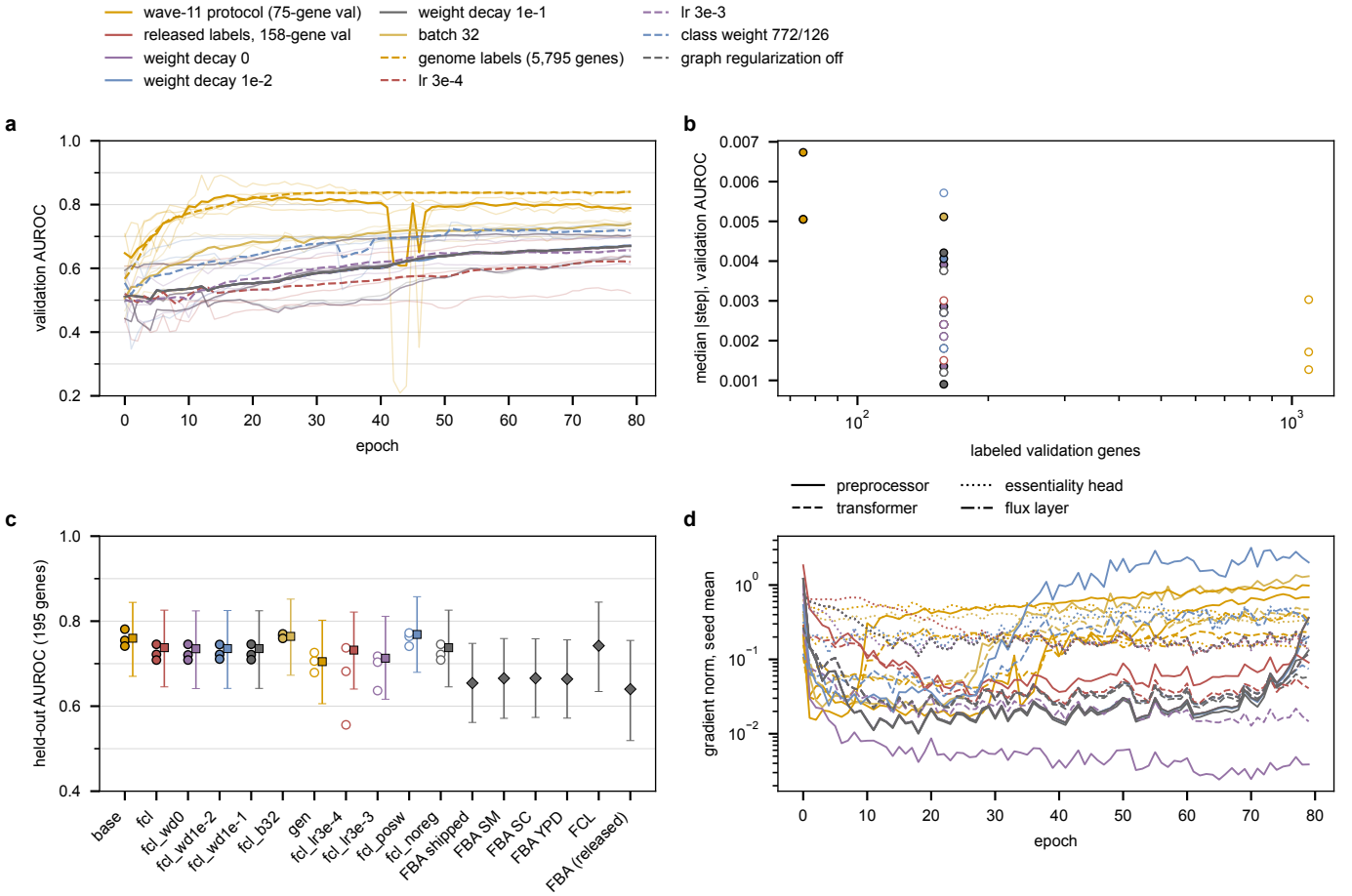


Figure 3: Round 1, the backbone. **a**, Validation AUROC per epoch, thin lines per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step of that curve against the number of labeled validation genes it was read on. **c**, Held-out AUROC on the 195 genes per cell, the seed mean, and the gene bootstrap of the pooled per-gene logits, beside FBA on each medium. **d**, Per-module gradient norm per epoch, seed mean, on the arms that logged it. Generated by `scripts/wave12_readout.py -round w12`.

3.1 The curve was noisy because the validation set was small ✗

The median epoch-to-epoch step of the validation AUROC is 0.0056 on the 75-gene draw, 0.0026 on the 158-gene split and 0.0020 on the 1,093-gene genome-wide split (Figure 3b). Two of the three width-24 cells of the previous protocol selected epochs 4 and 5 on validation peaks of 0.82 and 0.86 and scored 0.74 on the held-out genes; no cell on a split file selects before epoch 50. The old protocol's curve reads near 0.8 on its own 75 genes and the split cells read 0.65 to 0.72 on theirs, so the levels are not comparable, only the steps are.

3.2 The split file costs training genes, and the cells are under-trained ✗

The split-file cell scores 0.725 against the old protocol's 0.759 on the same held-out genes. It trains on 629 labeled genes where the old draw trained on roughly 740, and every one of its seeds selected epoch 66 to 77 of 80 with the validation curve still rising (Figure 3a). At batch 128 an epoch is five optimizer steps, so 80 epochs is 400 updates. The two levers that move all three seeds in the same direction are both more optimization: batch 32, four times the steps per epoch, lifts the seeds by 0.020, 0.048 and 0.051 and collapses their spread to 0.005; the class weight, which steepens the curve, lifts them by 0.019, 0.051 and 0.033. Round 3 combines the two and raises the epoch budget.

3.3 Weight decay and graph regularization do nothing here ✗

Weight decay at 0, 10^{-2} and 10^{-1} reproduces the base cell to the third decimal on every seed, and so does removing graph regularization. At 400 updates with learning rate 10^{-3}, a decay of 10^{-1} shrinks a weight by four percent, so it is not a lever at this budget. Graph regularization is a different case: its configuration pins the nine gene graphs to attention heads 0 to 8 of the *second* transformer block, so on a one-block cell the term is never evaluated, and the arm with it off reproduces the arm with it on because both are the same cell. The term was inert on every one-block cell in this document; a two-block cell with four heads indexes a head it does not have and fails (round 8, job 2261), so the two-block cells run with it off. The learning rate at 3×10^{-4} and 3×10^{-3} is worse than 10^{-3} on the seed mean, with seed spreads of 0.09 and 0.04 that put the pooled numbers inside the base interval.

3.4 More labels from the genome make the held-out ranking worse ✗

The genome-wide file adds SGD's essential list and the viable single-mutant fitness records as nonessential, 4,368 training genes against 629, and scores 0.704 against 0.725, three of three seeds down. Its validation AUROC on its own 1,093 genes is 0.84 while the released held-out genes read 0.70, the widest validation-to-held-out gap of any arm. The viable-fitness negatives are a different population from the released split's nonessential genes, and a model that ranks them well does not rank the released genes better. Hypothesis, untested: the released split's nonessential class is enriched for genes with a measurable fitness defect, which the genome-wide negatives are not.

3.5 Gradient norms ✗

On the split-file cells the preprocessor's gradient norm starts near 10^{-2} and rises about thirty-fold over 80 epochs while the head's stays flat near 0.4 (Figure 3d); the clip at 10 never binds. The rise tracks the validation curve's climb, which is the same under-training reading from the gradient side.

3.6 Round 3: the two levers combined, a longer budget, and the fold ensemble ✗

Round 3 (jobs 2209 and 2224 to 2236) combines the two round-1 levers that moved three of three seeds, batch 32 and the class weight, at 80 and at 200 epochs, runs batch 32 alone at 200 epochs as the class weight's control, and trains five fold models on the cross-validation files, which partition the 787 labeled non-test genes five ways with the held-out genes fixed; their logits are averaged on the held-out genes, the protocol behind the published Flux Cone Learning number (Table 15, Figure 4). With the loader workers removed a 200-epoch cell takes 11 minutes.

Table 15: Round 3, more optimization and the fold ensemble. Batch 32 with the class weight at 80 and 200 epochs, batch 32 alone at 200 epochs, and the five fold models of the cross-validation files, each on its own fold. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 80 epochs	0.750	0.043	0.780, 0.769, 0.700	0.774 [0.69, 0.86]	75, 68, 31	0.0026
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
batch 32, 200 epochs	0.761	0.005	0.766, 0.761, 0.755	0.764 [0.67, 0.85]	74, 100, 181	0.0026
5-fold, fold 0	0.780	0.000	0.780	0.780 [0.69, 0.86]	75	0.0033
5-fold, fold 1	**0.806**	0.000	0.806	**0.806** [0.72, 0.89]	50	0.0036
5-fold, fold 2	0.797	0.000	0.797	0.797 [0.72, 0.87]	14	0.0033
5-fold, fold 3	0.751	0.000	0.751	0.751 [0.66, 0.84]	184	0.0021
5-fold, fold 4	0.789	0.000	0.789	0.789 [0.70, 0.87]	108	0.0039

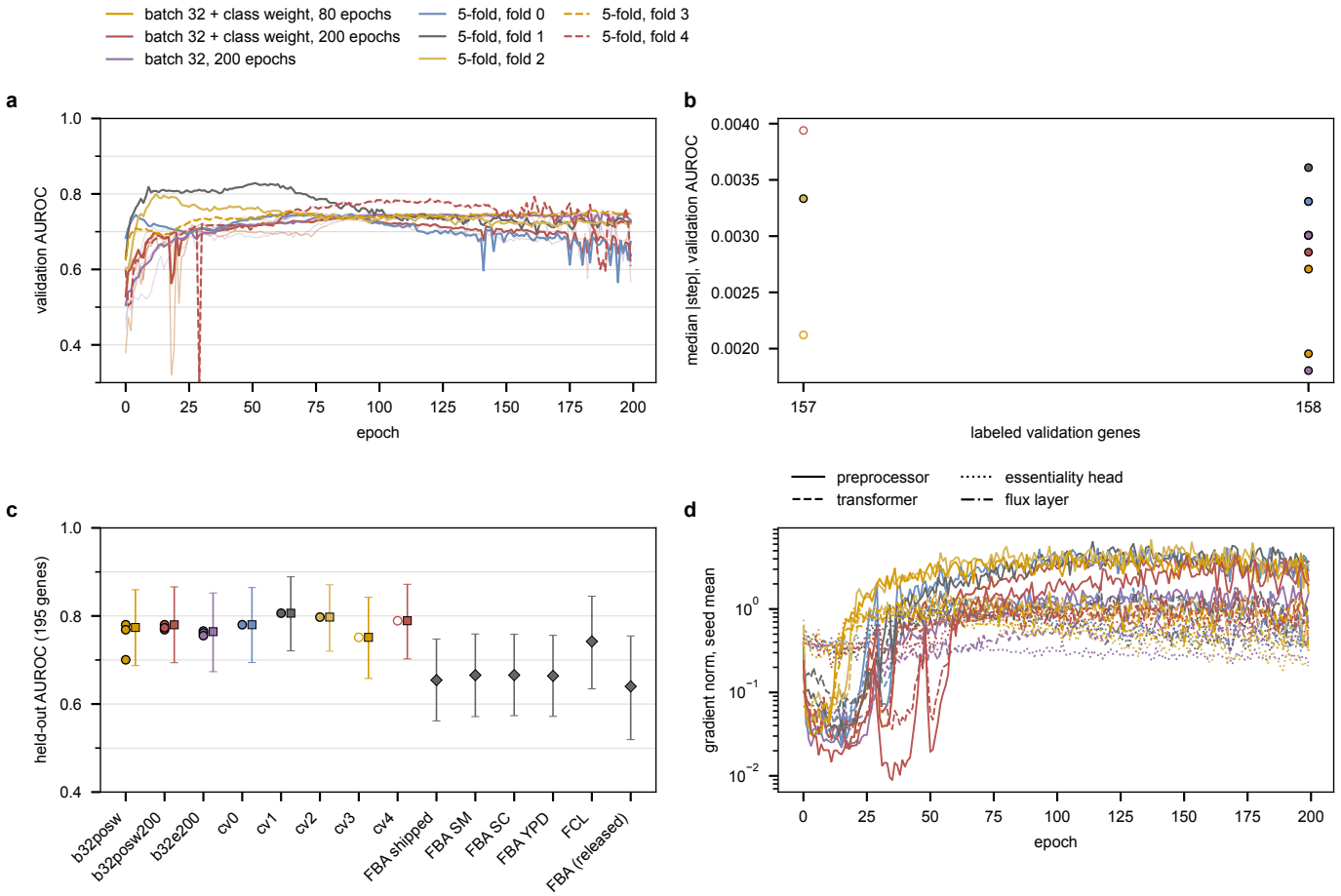


Figure 4: Round 3, the levers combined and the fold models. a, Validation AUROC per epoch, seed mean per arm; each fold model reads its own validation fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w14`.

Batch 32 with the class weight at 200 epochs scores 0.780, 0.769 and 0.773, a seed mean of 0.774 with a spread of 0.006, the tightest cell of any round; the pooled bootstrap is 0.780 [0.69, 0.87]. Batch 32 alone at 200 epochs scores 0.766, 0.761 and 0.755, so the class weight adds 0.01 to 0.02 on three of three seeds at the longer budget as it did at 80. The longer budget itself moves one seed: the 80- and 200-epoch cells share a trajectory through epoch 80 and select the same epoch on seeds 2700 and 2701 (75 and 68), while seed 2702 selects epoch 31 at 80 epochs and epoch 148 at 200, 0.700 against 0.773. Fold 0 of the cross-validation files is the released-label split file exactly (same 629 training, 158 validation and 195 held-out genes, both drawn at seed 0), so its fold model is the 200-epoch cell's first seed. The five fold models score 0.780, 0.806, 0.797, 0.751 and 0.789 on the held-out genes, a fold mean of 0.785, and their averaged logits score 0.795 [0.71, 0.88] (`cv_ensemble_readout.py`). That is the number to set beside the published 0.73 and beside 0.742 on these genes: the backbone under Flux Cone Learning's own protocol is 0.05 above it, and 0.01 to 0.04 below the ladder's rungs with FBA appended.

3.7 Round 5: FBA's growth ratios as an input ✗

The ladder's rungs cross 0.80 only with FBA's four growth ratios appended to the encoding, and no transformer cell has seen those numbers. Round 5 (jobs 2249 and 2250) gives the round-3 cell the same complement: the `fba` readout adds an affine of the deleted gene's knockout-to-wild-type growth ratio on the four media to the context head's logit, five parameters on top of the 16,561, with the four weights written into every epoch row. It runs at the three seeds, paired against the round-3 cell, and on the five cross-validation folds for the ensemble (Table 16, Figure 5). The read was fixed first: the transformer gains the ladder's complement if its seed mean moves from 0.774 toward 0.82 on three of three seeds and the fold ensemble from 0.795 toward the ladder's 0.81 to 0.83.

Table 16: Round 5, FBA's growth ratios as an input. The round-3 pooled cell and the same cell with an affine of FBA's four per-medium growth ratios for the deleted gene added to the context head, at three seeds, and that readout on the five cross-validation folds whose logits are averaged on the held-out genes. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844 ↑	0.838 [0.76, 0.91]	28, 70, 61	0.0036
context head + FBA ratios, fold 0	0.814	0.000	0.814	0.814 [0.73, 0.89]	28	0.0033
context head + FBA ratios, fold 1	0.824	0.000	0.824	0.824 [0.74, 0.90]	42	0.0042
context head + FBA ratios, fold 2	0.839	0.000	0.839	0.839 [0.77, 0.91]	37	0.0048
context head + FBA ratios, fold 3	0.775	0.000	0.775	0.775 [0.69, 0.86]	122	0.0042
context head + FBA ratios, fold 4	**0.853**	0.000	0.853	**0.853** [0.78, 0.92]	54	0.0036

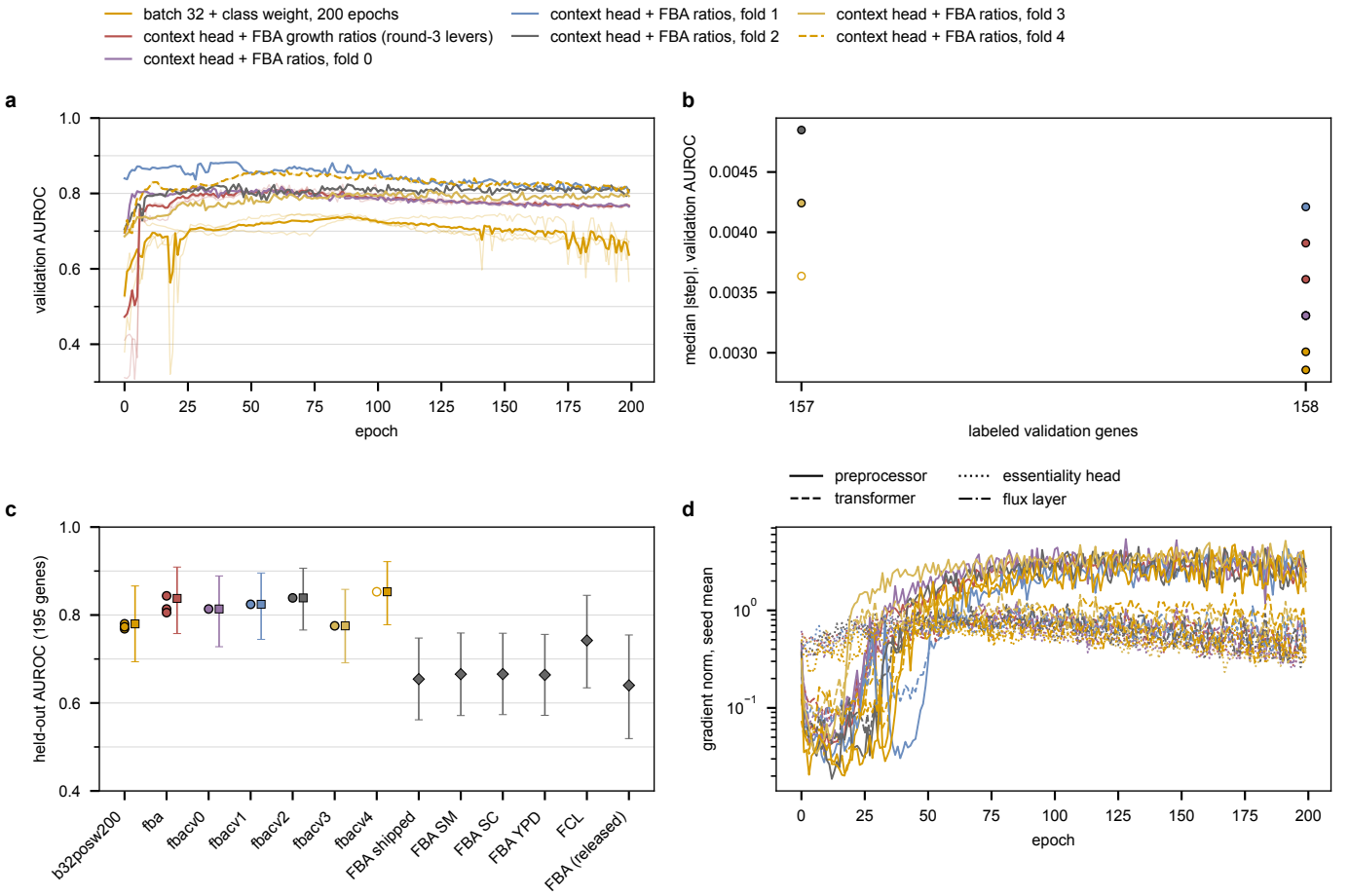


Figure 5: Round 5, FBA's growth ratios as an input. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; each fold model reads its own fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w16`.

Both halves of the read hold. The cell scores 0.814, 0.805 and 0.844 against 0.780, 0.769 and 0.773 on the same seeds, up by 0.034, 0.037 and 0.071, a seed mean of 0.821 and a pooled bootstrap of 0.838 [0.76, 0.91]. The five fold models score

0.814, 0.824, 0.839, 0.775 and 0.853, a fold mean of 0.821, and their averaged logits 0.822 [0.74, 0.90] on the held-out genes, against 0.795 without the ratios. The transformer with the solver's four numbers sits where the ladder's best rungs sit (0.806 to 0.830), 0.08 above Flux Cone Learning on the same genes and 0.09 above its published number. The four weights end negative on every cell, largest on the shipped-bound and YPD columns (about −0.5 to −0.7 at the selected epoch against −0.3 to −0.5 on SM and near zero to −0.4 on SC), which is the direction a growth ratio must carry (a deletion that stops growth is essential), and the cells select earlier, at epochs 28 to 70 against 68 to 148: the affine is learned in the first epochs and the context head fills in around it.

The two sources are complementary and the complement is the same for the transformer as for a logistic regression on a frozen embedding: the backbone alone reads 0.77, the solver alone 0.66, and the sum 0.82. What the flux layer could not supply in rounds 2 and 4, where its growth column matched the head and its k_{cat} content went unread, the FBA solve supplies as four numbers, which locates the module's value on this endpoint in the solve itself rather than in the learned layer.

3.8 Round 8: width and depth on the FBA cell ✗

The round-5 cell ties the ladder's best rungs and does not pass them, and it is the smallest transformer in the document. Round 8 (jobs 2259, 2260, 2262 and 2263) moves its size one axis at a time with the readout, batch, class weight and budget fixed: width 24 and width 48 at one block (40,902 and 112,902 parameters against 16,566), two blocks at width 12 (18,450), and the two-layer preprocessor (542,480, most of it in the preprocessor's first layer), each at the three seeds, three cells packed on one card (Table 17, Figure 6). The read was fixed first: a cell passes the ladder if its seed mean clears 0.83 with three of three seeds above the round-5 cell.

Table 17: Round 8, width and depth on the FBA cell. The round-5 cell (width 12, one block, one-layer preprocessor, context head plus the FBA affine) against the same readout at width 24 and 48, at two blocks, and with the two-layer preprocessor. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
context head + FBA ratios, width 24	0.817	0.005	0.818, 0.822, 0.812	0.823 [0.74, 0.90]	50, 41, 66	0.0045
context head + FBA ratios, width 48	**0.824**	0.004	0.827, 0.819, 0.826	0.831 [0.74, 0.91]	58, 86, 64	0.0037
context head + FBA ratios, two blocks	0.812	0.007	0.810, 0.820, 0.806	0.827 [0.74, 0.90]	57, 144, 41	0.0032
context head + FBA ratios, two-layer preprocessor	0.821	0.014	0.804, 0.829, 0.828	0.820 [0.74, 0.90]	82, 73, 103	0.0029

No cell passes. Width 48 has the highest seed mean of the four, 0.824 against 0.821, with a seed spread of 0.004, and it is up on seeds 2700 and 2701 by 0.013 and 0.014 and down on seed 2702 by 0.018; width 24 scores 0.817, the two-layer preprocessor 0.821, and two blocks 0.812, the last down on two of three seeds. Seed 2702 of the round-5 cell, 0.844, is the highest single seed in the document and every round-8 cell is below it on that seed, so no arm clears the bar on all three. The seed spread narrows with width, from 0.020 at width 12 to 0.005 at 24 and 0.004 at 48, and every round-8 seed mean lies inside the round-5 cell's pooled bootstrap of 0.84 [0.76, 0.91]. The validation curves peak at 0.80 to 0.82 between epochs 40 and 90 on every arm and drift down from there, most at width 24, and the cells select epochs 41 to 144. The two-block cells run with graph regularization off, since the term indexes a second-block head that a four-head block does not have (the failed job 2261); it was inert on every one-block cell, so the comparison is like for like.

A width-12 transformer with FBA's four numbers reads 0.82 on these genes and neither four times the width, a second block, nor a wider preprocessor moves it by more than the seed noise. Capacity is not the lever on this endpoint and this split; the levers left untested are the gene encoding the transformer reads and what the solve supplies beyond four growth ratios.

3.9 Round 9: the FBA cell on the ladder's other encodings ✗

Every transformer cell so far reads ProtT5, and the ladder's best selected rung reads something else: logistic regression on the 197-dimensional chromosome-pathway indicator with the four ratios scores 0.830. Round 9 (jobs 2280 to 2283) opens the round-5 cell on the ladder's other encodings through `-node-embeddings`, with the readout, width, batch, class weight and budget fixed: the chromosome-pathway indicator alone, the indicator concatenated with ProtT5, ESM2, and the ladder's seven-encoding concatenation (ProtT5, CaLM, the two species-LM embeddings, the 5,979-base window, codon frequency and the indicator), each at the three seeds (Table 18, Figure 7; *chrom-pathway* in the figure is the chromosome-pathway indicator). The read is the round-8 one: a cell passes the ladder if its seed mean clears 0.83 with three of three seeds above the round-5 cell.

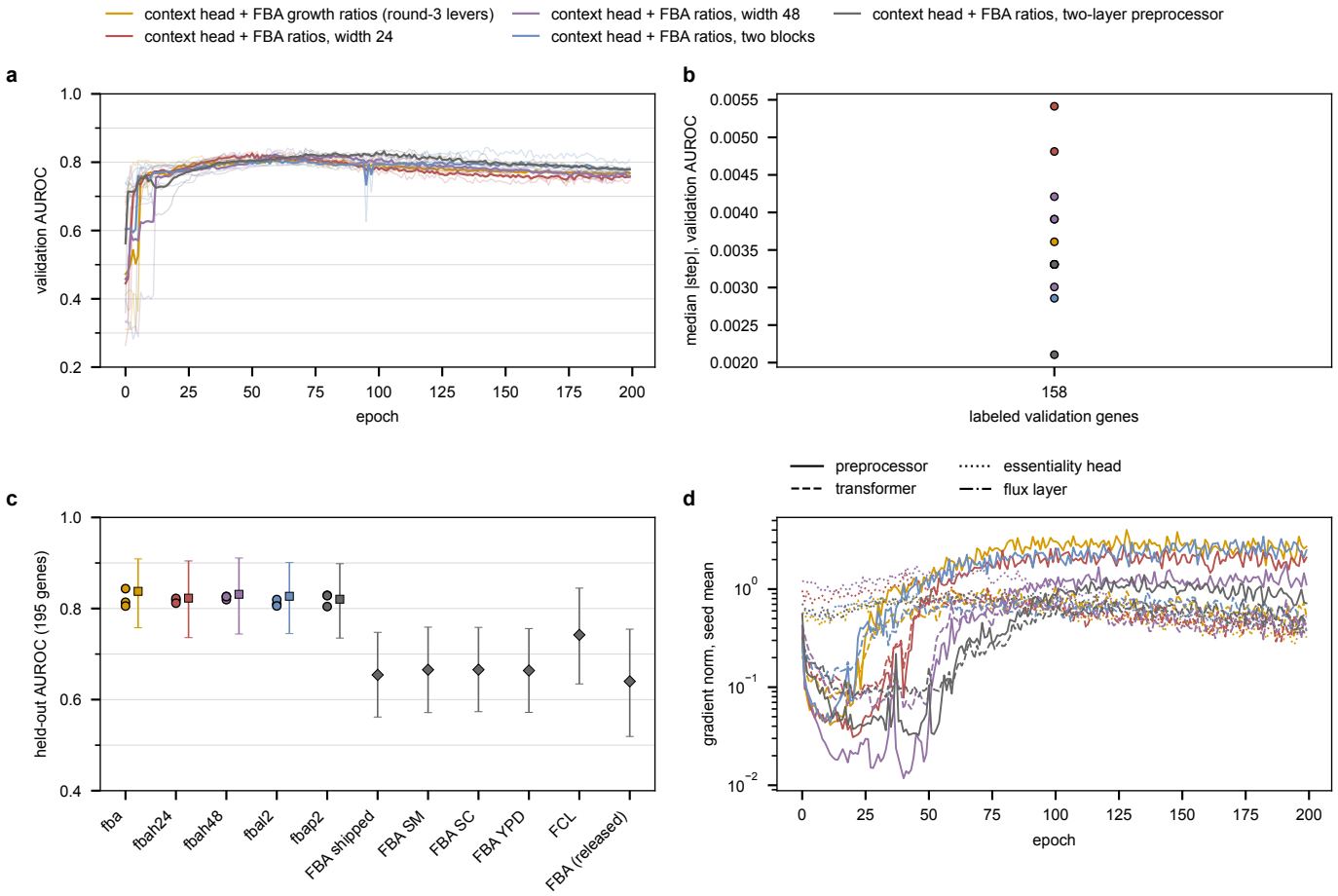


Figure 6: Round 8, width and depth on the FBA cell. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w19`.

Table 18: Round 9, the FBA affine on the ladder's other encodings. The round-5 cell (ProtT5 backbone, context head plus the FBA affine) against the same cell opened on the chromosome-pathway indicator, on the indicator concatenated with ProtT5, on ESM2, and on the ladder's seven-encoding concatenation. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway backbone + FBA ratios	0.781	0.027	0.810, 0.759, 0.774 ↓	0.823 [0.75, 0.90]	21, 2, 2	0.0083
chrom-pathway + ProtT5 backbone + FBA ratios	**0.871**	0.015	0.860, 0.866, 0.888 ↑	**0.868** [0.80, 0.93]	53, 51, 38	0.0030
ESM2 backbone + FBA ratios	0.740	0.008	0.734, 0.736, 0.749 ↓	0.737 [0.64, 0.83]	133, 192, 64	0.0060
seven-encoding backbone + FBA ratios	0.820	0.009	0.822, 0.810, 0.827	0.823 [0.74, 0.90]	91, 92, 55	0.0026

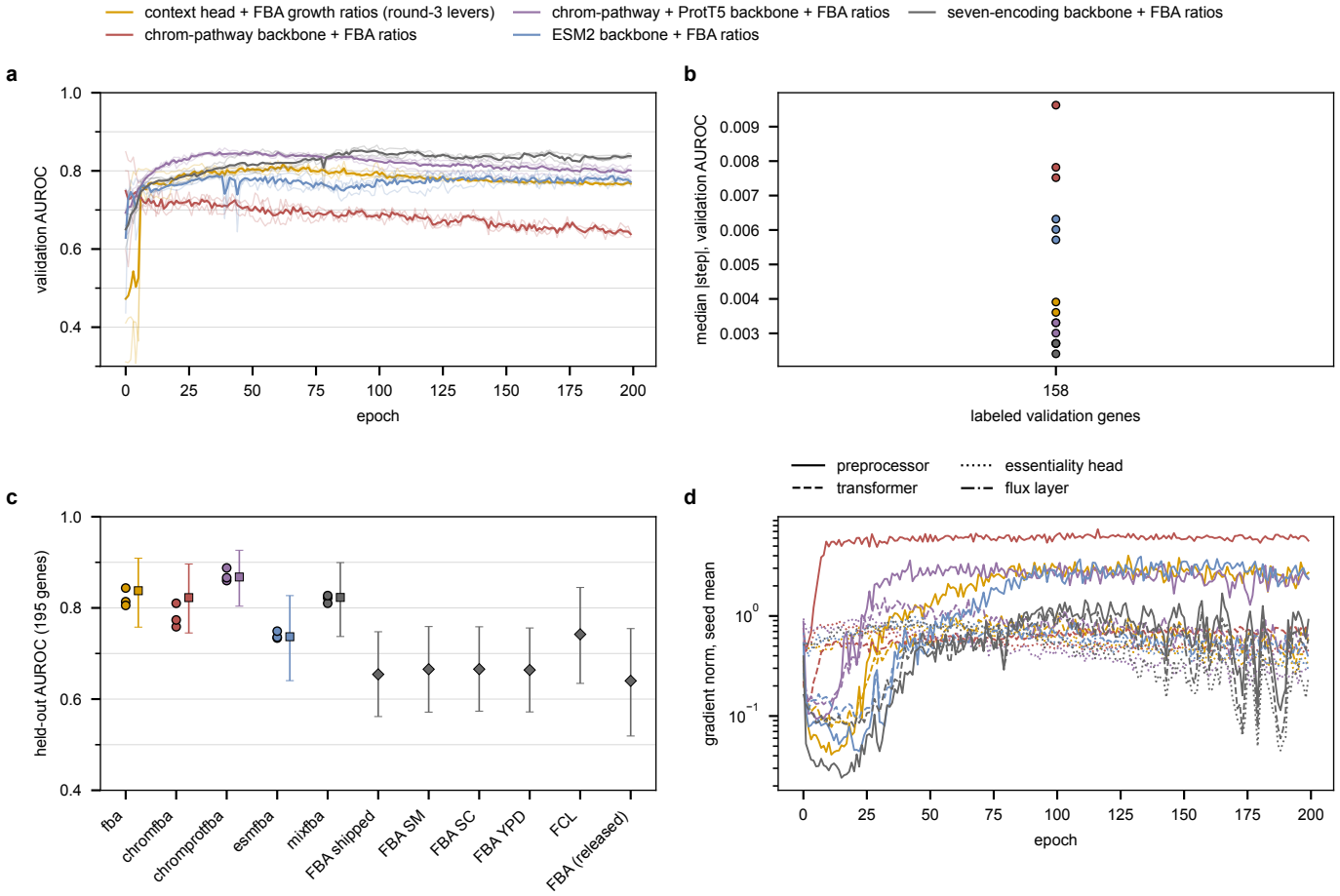


Figure 7: Round 9, the FBA cell on the ladder's other encodings. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w20`.

The indicator concatenated with ProtT5 passes. It scores 0.860, 0.866 and 0.888 against the round-5 cell's 0.814, 0.805 and 0.844 on the same seeds, up by 0.046, 0.061 and 0.044, a seed mean of 0.871 with a spread of 0.015 and a pooled bootstrap of 0.868 [0.80, 0.93]; the lower end of that interval sits at the ladder's best rung. It is the first cell in this document whose seed mean clears 0.83 and whose weakest seed clears the round-5 cell's strongest. Its cells select epochs 38 to 53 on a validation curve that peaks near 0.85 between epochs 25 and 50 and drifts to 0.80 by epoch 200 (Figure 7a), and the cell has 18,930 parameters, 2,364 more than the round-5 cell for the wider input. The indicator alone does not: 0.810, 0.759 and 0.774, a mean of 0.781, with two seeds selecting epoch 2 and a validation curve that falls from 0.73 from the first epochs while its preprocessor's gradient norm sits near 5 for the whole run (Figure 7d), against 0.1 to 3 on the other arms. The two encodings add through the transformer, 0.87 against 0.83 for the indicator's rung and 0.82 for ProtT5's; the ladder never fit the pair on its own. Beside the chromosome and pathway indicators the encoding holds five continuous gene properties, two of them the gene's expression level and dispersion, and round 14 below says which columns carry the gain. ESM2 with the affine scores 0.740, below ProtT5 with the affine by 0.08 and level with ESM2's own ladder rung, and the seven-encoding concatenation 0.820, the round-5 cell's number from 78,066

parameters: adding five encodings to the two that pass returns the concatenation to where ProtT5 alone sits. Round 12 confirms the passing cell under the fold protocol and on four more seeds before the number is reported, since 0.87 on 195 genes is one cell's draw.

3.10 Round 12: the passing cell confirmed, and the number to report ✗

Round 12 (jobs 2291 to 2293) runs the round-9 cell at seeds 2703 to 2706 on the split file and as five fold models on the cross-validation files, the protocol behind the published Flux Cone Learning number, beside the round-5 cell's own fold models (Table 19, Figure 8). The four new seeds score 0.846, 0.877, 0.868 and 0.878, so over seven seeds the cell reads 0.869 with a spread of 0.014 and a pooled bootstrap of 0.870 [0.81, 0.93]; the weakest of the seven is above the round-5 cell's strongest. The five fold models score 0.860, 0.899, 0.889, 0.876 and 0.884 on the held-out genes, a fold mean of 0.881, every fold above the best of the round-5 folds (0.853), and their averaged logits score 0.893 [0.84, 0.94] (`cv_ensemble_readout.py -prefix w23_chromprotcv`), against 0.822 for the round-5 folds and 0.795 for the backbone alone. The fold models select epochs 25 to 53.

Table 19: Round 12, the passing cell confirmed. The round-5 cell (ProtT5 backbone, context head plus the FBA affine) and its five fold models, against the round-9 cell (chromosome-pathway indicator concatenated with ProtT5, same readout) on seven seeds, 2700 to 2706, and on the five cross-validation folds whose logits are averaged on the held-out genes. Width 12, one block, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway + ProtT5 backbone + FBA ratios	0.869	0.014	0.846 to 0.888 (7)	0.870 [0.81, 0.93]	38 to 53	0.0027
context head + FBA ratios, fold 0	0.814	0.000	0.814	0.814 [0.73, 0.89]	28	0.0033
context head + FBA ratios, fold 1	0.824	0.000	0.824	0.824 [0.74, 0.90]	42	0.0042
context head + FBA ratios, fold 2	0.839	0.000	0.839	0.839 [0.77, 0.91]	37	0.0048
context head + FBA ratios, fold 3	0.775	0.000	0.775	0.775 [0.69, 0.86]	122	0.0042
context head + FBA ratios, fold 4	0.853	0.000	0.853	0.853 [0.78, 0.92]	54	0.0036
chrom-pathway + ProtT5 + FBA ratios, fold 0	0.860	0.000	0.860	0.860 [0.79, 0.92]	53	0.0030
chrom-pathway + ProtT5 + FBA ratios, fold 1	**0.899**	0.000	0.899	**0.899** [0.84, 0.95]	32	0.0030
chrom-pathway + ProtT5 + FBA ratios, fold 2	0.889	0.000	0.889	0.889 [0.83, 0.94]	25	0.0055
chrom-pathway + ProtT5 + FBA ratios, fold 3	0.875	0.000	0.875	0.875 [0.81, 0.93]	34	0.0045
chrom-pathway + ProtT5 + FBA ratios, fold 4	0.884	0.000	0.884	0.884 [0.82, 0.94]	27	0.0055

The number to set beside the published 0.73 and beside 0.742 on these genes is the fold ensemble, 0.893 [0.84, 0.94]: a width-12 transformer reading the chromosome-pathway indicator and ProtT5, with FBA's four growth ratios on its logit, scored under Flux Cone Learning's own protocol on its own held-out genes, is 0.15 above it, and its interval excludes the ladder's best rung (0.830). The cell has 18,930 parameters. Every part of the gain is accounted for by an input the ontology computes for every gene at no training cost: the four solver numbers (0.774 to 0.821 on the seed mean) and the pathway indicator concatenated with the protein embedding (0.821 to 0.869). The inputs are the number: a two-layer MLP on the same three inputs, with no graph and no perturbation, reads 0.870 per cell in six seconds of CPU and 0.886 [0.82, 0.94] as a fold ensemble (Section 2), so what the transformer's architecture adds on this endpoint is not measurable on 195 genes. The 0.893 is reported as the document's number because it is the cell the rounds selected under the fixed reads; the MLP's 0.886 and the graph convolution's 0.888 are the same number.

3.11 Round 11: the affine widened with the solve ✗

The ladder's solve-derived encodings (Section 2) carry 0.15 AUROC beyond the growth ratio. Round 11 (jobs 2294 to 2296) appends them to the round-5 cell's affine, standardized column by column over the labeled genes: the eight reaction statistics (12 columns with the ratios), the deletion's pFBA flux vector (4,135), and both (4,139), each at the three seeds on the ProtT5 backbone (Table 20, Figure 9), under the round-8 read.

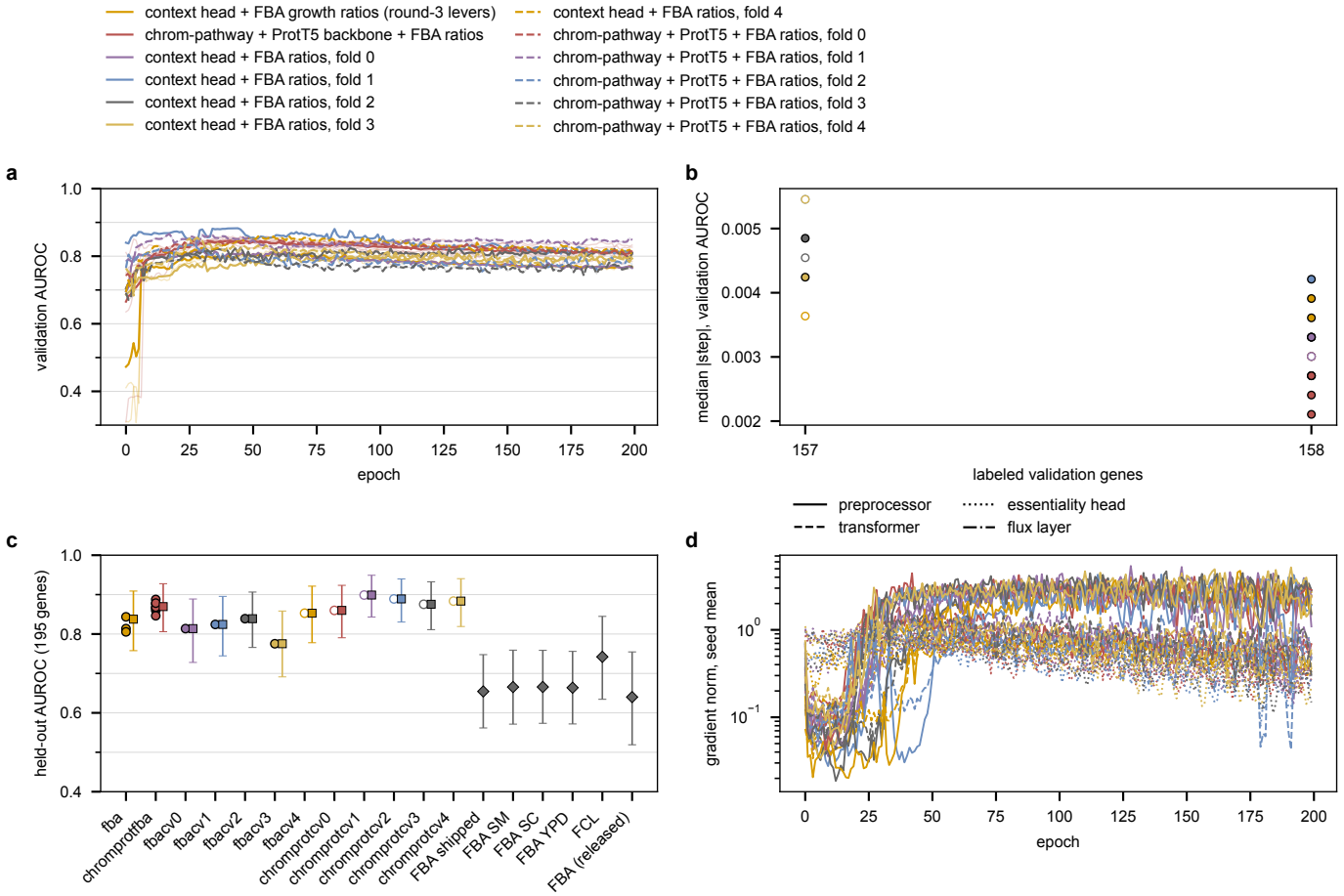


Figure 8: Round 12, the passing cell confirmed. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; each fold model reads its own fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w23`.

Table 20: Round 11, the affine widened with the solve. The round-5 cell (context head plus an affine of the four FBA growth ratios) against the same cell whose affine also reads the eight statistics of the deleted gene's reactions under the wild-type pFBA solve and its variability, the deletion's own pFBA flux vector (4,131 columns), or both; every column standardized over the labeled genes. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
context head + FBA ratios + eight reaction statistics	0.795	0.019	0.816, 0.790, 0.779	0.809 [0.70, 0.90]	55, 71, 124	0.0049
context head + FBA ratios + deletion pFBA vector	0.831	0.076	0.876, 0.743, 0.874	0.853 [0.76, 0.93]	70, 7, 99	0.0064
context head + FBA ratios + statistics + pFBA vector	**0.858**	0.008	0.850, 0.858, 0.867 ↑	**0.858** [0.77, 0.93]	188, 153, 162	0.0056

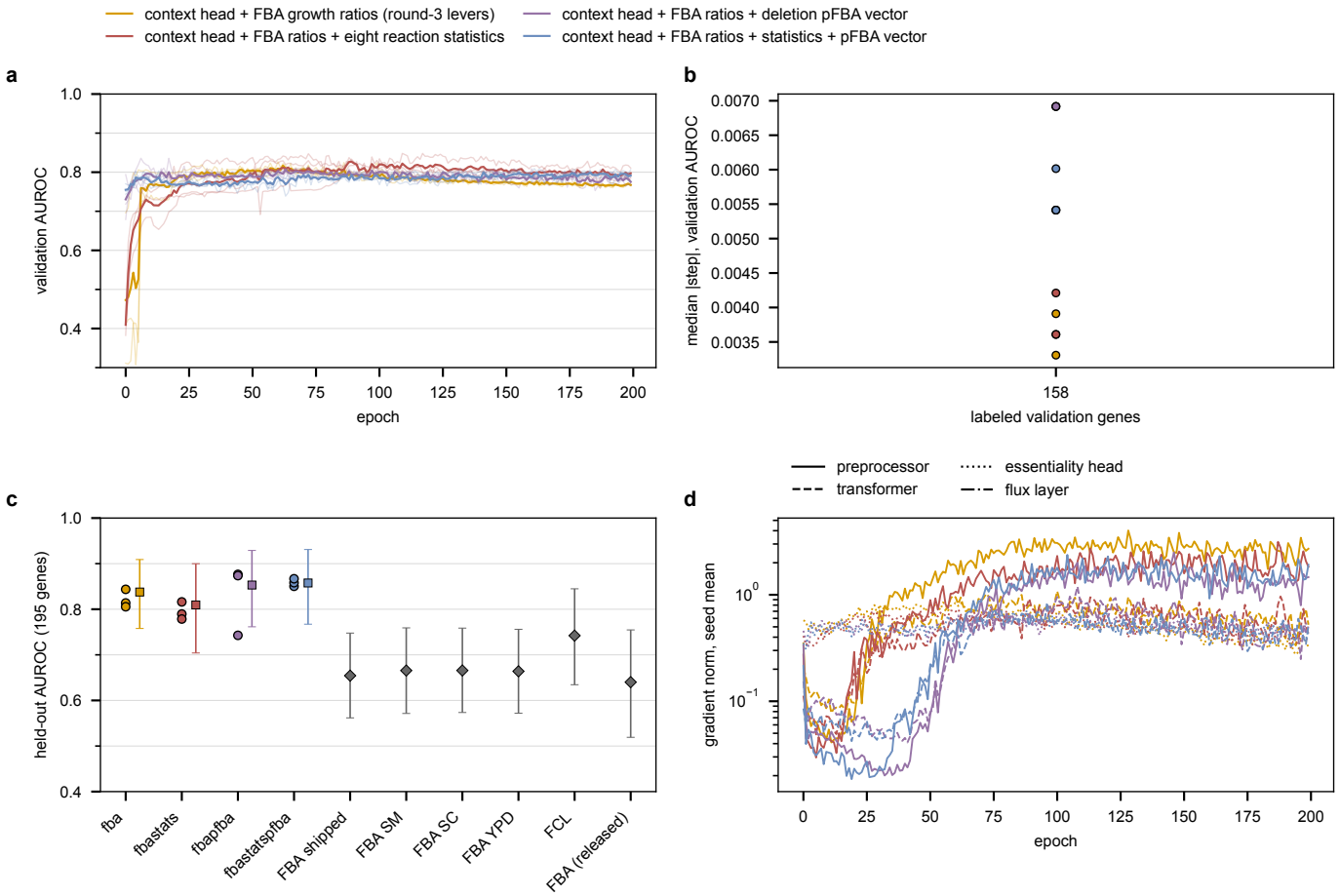


Figure 9: Round 11, the affine widened with the solve. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w22`.

The statistics and the vector together pass; neither does alone. With both, the cell scores 0.850, 0.858 and 0.867 against 0.814, 0.805 and 0.844, up on three of three seeds by 0.036, 0.053 and 0.023, a seed mean of 0.858 with a spread of 0.008, and it selects late, at epochs 153 to 188, on a validation curve that is still level at 200. The eight statistics alone read 0.816, 0.790 and 0.779, down on three of three seeds, though a random forest on the same twelve numbers scores 0.817 on the ladder: through a linear affine the statistics do not add to what the context head already carries, and what the random forest reads in them the affine cannot. The vector alone reads 0.877, 0.743 and 0.874, two seeds at the level of the pair and one collapsed at epoch 7, a collapse no seed shows with the eight statistics beside the vector. Round 16 puts the passing affine on the round-9 backbone.

3.12 Round 14: which columns of the indicator carry the gain ✗

The chromosome-pathway indicator is three things: five scaled gene properties, a chromosome one-hot, and a multi-hot over 173 SGD pathways. Round 14 (jobs 2307 and 2308) drops the pathway columns (`normalized_chrom`, 24

dimensions) and runs the round-9 cell with and without ProtT5 beside it (Table 21, Figure 10).

Table 21: Round 14, attribution inside the chromosome-pathway indicator. The round-5 cell, the round-9 cells on the indicator alone and with ProtT5, and the same two cells on the indicator without its pathway columns (`normalized_chrom`: the five gene properties and the chromosome one-hot), all with the FBA affine. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway backbone + FBA ratios	0.781	0.027	0.810, 0.759, 0.774 ↓	0.823 [0.75, 0.90]	21, 2, 2	0.0083
chrom-pathway + ProtT5 backbone + FBA ratios	**0.871**	0.015	0.860, 0.866, 0.888 ↑	**0.868** [0.80, 0.93]	53, 51, 38	0.0030
gene properties + chromosome backbone + FBA ratios	0.669	0.030	0.636, 0.676, 0.694 ↓	0.663 [0.54, 0.77]	32, 69, 12	0.0110
gene properties + chromosome + ProtT5 backbone + FBA ratios	0.812	0.003	0.811, 0.816, 0.810	0.812 [0.72, 0.89]	65, 55, 53	0.0039

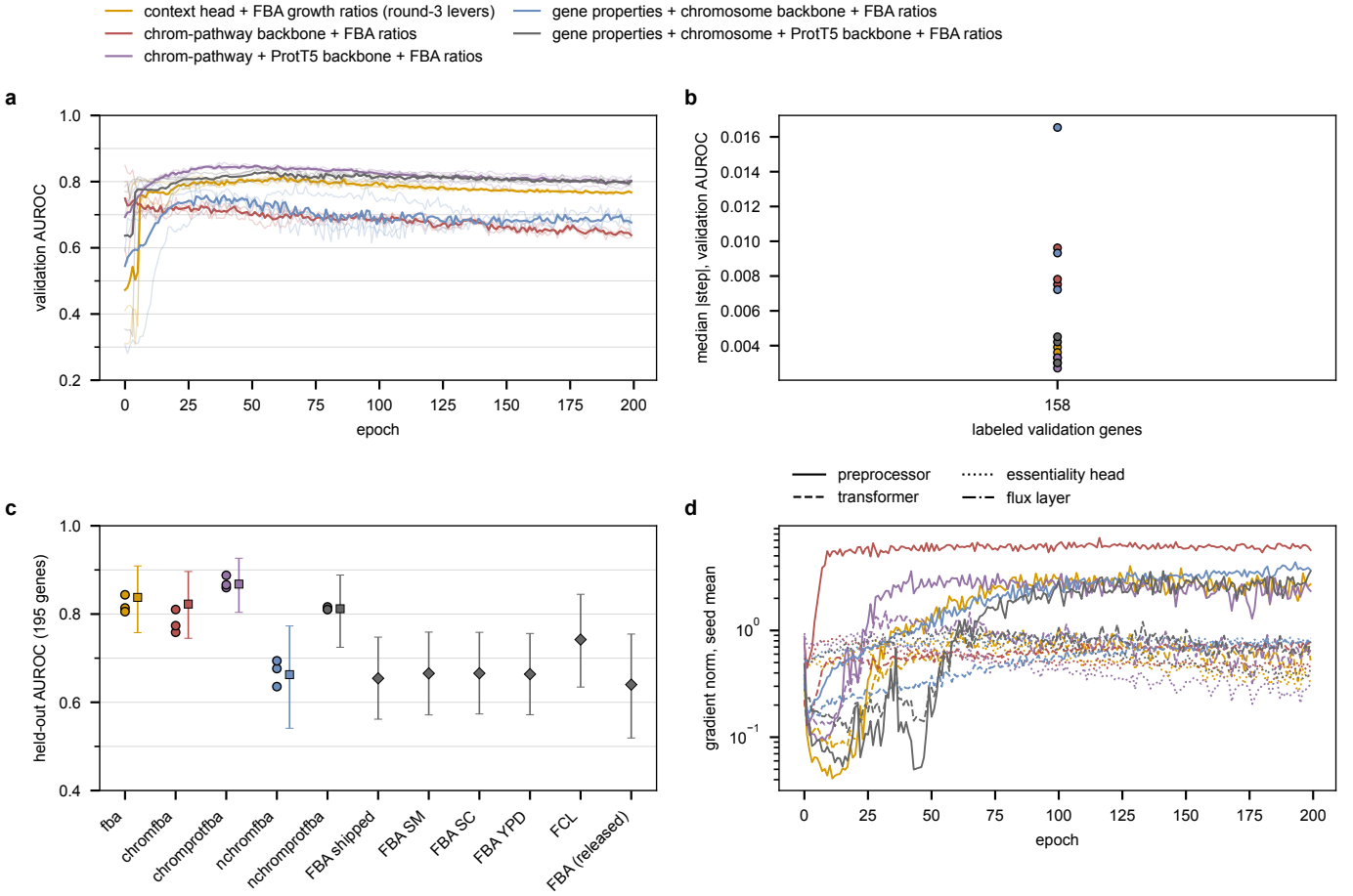


Figure 10: Round 14, attribution inside the chromosome-pathway indicator. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w25`.

The pathways carry it. Without them, the properties and the chromosome beside ProtT5 read 0.811, 0.816 and 0.810, a seed mean of 0.813 against 0.871 with the pathways and 0.821 for ProtT5 alone, so the five properties and the chromosome add nothing to the protein embedding, expression included; on their own they read 0.669, the solver's level. The 173 pathway columns are what lifts the cell from 0.82 to 0.87, and they do it only with ProtT5 beside them: the full indicator alone reads 0.78 (round 9). A gene's pathway membership and its protein sequence are complementary inputs on this endpoint, and each is cheap.

3.13 Round 22: the indicator split into its blocks ✗

Round 14 attributes the gain by subtraction, the indicator without its pathway columns; round 22 (job 2338) attributes it by addition, one block at a time on the round-9 recipe. The 197-column indicator is three blocks, the 173-column pathway multi-hot, the 17-column chromosome one-hot and the seven scaled gene properties, and each is built as an embedding of its own (`pathways`, `chromosome`, `normalized_props` in the embedding registry) and run beside ProtT5 with the FBA affine, three seeds each, with the pathway block also run without ProtT5. The read is paired against round 9's cell: the block whose three seeds match round 9's is the block that carries the gain, and a block that reads at the ProtT5-alone level (0.821) carries none. On the cheap models the same split runs on the MLP and on graph convolution (Table 7), where the pathway block beside ProtT5 and the ratios reads 0.876, 0.885 and 0.878 on the MLP, above the full indicator's 0.870.

Table 22: Round 22, the chromosome-pathway indicator split into its blocks. The round-9 cell (the paired comparator: the 197-column indicator beside ProtT5 with the FBA affine), the round-5 affine on ProtT5 alone, and the round-14 cell (properties and chromosome, no pathways), against the same recipe with one block at a time: the 173-column pathway multi-hot, the 17-column chromosome one-hot, the seven scaled properties, and the pathway block alone without ProtT5. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
chrom-pathway + ProtT5 backbone + FBA ratios	0.871	0.015	0.860, 0.866, 0.888	0.868 [0.80, 0.93]	53, 51, 38	0.0030
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844 ↓	0.838 [0.76, 0.91]	28, 70, 61	0.0036
gene properties + chromosome + ProtT5 backbone + FBA ratios	0.812	0.003	0.811, 0.816, 0.810 ↓	0.812 [0.72, 0.89]	65, 55, 53	0.0039
pathway multi-hot + ProtT5 backbone + FBA ratios	**0.890**	0.005	0.889, 0.895, 0.885	**0.890** [0.83, 0.94]	31, 45, 55	0.0054
chromosome one-hot + ProtT5 backbone + FBA ratios	0.811	0.019	0.824, 0.821, 0.789 ↓	0.818 [0.73, 0.90]	40, 44, 55	0.0039
seven gene properties + ProtT5 backbone + FBA ratios	0.824	0.020	0.816, 0.846, 0.809 ↓	0.829 [0.75, 0.90]	71, 63, 38	0.0035
pathway multi-hot backbone + FBA ratios	0.766	0.007	0.773, 0.760, 0.766 ↓	0.774 [0.68, 0.86]	123, 140, 96	0.0018

The transformer says what the MLP said. ProtT5 with the pathway block and the affine reads 0.889, 0.895 and 0.885 (mean 0.890; pooled 0.890 with a gene bootstrap of 0.834 to 0.942), against round 9's full indicator at 0.860, 0.866 and 0.888: above by 0.03 on two seeds and under by 0.003 on the third, the seed round 9 selected at its best. ProtT5 with the chromosome block reads 0.824, 0.821 and 0.789 and with the seven properties 0.816, 0.846 and 0.809, both at the level of ProtT5 with the affine alone (0.821). The pathway block without ProtT5 reads 0.773, 0.760 and 0.766 on the transformer where the MLP had it at 0.828, the one place the two models part: the transformer needs the sequence embedding to use the pathway columns, and the MLP does not. The block that carries the gain is the pathway multi-hot, and the 24 columns beside it carry none of it. The best single cell in this section is now the pathway block with ProtT5 and the affine at 0.895, which puts the input, not the architecture, 0.02 above the ensemble the document reports.

3.14 Round 26: the Gene Ontology multi-hot as a backbone input ✗

If 173 pathway memberships carry the gain over the protein embedding, the larger annotation of the same kind is the Gene Ontology, and the same question applies to it. The GO multi-hot (Section 1) is built as its own dataset beside the chromosome, pathway and property vectors (Table 23), so each is a starting vector any model can take. Round 26 (jobs 2363 and 2364) runs the round-9 recipe with it: GO beside ProtT5 with the affine, the biological-process namespace alone beside ProtT5, GO alone with the affine, and GO with the pathway multi-hot beside ProtT5, three seeds each (Table 24, Figure 12). The read pairs each cell against the round-22 pathway cell by seed: GO beside ProtT5 at or above it on three seeds of three says GO carries at least what the pathways carry; GO with the pathways above it on three of three says the two annotations add. The cheap models run the same inputs on the MLP and on graph convolution (Table 8), and the ladder gains a linear rung on each annotation vector alone.

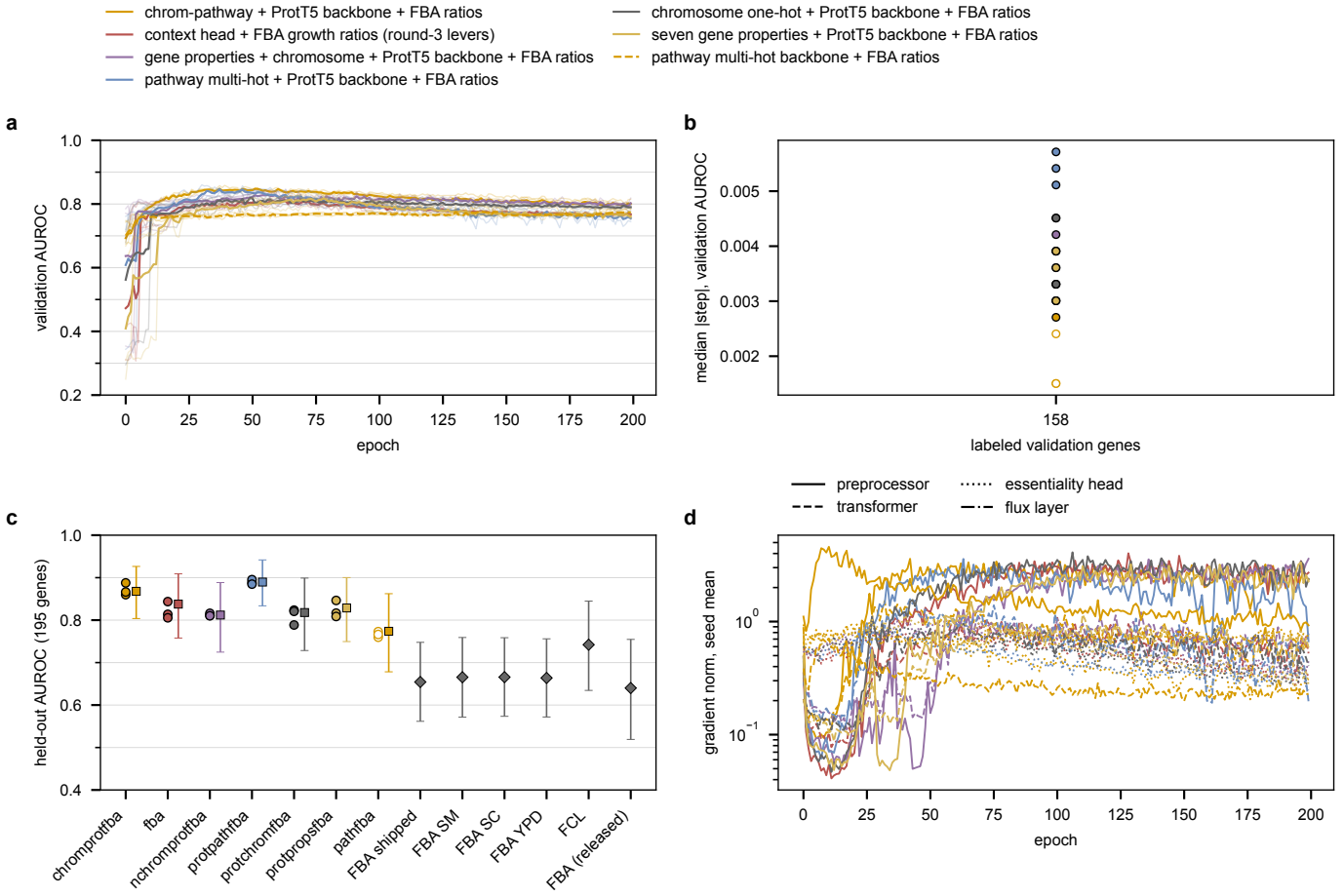


Figure 11: Round 22, the chromosome-pathway indicator split into its blocks on the transformer. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w38`.

Table 23: The per-gene starting vectors used in this document, each a dataset in the source tree registered under its name in `NodeEmbeddingBuilder` so any model opens it by name. Width is the column count of the built tensor; covered is the number of genes with a nonzero row over the genes the dataset holds. Written by `notes_tex_tables.py` from the built tensors.

name	what	width	covered	construction	module
`prot_T5_all`	protein language model embedding	1,024	6,607 / 6,607	ProtT5 mean-pooled over the protein sequence	`protT5.py`
`pathways`	pathway multi-hot	173	517 / 6,607	one column per SGD pathway a gene is annotated to	`sgd_gene_graph_hot.py`
`chromosome`	chromosome one-hot	17	6,607 / 6,607	one column per chromosome	`sgd_gene_graph_hot.py`
`normalized_props`	scaled gene properties	7	6,607 / 6,607	protein length, molecular weight, isoelectric point, expression median and median absolute deviation, scaled to [0, 1]	`sgd_gene_graph_hot.py`
`normalized_chrom_pathways`	chromosome-pathway indicator	197	6,607 / 6,607	the three blocks above concatenated	`sgd_gene_graph_hot.py`
`go`	GO multi-hot, propagated	2,362	5,501 / 6,607	SGD GO annotations propagated to ancestors (is_a, part_of), roots dropped, terms with ten or more genes	`go_term_hot.py`
`go_direct`	GO multi-hot, direct	2,887	5,492 / 6,607	the annotated terms only, terms with two or more genes	`go_term_hot.py`
`go_bp`	GO biological process	1,580	4,909 / 6,607	the propagated vector restricted to one namespace	`go_term_hot.py`
`go_mf`	GO molecular function	430	4,146 / 6,607	as above	`go_term_hot.py`
`go_cc`	GO cellular component	352	5,256 / 6,607	as above	`go_term_hot.py`

Table 24: Round 26, the Gene Ontology multi-hot as a backbone input. The round-5 affine on ProtT5, the round-9 indicator cell and the round-22 pathway cell (the comparators), against the round-9 recipe with the GO multi-hot beside ProtT5 (the SGD annotations propagated to their ancestors over terms with ten or more genes, 2,362 columns), the biological-process namespace alone beside ProtT5 (1,580 columns), the GO multi-hot alone, and GO with the pathway multi-hot beside ProtT5. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway + ProtT5 backbone + FBA ratios	0.871	0.015	0.860, 0.866, 0.888 ↑	0.868 [0.80, 0.93]	53, 51, 38	0.0030
pathway multi-hot + ProtT5 backbone + FBA ratios	**0.890**	0.005	0.889, 0.895, 0.885 ↑	**0.890** [0.83, 0.94]	31, 45, 55	0.0054
GO multi-hot (propagated, 2,362) + ProtT5 backbone + FBA ratios	0.833	0.008	0.830, 0.827, 0.842	0.839 [0.76, 0.91]	123, 48, 62	0.0044
GO biological process (1,580) + ProtT5 backbone + FBA ratios	0.821	0.014	0.805, 0.832, 0.826	0.827 [0.73, 0.91]	180, 160, 104	0.0034
GO multi-hot backbone + FBA ratios	0.800	0.012	0.810, 0.787, 0.804 ↓	0.810 [0.72, 0.89]	28, 11, 139	0.0048
GO + pathway multi-hots + ProtT5 backbone + FBA ratios	0.849	0.005	0.852, 0.852, 0.844	0.851 [0.78, 0.92]	131, 48, 126	0.0019

The transformer takes nothing from GO that it did not have from ProtT5, and beside the pathway block GO costs it. GO with ProtT5 and the affine reads 0.830, 0.827 and 0.842 (mean 0.833), level with ProtT5 and the affine alone (0.821) and 0.05 to 0.07 under the pathway cell on every seed; the biological-process namespace alone reads 0.821, and GO alone with the affine 0.800. GO added to the pathway cell reads 0.852, 0.852 and 0.844 against that cell's 0.889, 0.895 and 0.885, a cost of 0.04 on three seeds of three. Both clauses of the read fail. The same vector is the strongest single encoding on the ladder (0.808 linear with no solve, 0.844 under a random forest with the ratios) and, on the cheap models under the same protocol, lifts the MLP to 0.855 and graph convolution to 0.887 (Table 8), so what fails is not the vector but the transformer's use of it; why a width-12 cell reads 173 sparse pathway columns and not 2,362 dense GO columns is not measured. The one-column control that marks a gene as annotated to any pathway reads 0.774 on the MLP and 0.832 on graph convolution, the ProtT5-with-ratios level, so the pathway block's gain is which pathway, not whether there is one.

3.15 Round 28: the GO cells with a wider cell ✗

If the width-12 preprocessor is what keeps the transformer from reading 2,362 GO columns, a wider one reads them. Round 28 (job 2378) runs the two round-26 GO cells, GO beside ProtT5 and GO with the pathways beside ProtT5, at width 24 with a two-layer preprocessor (the round-8 cell's shape), three seeds each, against the width-12 cells and the pathway cell (Table 25). The read pairs each wide cell against its width-12 cell by seed: above on three of three and the preprocessor is the limit; level and it is not.

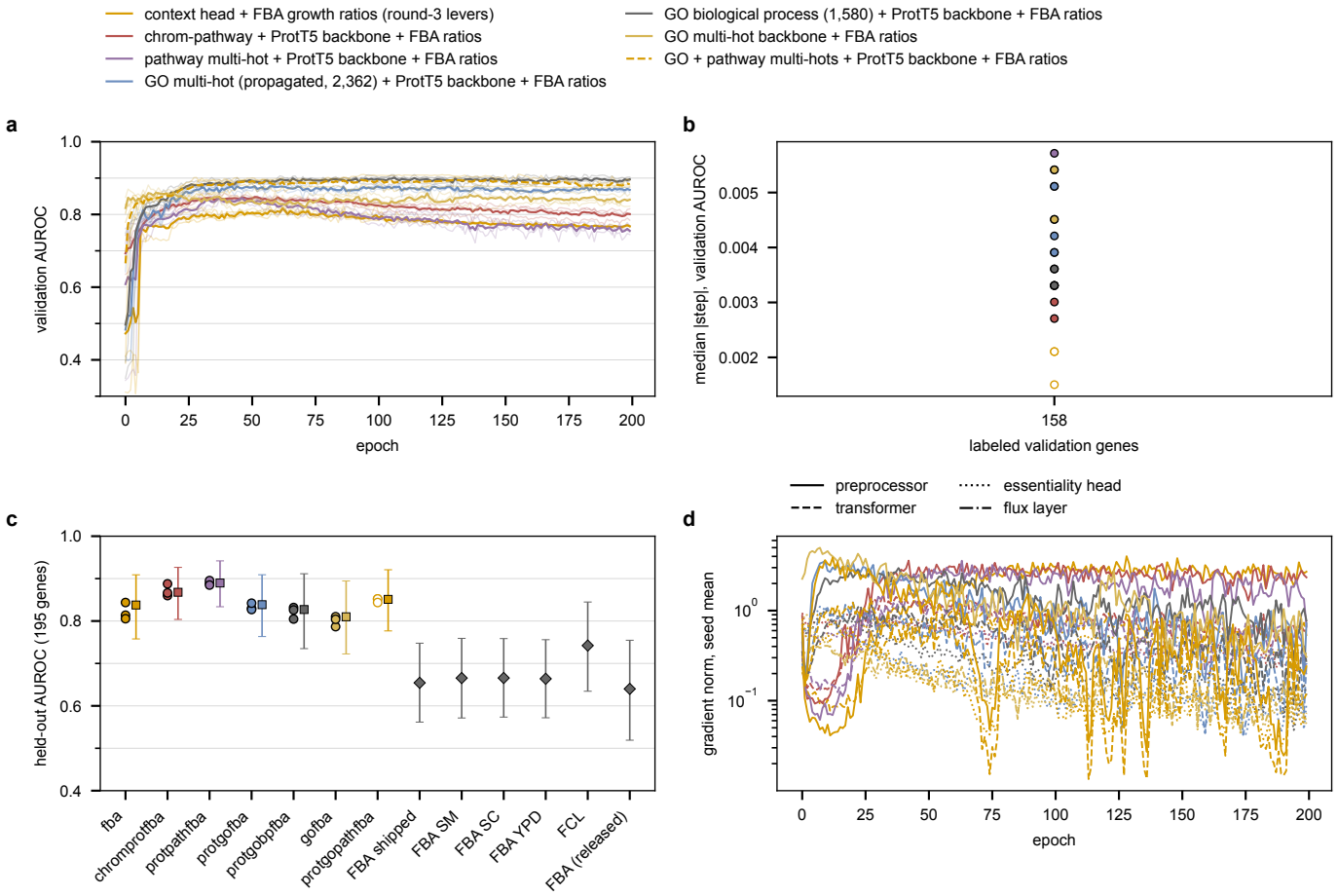


Figure 12: Round 26, the Gene Ontology multi-hot as a backbone input. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w42`.

Table 25: Round 28, the GO cells with a wider cell. The round-5 affine on ProtT5, the round-22 pathway cell and the two round-26 GO cells at width 12 with a one-layer preprocessor (the comparators), against the same GO cells at width 24 with a two-layer preprocessor. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
pathway multi-hot + ProtT5 backbone + FBA ratios	**0.890**	0.005	0.889, 0.895, 0.885 ↑	**0.890** [0.83, 0.94]	31, 45, 55	0.0054
GO multi-hot (propagated, 2,362) + ProtT5 backbone + FBA ratios	0.833	0.008	0.830, 0.827, 0.842	0.839 [0.76, 0.91]	123, 48, 62	0.0044
GO + pathway multi-hots + ProtT5 backbone + FBA ratios	0.849	0.005	0.852, 0.852, 0.844	0.851 [0.78, 0.92]	131, 48, 126	0.0019
GO + ProtT5 + FBA ratios, width 24, two-layer preprocessor	0.842	0.005	0.847, 0.841, 0.838	0.847 [0.77, 0.92]	152, 65, 129	0.0037
GO + pathways + ProtT5 + FBA ratios, width 24, two-layer preprocessor	0.852	0.003	0.849, 0.855, 0.853 ↑	0.858 [0.78, 0.93]	42, 182, 76	0.0028

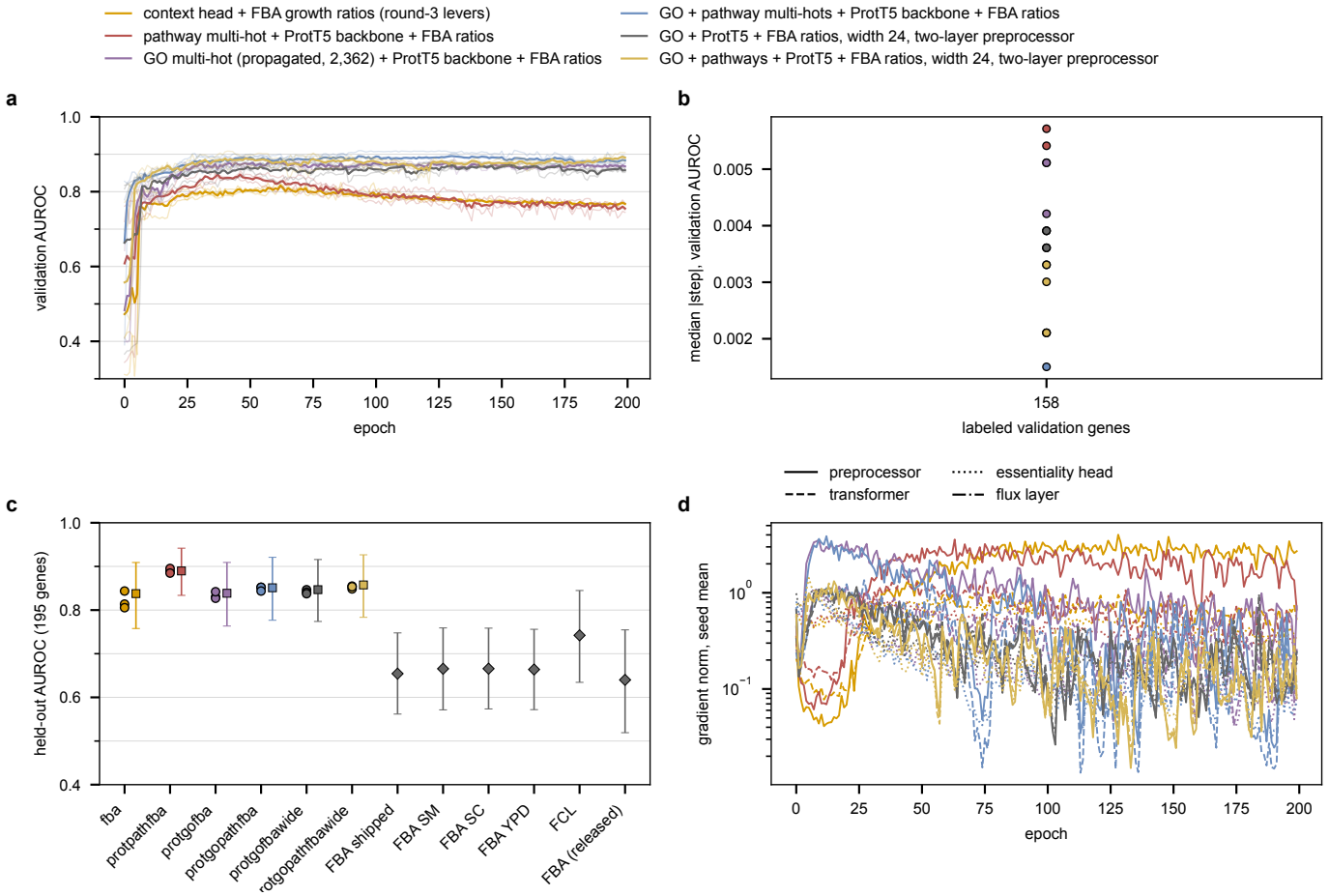


Figure 13: Round 28, the GO cells with a wider cell. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w44`.

The preprocessor is not the limit. At width 24 with a two-layer preprocessor, 5.8 million parameters against 17 thousand, GO beside ProtT5 reads 0.847, 0.841 and 0.838 against the width-12 cell's 0.830, 0.827 and 0.842, above on two seeds by 0.01 to 0.02 and level on the third; GO with the pathways beside ProtT5 reads 0.849, 0.855 and 0.853 against 0.852, 0.852 and 0.844, level. A cell 350 times larger reads the GO vector the same, so whatever keeps the transformer at 0.84 on an input that gives graph convolution 0.91 is not capacity. What it is, is not measured; the pooled readout, which averages the perturbed tokens, and the density of the GO rows against the pathway rows are the untested candidates.

3.16 Which genes the best cells misrank ✗

At 195 genes the AUROC is a sum over a few dozen misrankings, so the genes behind it can be named. For every cell of the six best arms, `hard_genes.py` reads the per-gene logits on the held-out genes and scores each gene's *misrank*,

the fraction of the other class on the wrong side of it: for an essential gene the nonessential genes scoring above it, for a nonessential gene the essential genes scoring below it. One minus the class mean of the misrank is that class's share of the AUROC, and averaging over an arm's cells separates the genes an arm gets wrong from the ones a seed gets wrong (Figure 14, Table 26).

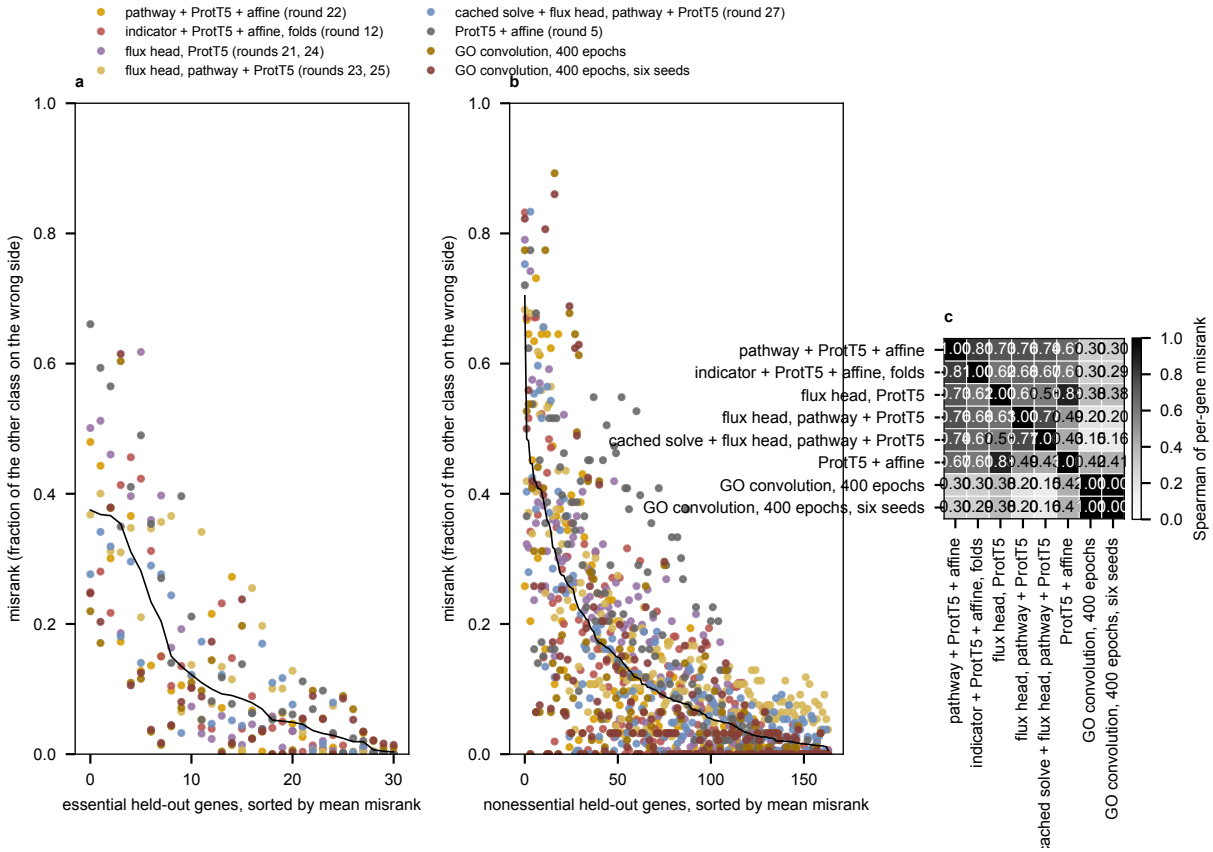


Figure 14: Which held-out genes the best cells misrank. a, The 31 essential held-out genes sorted by their mean misrank over six arms, one marker per arm (the mean over that arm's cells), the line the mean over arms. **b**, The same for the 164 nonessential genes. **c**, The Spearman correlation of the per-gene misrank between arms. Arms: the pathway multi-hot with ProtT5 and the affine (round 22, three seeds), the round-12 fold models (five), the head over the solved flux on ProtT5 (rounds 21 and 24, six seeds) and on the pathway input (rounds 23 and 25, six), the same head over the cached GEM-box solve on the pathway input (round 27, six), and ProtT5 with the affine (round 5, three). Written by `experiments/028-gene-essentiality/scripts/hard_genes.py`.

Table 26: The fifteen held-out genes of each class the best cells misrank most, sorted by the mean misrank over the arms. A gene's misrank in one cell is the fraction of the other class on the wrong side of it (for an essential gene, the nonessential genes scoring above it; for a nonessential gene, the essential genes scoring below it), averaged over the arm's cells; one minus the class mean of the misrank is that class's share of the AUROC. FBA is the knockout-to-wild-type growth ratio on YPD; pathways and GO the gene's counts in the pathway multi-hot and the propagated GO vector. Arms: A1: pathway + ProtT5 + affine (round 22); A2: indicator + ProtT5 + affine, folds (round 12); A3: flux head, ProtT5 (rounds 21, 24); A4: flux head, pathway + ProtT5 (rounds 23, 25); A5: cached solve + flux head, pathway + ProtT5 (round 27); A6: ProtT5 + affine (round 5). Written by `hard_genes.py` from the per-gene prediction dumps.

gene	A1	A2	A3	A4	A5	A6	FBA	pathways	GO
essential genes									
YKL004W	0.48	0.25	0.50	0.37	0.28	0.66	1.00	1	39
YDR373W	0.44	0.28	0.51	0.40	0.34	0.59	1.00	0	25
YBL030C	0.30	0.22	0.46	0.31	0.32	0.57	1.00	0	68
YIL048W	0.17	0.41	0.19	0.31	0.18	0.35	1.00	0	52
YML125C	0.37	0.46	0.40	0.35	0.30	0.41	1.00	0	27
YAL038W	0.13	0.42	0.62	0.21	0.14	0.49	1.00	2	81
YMR296C	0.14	0.31	0.36	0.35	0.29	0.35	1.00	1	36
YDR062W	0.15	0.15	0.40	0.36	0.28	0.27	1.00	1	36
YBR011C	0.10	0.10	0.16	0.37	0.24	0.14	1.00	0	18
YDR050C	0.11	0.14	0.19	0.07	0.03	0.40	1.00	2	85
YDL120W	0.12	0.10	0.07	0.14	0.12	0.21	1.00	0	50
YGR277C	0.05	0.04	0.03	0.34	0.17	0.05	1.00	1	64
YGL055W	0.22	0.04	0.02	0.08	0.05	0.08	0.00	3	51
YMR281W	0.07	0.23	0.06	0.13	0.12	0.12	1.00	0	64
YDR454C	0.27	0.15	0.03	0.11	0.05	0.08	0.00	5	65
nonessential genes									
YGL012W	0.26	0.83	0.79	0.68	0.75	0.72	0.00	2	48
YLR450W	0.67	0.67	0.32	0.42	0.40	0.37	1.00	3	63
YBR121C	0.65	0.52	0.32	0.37	0.28	0.62	1.00	0	66
YOL049W	0.20	0.34	0.74	0.68	0.83	0.77	0.00	2	27
YDR315C	0.45	0.41	0.58	0.34	0.42	0.54	1.00	0	73
YGR192C	0.63	0.67	0.24	0.54	0.55	0.16	1.00	3	125
YJR121W	0.73	0.67	0.35	0.40	0.38	0.68	0.34	0	121
YGR240C	0.30	0.37	0.39	0.62	0.60	0.14	1.00	2	105
YHR106W	0.61	0.41	0.57	0.52	0.56	0.45	1.00	1	30
YJR009C	0.65	0.52	0.25	0.56	0.52	0.17	1.00	3	113
YMR205C	0.31	0.15	0.38	0.66	0.66	0.15	1.00	2	119
YMR202W	0.16	0.25	0.26	0.25	0.13	0.35	1.00	2	41
YLR377C	0.65	0.43	0.31	0.53	0.56	0.39	1.00	3	26
YGR106C	0.54	0.48	0.59	0.34	0.37	0.52	1.00	0	30
YJL052W	0.62	0.47	0.24	0.56	0.54	0.16	1.00	3	107

The solve's own call splits the essential genes. Nineteen of the 31 essential held-out genes have a knockout growth ratio of 1.0 on YPD, the solve calling them dispensable, and thirteen of the fifteen hardest are among those nineteen, while nine of the fifteen easiest have a ratio of 0. The hardest essential genes are the ones FBA misses, and every model misses them with it: AUR1, FRQ1, PET9 (the ADP/ATP carrier, covered in the GEM by AAC1 and AAC3), NEO1, PGA3, CDC19 (pyruvate kinase, covered by its paralog PYK2), LCB1 and LCB2 (serine palmitoyltransferase), IPP1, TPI1. The hardest nonessential genes are metabolic enzymes the models call essential: ERG4, HMG2, GSH2, the three glyceraldehyde-3-phosphate dehydrogenase isozymes TDH3, TDH2 and TDH1, ATP2, the two phosphofructokinase subunits PFK1 and PFK2, TRR2, ERG2 and FBP1, twelve of the fifteen with a ratio of 1.0 and most with 26 to 125 GO terms. Both failure sets are isozyme redundancy read the wrong way: an essential gene whose paralog covers it in the solve, and a nonessential paralog of an essential metabolic function that the annotation vectors make look essential. The arms agree on the genes they fail: the two affine cells at a Spearman of 0.81, the flux head on ProtT5 with the affine on ProtT5 at 0.88, the two flux heads on the pathway input with each other at 0.77, and those two least aligned with the rest (0.43 to 0.76), the arms whose means carry an unfit seed. Hypothesis, untested: a paralog-aware input, the GPR disjunction with an expression weight or the isozyme count of a gene's reactions, is what would move the fourteen; the solve's own vector cannot, since the solve reroutes through the paralog.

Graph convolution on GO fails on different genes. Its per-gene misrank correlates with the transformer arms' at 0.15 to 0.42 where those arms correlate with one another at 0.4 to 0.9, the same on three seeds and on six, and its class mean misrank on the essential genes is 0.091 against the pathway cell's 0.110. On the hardest essential genes it halves the misrank the solve leaves: FRQ1 from 0.44 to 0.17, AUR1 from 0.48 to 0.22, PGA3 from 0.37 to 0.11, LCB1 from 0.14 to 0.03, LCB2 from 0.15 to 0.01, while it does worse on NEO1 (0.17 to 0.60) and PET9 (0.30 to 0.38). Among the nonessential genes it repairs twelve the pathway cell misranks by more than 0.3 (TDH1, ACS1, ATP2, FBP1, TRR2 among them) and breaks six the pathway cell had right. Two models that read the same genes wrong for different reasons are the case for averaging them.

3.17 Averaging the two families ✗

The average is a post hoc read of predictions already made, with nothing fitted: each family's per-cell logits on the held-out genes are standardized and averaged over its seeds, the families are averaged with equal weight, and the mean is scored once (Table 27). The GO convolution and the pathway cell with the affine average to 0.933 [0.89, 0.97] from 0.909 and 0.891 alone; the GO convolution and the flux head on the pathway input, six seeds with two unfit among them, average to 0.941 [0.90, 0.98] from 0.909 and 0.911; all three to 0.935. With the six-seed families of the later rounds, the GO convolution on six seeds (0.905 as a family) and the scalable module form of round 27 (0.918 as a family, the highest single family) average to 0.946 [0.91, 0.98], and with the pathway affine beside them to 0.939. Every average tried is in the table, so 0.946 is the maximum of eleven post hoc choices and carries that bias, and the held-out genes were used to choose nothing else. It says what the per-gene analysis said: the two families carry different information about the same genes, and the sum of the annotation read through the graph and the solve read through the transformer is more than either. Under the reported protocol the same average holds (Table 28): the five GO fold models and the five pathway flux-head fold models, each family 0.917 and 0.897 as its own ensemble, average to 0.949 [0.91, 0.98]; the GO fold models with the reported cell's fold models (0.893 alone) to 0.944; all three to 0.945. Six averages tried, all listed, the best the maximum of six. The interval of 0.949 excludes the reported 0.893 point, and the two families' own intervals overlap it, which is what two models that fail on different genes produce when averaged and what neither produces alone.

Table 27: Post hoc averages of existing predictions on the 195 held-out genes: each family's per-cell logits are standardized and averaged over its seeds, families are averaged with equal weight, and the mean is scored once with a 2,000-draw gene bootstrap. Nothing is fitted and every average tried is listed, so the best row is the maximum of seven post hoc choices, not a selected model. Written by `family_ensembles.py`. Bold marks the highest value in a column.

average of	cells	AUROC [95% CI]
GO convolution (400 epochs, 3 seeds)	3	0.909 [0.85, 0.95]
GO convolution (400 epochs, 6 seeds)	6	0.905 [0.85, 0.95]
pathway + ProtT5 + affine (round 22, 3 seeds)	3	0.891 [0.83, 0.94]
flux head, pathway + ProtT5 (rounds 23 and 25, 6 seeds)	6	0.911 [0.86, 0.95]
cached solve + flux head, pathway + ProtT5 (round 27, 6 seeds)	6	0.918 [0.87, 0.96]
GO convolution (400 epochs, 6 seeds) + cached solve + flux head, pathway + ProtT5 (round 27, 6 seeds)	12	**0.946** [0.91, 0.98]
GO convolution (400 epochs, 6 seeds) + pathway + ProtT5 + affine (round 22, 3 seeds) + cached solve + flux head, pathway + ProtT5 (round 27, 6 seeds)	15	0.939 [0.90, 0.97]
GO convolution (400 epochs, 3 seeds) + pathway + ProtT5 + affine (round 22, 3 seeds)	6	0.933 [0.89, 0.97]
GO convolution (400 epochs, 3 seeds) + flux head, pathway + ProtT5 (rounds 23 and 25, 6 seeds)	9	0.941 [0.90, 0.97]
GO convolution (400 epochs, 3 seeds) + flux head, ProtT5 (rounds 21 and 24, 6 seeds)	9	0.923 [0.87, 0.96]
GO convolution (400 epochs, 3 seeds) + pathway + ProtT5 + affine (round 22, 3 seeds) + flux head, pathway + ProtT5 (rounds 23 and 25, 6 seeds)	12	0.935 [0.89, 0.97]

Table 28: The same post hoc averages under the reported protocol: each family is its five fold models (one seed per fold, the cross-validation files of Section 3), standardized per fold and averaged, families averaged with equal weight, scored once on the 195 held-out genes with a 2,000-draw gene bootstrap. Six averages tried, all listed. Written by `family_ensembles.py -folds`. Bold marks the highest value in a column.

average of	cells	AUROC [95% CI]
GO convolution, five folds (400 epochs)	5	0.917 [0.87, 0.95]
indicator + ProtT5 + affine, five folds (round 12)	5	0.897 [0.84, 0.94]
flux head, pathway + ProtT5, five folds (round 25)	5	0.897 [0.84, 0.94]
GO convolution, five folds (400 epochs) + indicator + ProtT5 + affine, five folds (round 12)	10	0.944 [0.90, 0.98]
GO convolution, five folds (400 epochs) + flux head, pathway + ProtT5, five folds (round 25)	10	**0.949** [0.91, 0.98]
GO convolution, five folds (400 epochs) + indicator + ProtT5 + affine, five folds (round 12) + flux head, pathway + ProtT5, five folds (round 25)	15	0.945 [0.91, 0.97]

3.18 Round 30: the genome-wide labels as a larger held-out set ✗

Every number above sits on 195 genes, and the open list's first question was whether the graph convolution's margin survives a larger held-out set. The genome-wide labels of round 1 give one: 4,674 genes the released file never labeled, 974 essential by SGD annotation and 3,700 nonessential by a deletion-collection fitness record. Round 30 trains on the released labeled genes outside the released held-out set (the split `ess_genome_heldout_val0.2_s0`: 629 train, 158 validation, the 195 released held-out genes dropped so the read is not on them) and reads once on the 4,674 (Table 29). Two properties of the set bound what it can say. None of the 4,674 carries a yeast-GEM reaction, since the released file had labeled the GEM's genes, so the read is a transfer from metabolic to non-metabolic genes and FBA's knockout

ratio is 1.0 on every one of them, an AUROC of exactly 0.5. And label origin coincides with class: every essential gene is an SGD annotation and every nonessential one a fitness record, so a model that can tell the two sources apart scores on this set without reading essentiality.

Table 29: Round 30, the genome-wide labels as a larger held-out set. Every model is trained on the released labeled genes outside the released held-out set (629 train, 158 val, the split `ess_genome_heldout_val0.2_s0`) and read once on the 4,674 genes the released file never labeled, 974 of them essential by SGD annotation and the rest nonessential by a fitness record, so label origin coincides with class on this set. None of these genes carries a yeast-GEM reaction, so FBA's knockout ratio is 1.0 on every one and the solve has no call. Cells are seeds 2700 to 2702; the pooled column scores the seed-mean logit with a 2,000-draw gene bootstrap. Bold marks the highest value in a column.

arm	cells	per cell	mean	sd	pooled [95% CI]	selected epochs
pathway multi-hot + ProtT5 + FBA affine (round 22 recipe)	3	0.571, 0.571, 0.579	0.574	0.004	0.572 [0.55, 0.59]	–
MLP on ProtT5 alone	3	0.539, 0.531, 0.532	0.534	0.003	0.534 [0.51, 0.55]	62, 58, 56
MLP on ProtT5 + FBA ratios	3	0.494, 0.476, 0.488	0.486	0.008	0.481 [0.46, 0.50]	71, 197, 76
MLP on the three inputs	3	0.511, 0.508, 0.507	0.509	0.002	0.509 [0.49, 0.53]	55, 58, 55
graph convolution, physical graph, all-ones input	3	0.684, 0.683, 0.692	0.686	0.004	0.686 [0.67, 0.70]	91, 100, 77
graph convolution, physical graph, degree and clustering	3	0.757, 0.752, 0.749	0.753	0.003	0.759 [0.74, 0.78]	28, 31, 3
graph convolution, physical graph, ProtT5	3	0.545, 0.542, 0.544	0.544	0.001	0.544 [0.52, 0.57]	192, 197, 188
graph convolution, union of nine, all-ones input	3	0.643, 0.643, 0.643	0.643	0.000	0.643 [0.62, 0.66]	20, 5, 3
graph convolution, union of nine, degree and clustering	3	0.660, 0.667, 0.660	0.663	0.003	0.663 [0.64, 0.68]	197, 196, 197
graph convolution, union of nine, ProtT5	3	0.755, 0.753, 0.756	0.755	0.001	0.755 [0.74, 0.77]	197, 197, 197
graph attention, physical graph, degree and clustering	3	0.709, 0.692, 0.720	0.707	0.011	0.708 [0.69, 0.73]	179, 190, 119
graph attention, union of nine, degree and clustering	3	0.715, 0.704, 0.719	0.713	0.006	0.717 [0.70, 0.74]	197, 197, 197
graph convolution on ProtT5 alone, 400 epochs	3	0.746, 0.753, 0.749	0.749	0.003	0.749 [0.73, 0.77]	373, 380, 393
graph convolution on ProtT5 + FBA ratios	3	0.759, 0.758, 0.758	**0.759**	0.000	0.759 [0.74, 0.78]	196, 197, 197
graph convolution on the three inputs, 400 epochs	3	0.742, 0.744, 0.744	0.744	0.001	0.744 [0.72, 0.76]	380, 396, 380
graph convolution on GO alone, 400 epochs	3	0.716, 0.697, 0.694	0.702	0.010	0.703 [0.68, 0.72]	397, 395, 379
graph convolution on GO + FBA ratios, 400 epochs	3	0.699, 0.693, 0.707	0.700	0.006	0.700 [0.68, 0.72]	397, 392, 397
graph convolution on ProtT5 + GO + FBA ratios, 400 epochs	3	0.700, 0.697, 0.707	0.701	0.004	0.702 [0.68, 0.72]	397, 368, 391
FBA knockout ratio on YPD (0 of 4674 genes carry a reaction)	1	0.500	0.500	0.000	0.500 [0.50, 0.50]	–

Table 30: The gene's degree in each graph as the essentiality score, nothing trained, on the 4,674 genome-wide held-out genes of round 30, the 195 released held-out genes, and the 787 released genes the models train and validate on. AUROC of degree with a 2,000-draw gene bootstrap, and the median degree of the essential against the nonessential genes in that set. Bold marks the highest AUROC in a column. Written by `experiments/028-gene-essentiality/scripts/degree_by_class.py`.

	genome held-out (4,674)		released held-out (195)		released train + val (787)	
graph	AUROC	median ess / non	AUROC	median ess / non	AUROC	median ess / non
physical	**0.740** [0.72, 0.76]	50 / 16	0.594 [0.47, 0.71]	27 / 19	0.591 [0.53, 0.65]	21 / 17
regulatory	0.478 [0.46, 0.50]	4 / 5	0.513 [0.41, 0.62]	6 / 7	0.474 [0.42, 0.53]	6 / 6
tflink	0.562 [0.55, 0.58]	30 / 27	0.424 [0.31, 0.53]	37 / 46	0.424 [0.37, 0.48]	37 / 42
STRING neighborhood	0.514 [0.50, 0.53]	0 / 0	**0.618** [0.50, 0.73]	175 / 67	**0.629** [0.57, 0.68]	142 / 68
STRING fusion	0.625 [0.61, 0.64]	1 / 0	0.605 [0.49, 0.71]	7 / 4	0.555 [0.50, 0.61]	6 / 5
STRING cooccurence	0.497 [0.48, 0.51]	0 / 0	0.356 [0.27, 0.45]	0 / 1	0.368 [0.32, 0.42]	0 / 1
STRING coexpression	0.681 [0.66, 0.70]	394 / 182	0.617 [0.49, 0.73]	477 / 292	0.585 [0.52, 0.64]	385 / 300
STRING experimental	0.638 [0.62, 0.66]	263 / 121	0.594 [0.47, 0.70]	203 / 158	0.517 [0.46, 0.57]	139 / 149
STRING database	0.714 [0.70, 0.73]	22 / 1	0.577 [0.48, 0.68]	41 / 29	0.485 [0.43, 0.54]	24 / 29
union of nine	0.655 [0.64, 0.67]	534 / 297	0.611 [0.49, 0.73]	660 / 412	0.572 [0.51, 0.63]	540 / 449

The inputs do not transfer and the graph does. The MLP on the transformer's three inputs reads 0.509 on the 4,674 genes (0.486 on ProtT5 with the ratios), chance, from 0.870 on the released 195; the same inputs through two layers of graph convolution read 0.744, ProtT5 alone through the graph 0.749 and with the ratios 0.759, and GO alone through the graph 0.702, under the three-input convolution here where on the 195 it was 0.02 above it. The graph convolution on GO does not keep its margin off the metabolic genes; the graph itself carries what transfers, and the ratios add 0.01 to a convolution on ProtT5 when every ratio in the test set is 1.0, which measures the training set and not the test set. What the graph carries is the gene's neighborhood in the physical graph, measured directly with the ladder's structure-only inputs (job 2401): degree and clustering alone, through two layers of convolution over the physical graph, read 0.753 [0.74, 0.78], level with ProtT5 through the union graph (0.755) and with the best number on the set, while the all-ones input reads 0.686 on the physical graph and 0.643 on the union, ProtT5 without a graph 0.534, and ProtT5 through the physical graph at 200 epochs 0.544. On these genes the number is the degree of the gene in the protein interaction graph, which a convolution reads from any input that lets it count neighbors and a model without the graph cannot; whether that is essentiality or the deletion collection's own coverage the label origin does not let this set decide. Degree needs no model to say so (Table 30): the gene's degree in the physical graph, ranked as it is, reads 0.740 [0.72,

0.76] on the 4,674 genes, a median of 50 interactions for an essential gene against 16 for a nonessential one, and 0.594 [0.47, 0.71] on the released 195 and 0.591 on the 787 released training genes, where the medians are 27 against 19 and 21 against 17. The STRING database channel's degree reads 0.714 on the genome-wide genes and 0.577 on the released ones, the union's 0.655 and 0.611, and no other graph's degree passes 0.69 anywhere. So the physical graph's degree is a genome-wide essentiality signal that the metabolic genes barely vary in, which is why no model in this document read it on the 195 and every graph model read it on the 4,674. The transformer does not transfer either: the round-22 recipe, the pathway multi-hot with ProtT5 and the FBA affine, trained on the same 629 genes reads 0.574 [0.55, 0.59] on the 4,674 (0.570 to 0.578 over three seeds, job 2396), 0.07 above chance and 0.17 under the graph convolution on ProtT5 alone, from 0.890 on the released 195. Its inputs are the MLP's, and in this recipe the gene graphs enter training only through the graph regularizer, not as a message-passing input, so the three models that read 0.87 to 0.89 on the metabolic genes without the graph as an input all fall to chance or near it off them, and the one that convolves over the graph does not. The scalable module form on the same split runs at the time of writing; its solved flux is the wild-type vector on every one of these genes, a constant the head cannot rank by, so what it can add over the backbone's 0.574 is nothing by construction.

The other direction says the larger label set is not the missing training data for the metabolic genes (Tables 31 and 32). Trained on the 4,368 genome-labeled genes and read on the released 195, the graph convolution on GO reads 0.913 on every seed (0.912, 0.913, 0.913), the number it reads from 629 genes with the seed spread closed, and the cheap models on the transformer's three inputs lose to their 629-gene runs: the MLP 0.837 from 0.870, the 400-epoch convolution 0.843 from 0.888. Seven times the labels, of a different origin for six sevenths of them, buy the GO convolution nothing and cost the three-input models 0.03 to 0.05, which is what round 1 measured on the transformer.

Table 31: The three rows of Table 9 trained on the genome-wide labels (split `ess_genome_val0.2_s0`: 4,368 train and 1,093 validation genes, the released labels where the file has them and SGD or a fitness record elsewhere) and read on the same 195 released held-out genes, seeds 2700 to 2702. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
GCN	union of nine	GO (400 epochs)	2,362	155,457	**0.913**	0.000	0.912, 0.913, 0.913	397, 397, 397
GCN	union of nine	GO + FBA ratios (400 epochs)	2,366	155,713	0.911	0.002	0.912, 0.914, 0.908	397, 397, 395
GCN	union of nine	ProtT5 + GO + FBA ratios (400 epochs)	3,390	221,249	0.907	0.002	0.906, 0.910, 0.906	396, 397, 397

Table 32: The MLP and graph convolution on the transformer's inputs (Table 5) trained on the genome-wide labels (split `ess_genome_val0.2_s0`) and read on the same 195 released held-out genes, seeds 2700 to 2702. Bold marks the highest value in a column.

model	graph	inputs	dim	params	mean	sd	seeds	selected epochs
MLP	none	ProtT5 + chromosome-pathway	1,221	82,433	0.808	0.004	0.812, 0.803, 0.810	101, 71, 71
GCN	union of nine	ProtT5 + chromosome-pathway	1,221	82,433	0.824	0.001	0.825, 0.825, 0.823	197, 197, 197
MLP	none	ProtT5 + FBA ratios	1,028	70,081	0.770	0.000	0.769, 0.770, 0.770	79, 80, 79
GCN	union of nine	ProtT5 + FBA ratios	1,028	70,081	0.818	0.002	0.819, 0.820, 0.815	197, 196, 197
MLP	none	ProtT5 + chromosome-pathway + FBA ratios	1,225	82,689	0.837	0.002	0.837, 0.840, 0.834	68, 80, 71
GCN	union of nine	ProtT5 + chromosome-pathway + FBA ratios	1,225	82,689	0.824	0.003	0.822, 0.822, 0.827	197, 197, 197
GCN	union of nine	ProtT5 (400 epochs)	1,024	69,825	0.831	0.002	0.828, 0.831, 0.833	397, 397, 396
GCN	union of nine	ProtT5 + chromosome-pathway + FBA ratios (400 epochs)	1,225	82,689	**0.843**	0.002	0.841, 0.843, 0.846	395, 395, 395

3.19 Round 16: the round-9 backbone with the round-11 affine ✗

Round 16 (jobs 2309 and 2310) puts the widened affine on the round-9 backbone: the chromosome-pathway indicator with ProtT5, and the four ratios with the eight statistics and the deletion flux vector on the logit, and the same without the statistics, three seeds each, against round 9 (Table 33, Figure 15).

Table 33: Round 16, the round-9 backbone with the round-11 affine. The round-5 cell, the round-9 cell (chromosome-pathway indicator with ProtT5), the round-11 cell (ProtT5 with the widened affine), and the round-9 backbone with the widened affine: the four ratios and the pFBA vector, and those plus the eight statistics. Width 12, one block, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway + ProtT5 backbone + FBA ratios	0.871	0.015	0.860, 0.866, 0.888 ↑	0.868 [0.80, 0.93]	53, 51, 38	0.0030
context head + FBA ratios + statistics + pFBA vector	0.858	0.008	0.850, 0.858, 0.867 ↑	0.858 [0.77, 0.93]	188, 153, 162	0.0056
chrom-pathway + ProtT5 backbone + FBA ratios + pFBA vector	0.828	0.068	0.769, 0.812, 0.902	0.867 [0.79, 0.93]	6, 13, 68	0.0027
chrom-pathway + ProtT5 backbone + ratios + statistics + pFBA	**0.880**	0.018	0.863, 0.878, 0.898 ↑	**0.889** [0.83, 0.94]	60, 38, 42	0.0029

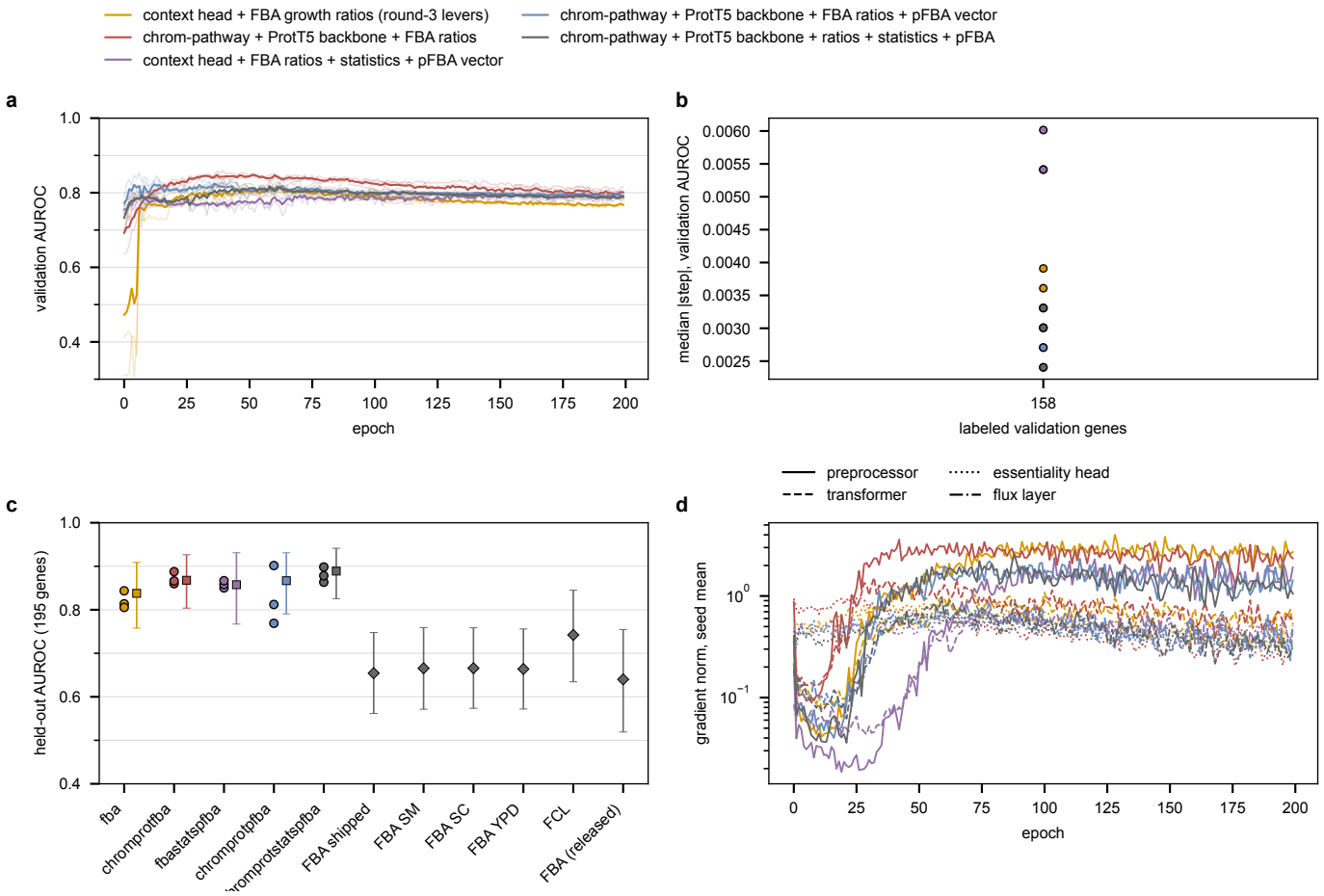


Figure 15: Round 16, the round-9 backbone with the round-11 affine. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w27`.

The two gains add, by a little. With the statistics and the vector the cell reads 0.863, 0.878 and 0.898 against round 9's 0.860, 0.866 and 0.888, up by 0.003, 0.012 and 0.010 on three of three seeds, a seed mean of 0.880 with a spread of 0.018 and a pooled bootstrap of 0.889 [0.83, 0.94]. The step is a fifth of the one the indicator gave and inside the seed spread, so it is the direction rather than the size that holds. Without the statistics the vector alone reads 0.769, 0.812 and 0.902, two seeds selecting epochs 6 and 13, the instability round 11 showed on the ProtT5 backbone, and the same cure. Round 17 runs the passing cell on the five folds, with the read fixed first: its ensemble replaces the round-12 ensemble's 0.893 as the reported number only if its fold models are above round 12's on at least four of the five folds; otherwise 0.893 stands.

3.20 Round 17: the round-16 cell on the five folds ✗

Round 17 (jobs 2312 and 2313) trains the round-16 cell on the five cross-validation files (Table 34, Figure 16). Its fold models score 0.863, 0.860, 0.887, 0.841 and 0.882 on the held-out genes, a fold mean of 0.867, against round 12's 0.860, 0.899, 0.889, 0.876 and 0.884: above on fold 0 by 0.003 and below on the other four, by 0.002 to 0.039. Their averaged logits score 0.901 [0.85, 0.95], above the round-12 ensemble's 0.893 by 0.008, with the two intervals almost coincident.

Table 34: Round 17, the round-16 cell on the five folds. The round-9 and round-16 cells on the split file, the round-12 fold models of the round-9 cell, and the five fold models of the round-16 cell (chromosome-pathway indicator with ProtT5, the affine reading the four ratios, the eight statistics and the pFBA vector), whose logits are averaged on the held-out genes. Width 12, one block, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
chrom-pathway + ProtT5 backbone + FBA ratios	0.871	0.015	0.860, 0.866, 0.888	0.868 [0.80, 0.93]	53, 51, 38	0.0030
chrom-pathway + ProtT5 backbone + ratios + statistics + pFBA	0.880	0.018	0.863, 0.878, 0.898 ↑	0.889 [0.83, 0.94]	60, 38, 42	0.0029
chrom-pathway + ProtT5 + FBA ratios, fold 0	0.860	0.000	0.860	0.860 [0.79, 0.92]	53	0.0030
chrom-pathway + ProtT5 + FBA ratios, fold 1	**0.899**	0.000	0.899	**0.899** [0.84, 0.95]	32	0.0030
chrom-pathway + ProtT5 + FBA ratios, fold 2	0.889	0.000	0.889	0.889 [0.83, 0.94]	25	0.0055
chrom-pathway + ProtT5 + FBA ratios, fold 3	0.875	0.000	0.875	0.875 [0.81, 0.93]	34	0.0045
chrom-pathway + ProtT5 + FBA ratios, fold 4	0.884	0.000	0.884	0.884 [0.82, 0.94]	27	0.0055
chrom-pathway + ProtT5 + widened affine, fold 0	0.863	0.000	0.863	0.863 [0.79, 0.93]	60	0.0024
chrom-pathway + ProtT5 + widened affine, fold 1	0.860	0.000	0.860	0.860 [0.77, 0.93]	135	0.0078
chrom-pathway + ProtT5 + widened affine, fold 2	0.887	0.000	0.887	0.887 [0.81, 0.95]	26	0.0085
chrom-pathway + ProtT5 + widened affine, fold 3	0.840	0.000	0.840	0.840 [0.75, 0.92]	38	0.0052
chrom-pathway + ProtT5 + widened affine, fold 4	0.882	0.000	0.882	0.882 [0.82, 0.94]	42	0.0082

By the rule fixed before the round, 0.893 stands: the widened affine's fold models are below round 12's on four of five folds, and an ensemble 0.008 higher from fold models 0.014 lower on average is the kind of draw the rule exists to refuse. The two cells are one cell to the resolution of 195 genes, and the reported number for this document is the round-12 ensemble, 0.893 [0.84, 0.94], from the chromosome-pathway indicator with ProtT5 and the four FBA growth ratios.

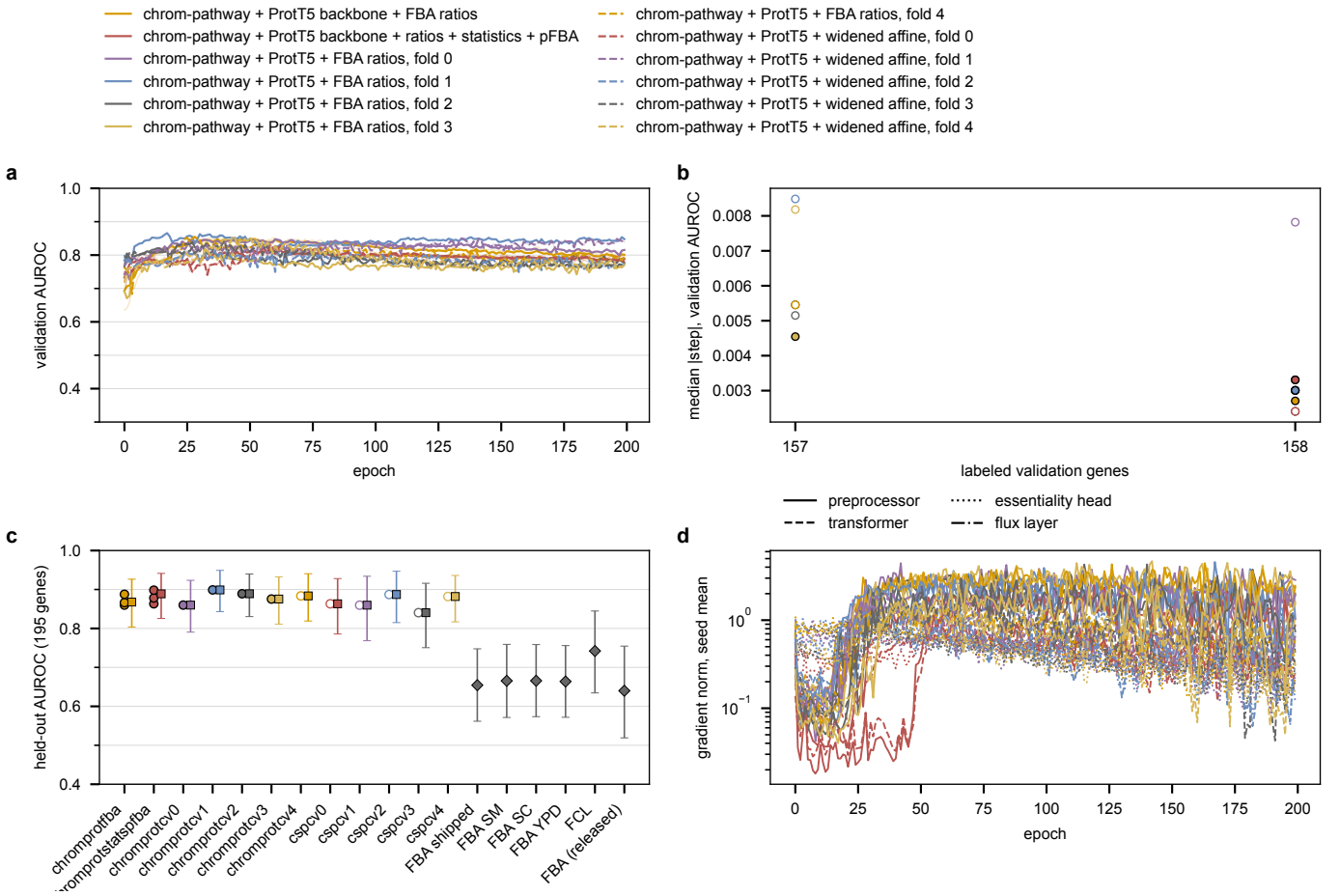


Figure 16: Round 17, the round-16 cell on the five folds. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; each fold model reads its own fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w28`.

4 The metabolic module: the solve helps, so why not the layer? ✗

One linear program per deletion on yeast-GEM ranks essential genes at 0.65 on its own and lifts every model in this document by 0.05 when its four growth ratios are appended (Sections 2 and 3). The metabolic module is built on the same genome-scale model: the same stoichiometry, the same bounds, the same gene-protein-reaction rules closing the deleted gene's reactions. The question of this section is why routing the prediction through it does not do what the solve does. The rounds answer it in steps: the module neither helps nor hurts on the backbone (rounds 2 and 4); it adds nothing on top of the solve (round 6); its growth column is not growth but a free coordinate, because the layer never reaches the polytope the solve optimizes over (the probe, round 7); handed the solve's own point on that polytope, the anchored layer's constraint set discards it (round 10) while the GEM's own box keeps it, and the gated module cell then reaches the level of the solve appended as an input (round 13); a layer that solves the program instead of projecting toward it is infeasible on the anchored box, whose capacity ceilings exclude growth (round 15), and on the GEM's box it is feasible, ranks above FBA before training, and gated reaches the same level (round 15b); with the ceilings calibrated to admit growth, the growth column alone ranks the held-out genes at 0.84 on one seed and the program's infeasible region absorbs the other two (round 15c). The gradient reaches the alternating layer throughout, so its null is not a gradient-flow failure; the layer that solves has the linear program's own gradient, the dual of the binding bounds. The section closes with what the comparison says a layer has to compute to help.

The pooled backbone reads its logit from an MLP on the perturbed context (Figure 18). A flux arm builds the enzyme-constrained thermodynamic flux layer on the same context and exposes one physically named quantity, the flux through the growth reaction `r_2111`. Three readouts put that quantity to different uses, and they differ in what the backbone's gradient has to pass through.

biomass (forced) $\ell = a\, v_{\text{growth}} + b$. The growth column is the only path from the context to the logit, so the module carries the whole prediction or the cell scores chance (Figure 19). This is FBA's own reading of essentiality with a learned threshold, and it is the wave-9 design.

context + growth $\ell = \text{MLP}(h) + a\, v_{\text{growth}} + b$. Both paths from the first step; the per-module gradient norms say how much of each update the module carries when it is not forced to carry all of it.

gated $\ell = \text{MLP}(h) + g\,(a\, v_{\text{growth}} + b)$ with $g_0 = 0$ (Figure 20). At initialization the cell is the pooled model exactly; the gate's trajectory, logged every epoch, says whether the optimizer chose the flux path, and the `flux_layer` gradient line says whether gradient reached the layer, which at $g = 0$ it cannot.



Figure 17: Color key for the readout diagrams. White is the input; orange a learned part of the backbone (the context, the abundance head); yellow the flux layer and its capacity terms, dashed where the box is a fixed table (the k_{cat} table); purple the fixed stoichiometry of the GEM; red the prediction and the loss. The key holds for Figures 18 to 20. Rendered from `notes/experiments.028-gene-essentiality.mermaid.readouts.md`.

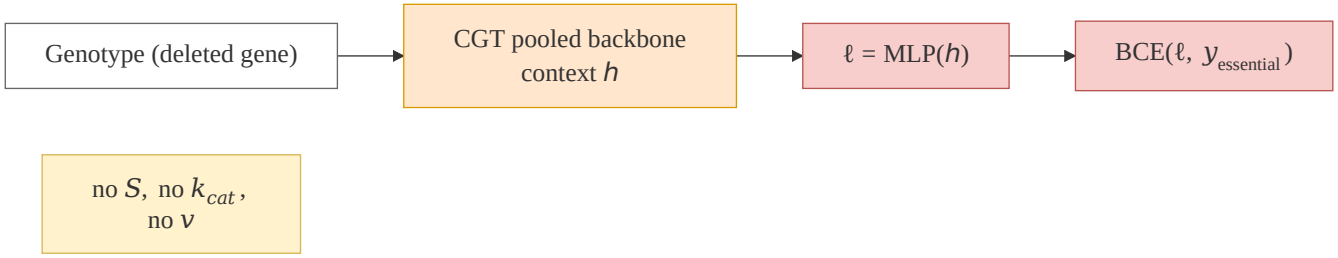


Figure 18: The pooled readout. The backbone alone: one MLP from the perturbed context to a logit, binary cross entropy against the released label. No stoichiometry, no kinetic parameters, no flux. Colors as in Figure 17.

4.1 What the layer's flux violates ✗

Every flux arm in this document builds the anchored layer: the predicted k_{cat} tier (97 measured, 782 homolog, 2,831 predicted, 18 default catalytic units), the union-external reaction energies (3,779 of 4,131 reactions with a $\Delta_r G'$), the

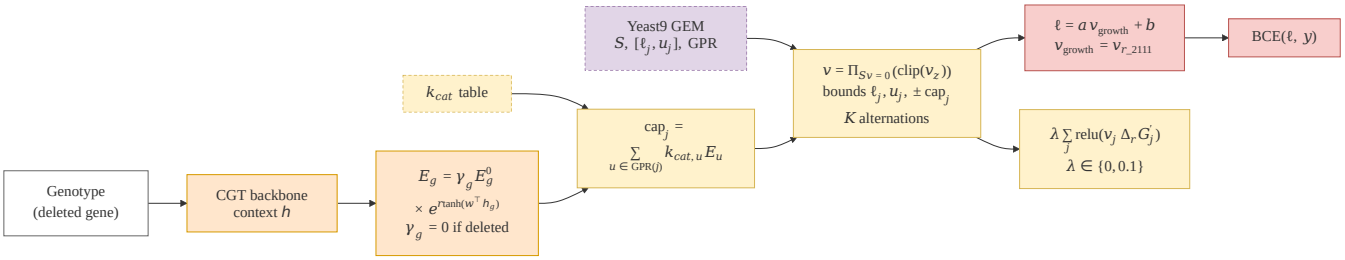


Figure 19: The forced biomass readout. The context sets each enzyme's abundance (a deleted gene's is zero), the k_{cat} table turns abundances into capacities, and the flux is the projection onto $Sv = 0$ of the capacity-clipped proposal; the logit is an affine of the growth column alone. The thermodynamic penalty is a loss on the flux with no path to the prediction, at scale 0.1 or 0. Colors as in Figure 17.

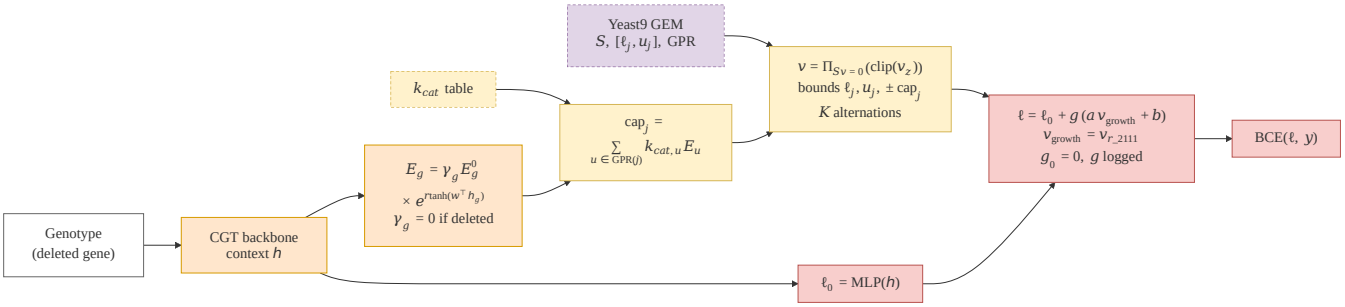


Figure 20: The gated readout. The pooled MLP's logit plus the growth affine behind a gate initialized at zero. The gate is a parameter of the readout; its value is written into every epoch row as `ess_gate`. Colors as in Figure 17.

enzyme-capacity ceiling from the abundance head in the box, and the second law, the protein budget and the Gibbs dissipation as penalty terms at the constraint scale. The layer logs every one of those residuals each epoch, and Table 35 reads them at the selected epoch across the module arms. With the balance exact, the flux is outside its bounds on 58 to 63 percent of reactions, runs against its reaction energy on 44 to 51 percent, uses 0.2 to 7 times the protein pool, and dissipates 5,000 to 12,000 J/gDW/h against the 3,700 limit, with the penalty on or off; with the bounds exact (round 7) the second-law violation falls to a third and the dissipation rises to 100,000. No arm's flux satisfies the constraints the layer was built to hold, so on this endpoint the k_{cat} table and the reaction energies are inputs to a projection that does not honor them, which is the precise sense in which the module is not the solve.

The Gibbs dissipation is reported here as a diagnostic and nothing else. The Niebel limit is an empirical plateau of one organism's physiology, not a constraint of the network, and a model that stays under it because it was penalized for exceeding it has been told the answer. The anchored arm's registry entry carries the dissipation hinge, so the round-2 and round-4 cells at constraint scale 0.1 trained against it; every cell of the layer that solves ran at constraint scale 0, where the hinge has weight zero, and from round 18 the module cells run on the `nodiss` entry of the same arm, where the term is absent by construction and the rate is logged only.

Table 35: What the layer's flux violates at the selected epoch, seed mean per module arm: the median mass-balance residual $|Sv|$ over metabolites; the fraction of reactions outside their bounds; the fraction of reactions running against their reaction energy ($v_j \Delta_r G'_j > 0$); the enzyme budget used as a multiple of the protein pool; and the Gibbs dissipation in J/gDW/h against the 3,700 limit. Rounds are abbreviated r2 to r13. Every arm builds the anchored layer (predicted k_{cat} tier, union-external ΔG) unless it names the GEM box. Constraint scale 0 means no penalty on any of these terms; 0.1 penalizes all of them.

arm	cells	$\|Sv\|$ median	box	second law	budget	dissipation
r2 forced, scale 0.1	3	1.4e-08	0.63	0.48	0.2	-8,375
r2 forced, scale 0	3	1.3e-08	0.62	0.51	0.8	7,099
r2 gated, scale 0.1	3	1.1e-08	0.63	0.44	0.3	-9,854
r4 forced, scale 0	3	7.9e-09	0.59	0.51	2.4	10,264
r4 gated, scale 0	3	1.1e-08	0.58	0.51	3.6	12,277
r6 forced + FBA affine	3	7.5e-09	0.59	0.51	2.2	10,115
r7 forced, clip last	3	9.0e-01	0.00	0.32	1.3	102,065
r7 forced, box form	3	2.0e+00	0.00	0.38	1.4	120,815
r10 forced, warm start	3	1.3e-09	0.33	0.49	6.9	5,815
r10 gated, warm start	3	9.3e-10	0.36	0.49	1.3	5,104
r13 forced, GEM box, warm start	3	2.8e-09	0.26	0.00	nan	nan
r13 gated, GEM box, warm start	3	2.5e-09	0.27	0.00	nan	nan

4.2 What holds by construction, and what does not ✗

Three constructions of the module appear in this section, and they differ in which constraints their flux satisfies exactly, which it satisfies through a penalty, and which it satisfies only when the backbone finds a way to. The alternating layer (Figure 21) writes a proposal in the null space of S, so mass balance holds at the start, and alternates the balance projection with the box clip K times, so whichever of the two runs last holds exactly and the other is a residual; the second law, the protein pool and the capacity ceilings are hinges at the constraint scale, and Table 35 measures what the hinges leave. The layer that solves (Figure 28) returns the growth optimum of the linear program under the layer's bounds, so mass balance, the box and every capacity ceiling hold exactly at once, with no penalty in the loss; what it does not guarantee is that a program exists, and rounds 15 to 15d measure what happens when it does not. The elastic layer that solves (Figure 33) keeps all of that and adds two things by construction: the protein pool, by projecting the abundance head's output onto the budget before the ceilings are formed, and feasibility, by letting the one forced flux in the model, the ATP maintenance reaction's lower bound of 0.7 mmol/gDW/h, go unmet at a price. Its zero flux with the maintenance fully unmet satisfies every constraint, so the program solves for every genotype at every ceiling, and a genotype that cannot meet its maintenance is not absent from the gradient but priced into it. Under every construction the second law enters only through the anchored reaction energies and the dissipation only as a logged rate. What "the enzyme-capacity box contains no growing flux" means, stated exactly with the elastic program in hand: at the ceilings as parameterized the maintenance flux the model forces cannot be carried, the shortfall is 0.54 of 0.70 with zero growth (`capacity_calibration.py -lp-slack-penalty 1`, 32 held-out genotypes), the hard program's feasible set is therefore empty, and the solver returns nothing; from a ceiling scale of 10 the shortfall is zero and growth is 0.01, and from 30 the box admits the growth the rounds train on.

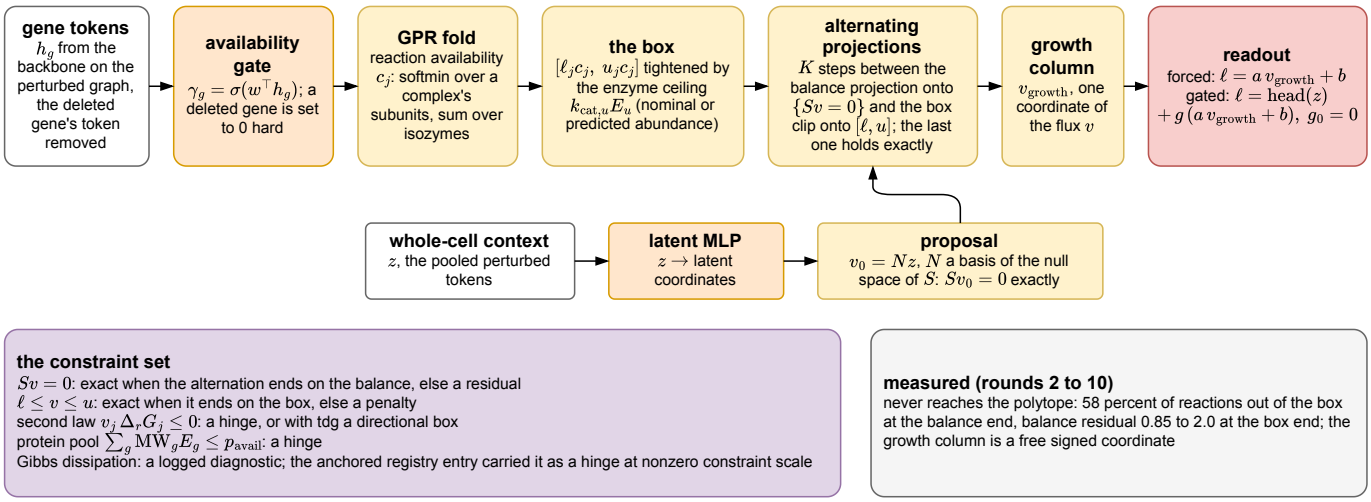


Figure 21: The alternating layer, rounds 2 to 13. The backbone's gene tokens set a learned availability gate, which the gene-protein-reaction rule folds to a per-reaction availability and a box tightened by the enzyme-capacity ceiling; a proposal written in the null space of S from the whole-cell context is alternately projected onto the balance and clipped onto the box, and the growth column of the result enters the readout, alone or behind a zero-initialized gate. Of the constraint set only the projection the alternation ends on holds exactly; the second law and the protein pool are hinges, and the Gibbs dissipation is a logged diagnostic. Colors as in Figure 17, the same in Figures 28 and 33. Drawn in draw.io from `notes/assets/drawio/028-module.gen.py`.

4.3 Round 2: the module on the stable backbone, with and without the penalties ✗

Round 2 (jobs 2182 to 2208) puts the three readouts on the width-12 cell of Section 3, on the released-label split, at three seeds each: the forced and gated readouts at constraint scale 0.1 and 0, the context readout at both scales, the forced readout at scale 0 with the k_{cat} vector permuted across catalytic units (same architecture, no unit-to-value correspondence), and the stoichiometry-only layer (`flux_off`: no thermodynamics, no capacity) with the forced and context readouts. Every flux arm carries the wave-9 flux configuration otherwise: the predicted k_{cat} tier, the union-external thermodynamic source, twenty warmup epochs, the dataset medium, the mixture concentration space. The paired comparator is the round-1 pooled cell on the same split at the same batch, weight decay and budget (Table 36, Figure 22).

The reads were fixed before the numbers: *not hurt* is the gated arm against the pooled cell paired by seed, with the gate's trajectory beside it; *help* is the forced arm above the pooled cell on three of three seeds and above its shuffled-k_{cat} control; the *collapse's cause* is the forced arm at scale 0.1 against scale 0, since on the betaxanthin targets the penalty scale was the dose that starved the layer's input.

Table 36: Round 2, the metabolic module. The round-1 pooled cell on the released-label split (`fcl`) and the same cell with the prediction routed through the flux layer: the growth column alone (forced), the context head plus the growth affine, and that affine behind a zero-initialized gate, at constraint scale 0.1 and 0, with the shuffled-kcat control and the stoichiometry-only layer. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
released labels, 158-gene val	0.725	0.019	0.745, 0.722, 0.708	0.738 [0.65, 0.83]	66, 77, 77	0.0026
growth column only, constraints 0.1	0.706	0.105	0.759, 0.585, 0.774	0.765 [0.67, 0.85]	77, 26, 76	0.0037
growth column only, constraints 0	0.732	0.068	0.770, 0.653, 0.772	**0.773** [0.68, 0.86]	72, 72, 72	0.0047
growth column only, kcat shuffled, constraints 0	**0.751**	0.029	0.771, 0.718, 0.765	0.769 [0.68, 0.86]	74, 77, 75	0.0029
context + growth column, constraints 0.1	0.666	0.086	0.566, 0.709, 0.722	0.717 [0.61, 0.82]	49, 77, 77	0.0053
context + growth column, constraints 0	0.667	0.096	0.556, 0.727, 0.718	0.724 [0.62, 0.82]	49, 73, 77	0.0068
gated growth column, constraints 0.1	0.719	0.029	0.721, 0.747, 0.690	0.738 [0.64, 0.83]	41, 77, 77	0.0027
gated growth column, constraints 0	0.735	0.018	0.714, 0.744, 0.747	0.747 [0.65, 0.84]	40, 77, 77	0.0031
growth column only, stoichiometry only	0.606	0.072	0.524, 0.631, 0.661 ↓	0.664 [0.56, 0.77]	19, 73, 65	0.0033
context + growth column, stoichiometry only	0.705	0.023	0.719, 0.717, 0.678 ↓	0.755 [0.67, 0.84]	75, 76, 61	0.0042

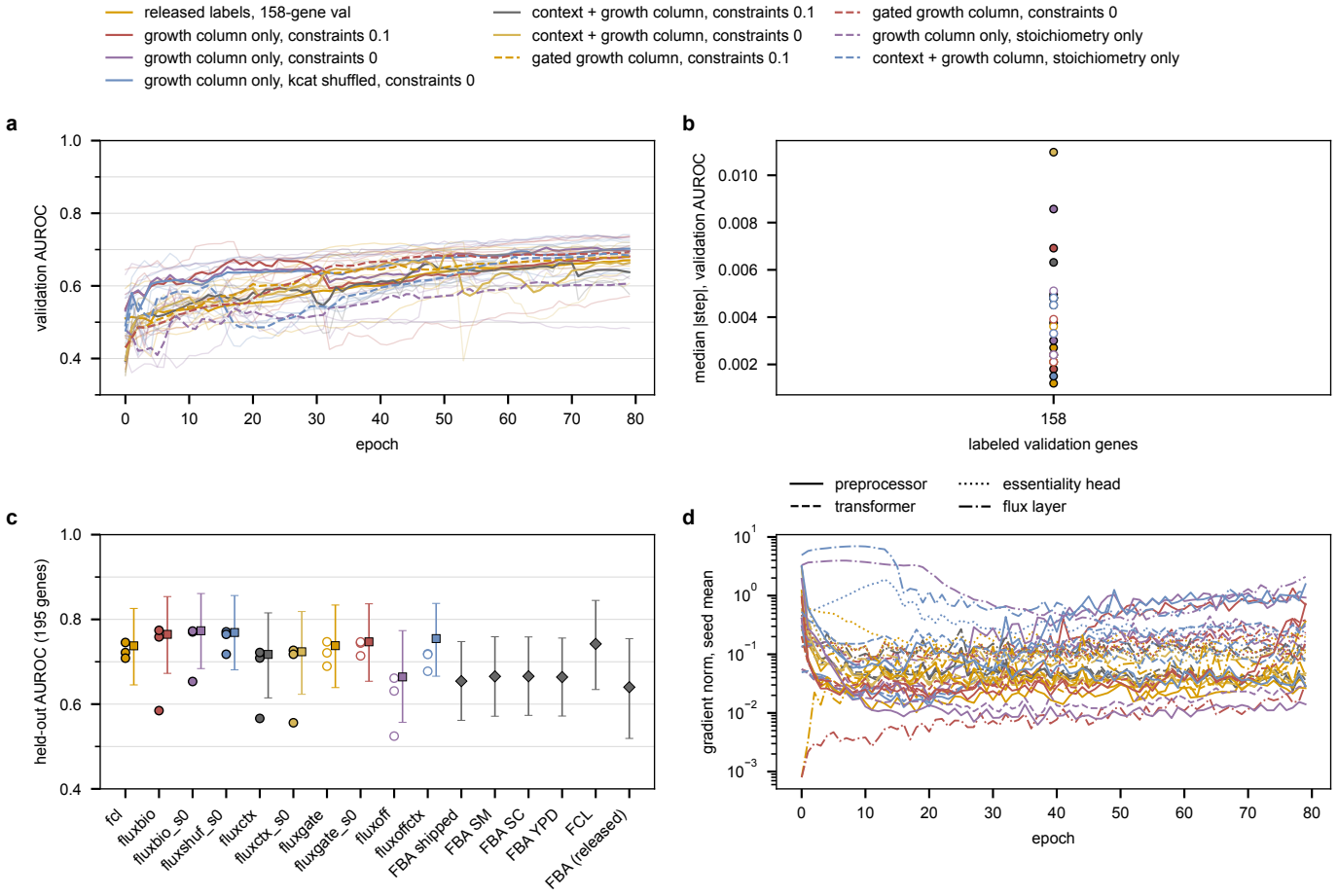


Figure 22: Round 2, the metabolic module. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step of that curve per cell, against the number of labeled validation genes, 158 for every cell in this round. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w13`.

Not hurt holds. The gated arm at scale 0 scores 0.714, 0.744 and 0.747 against the pooled cell's 0.745, 0.722 and 0.708 on the same seeds, two of three above by 0.02 to 0.04 and one below by 0.03, a seed mean of 0.735 against 0.725; at scale 0.1 the same comparison is one of three above. The gate leaves zero and stops: at epoch 80 it reads −0.056 to −0.063 on all six gated cells, and the flux layer's gradient norm sits at 0.01 to 0.07 against 0.22 to 0.36 on the forced arm's two seeds that train. The optimizer takes a small negative weight on the growth affine and the layer receives a tenth of the forced arm's gradient, so the gated cell is a pooled cell with a small correction, and it scores like one.

Help does not hold. The forced arm at scale 0 scores 0.770, 0.653 and 0.772, two of three seeds above the pooled cell and

one seed 0.07 below; at scale 0.1 the split is the same and the low seed reads 0.585. The shuffled-k_{cat} control, the same layer with the catalytic constants permuted across units, scores 0.771, 0.718 and 0.765, a seed mean of 0.751 against the real table's 0.732, so whatever the growth column carries here does not come from the correspondence between units and constants. The forced arm's pooled bootstrap, 0.773 [0.68, 0.86], is above Flux Cone Learning's 0.742 and FBA's 0.640 to 0.666 on the same genes, level with the pooled cell's 0.738, and below the ladder's 0.80 line.

The collapse on this endpoint is not the penalty's. Seed 2701 falls at both scales, to 0.585 at 0.1 and 0.653 at 0, and the two runs are the same run: the gradient norms start where the other seeds' do, decay within ten epochs to a hundredth of them (preprocessor 0.002, transformer 0.003 at epoch 80, against 1.0 and 0.5 on seed 2700) and stay there, with the training loss flat at 0.46 against 0.34, which is a stalled seed rather than a starved layer. The stoichiometry-only layer collapses on all three seeds (0.524, 0.631, 0.661), so the capacity bound and the thermodynamic term are what keep the growth column informative rather than what starve it, the opposite of the betaxanthin reading. Adding the context head beside the growth affine without a gate costs 0.06 on the seed mean (0.666 and 0.667 against 0.725) and puts one seed at 0.56; the same sum behind the zero-initialized gate loses nothing.

4.4 Round 4: the module on the best backbone ✗

Round 4 (jobs 2246 to 2248, three cells per GPU) repeats the gated readout, the forced readout and the shuffled-k_{cat} control at constraint scale 0 on the round-3 cell, batch 32 with the class weight at 200 epochs, paired by seed against it (Table 37, Figure 23). The reads are the round-2 reads.

Table 37: Round 4, the module on the round-3 levers. The round-3 pooled cell (batch 32, class weight, 200 epochs) and the same cell with the prediction routed through the flux layer at constraint scale 0: the gated readout, the forced growth column, and the forced growth column with the kcat vector shuffled. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
growth column only, kcat shuffled (round-3 levers)	**0.779**	0.006	0.785, 0.781, 0.772	**0.781** [0.69, 0.87]	66, 71, 74	0.0048

Every arm lands within 0.015 of the pooled cell on every seed. The gated readout scores 0.766, 0.782 and 0.765 against 0.780, 0.769 and 0.773, one of three above; the forced readout 0.772, 0.781 and 0.778, two of three above by 0.005 and 0.012 and one below by 0.008; the shuffled-k_{cat} control 0.785, 0.781 and 0.772, two of three above, a seed mean of 0.779 against the real table's 0.777. The gate now moves: it leaves zero after epoch 10 and climbs steadily to -0.17, $+0.20$ and $+0.11$ at epoch 200, of either sign, with the flux layer's gradient at 0.06 to 0.12 against 0.22 to 0.65 in the forced arm. The stalled seed of round 2 does not recur under batch 32 with the class weight. On every arm the validation curve peaks between epochs 60 and 110 and falls afterwards (Figure 23a), so the 200-epoch budget is more than the cell uses and the smoothed selection rule is what the number rests on.

The two rounds together answer the two questions. The module does not hurt: on the stable backbone and on the best one, the gated cell is the pooled cell to within seed noise. The module does not help: the forced growth column, which carries the whole prediction, matches the pooled MLP head to within 0.01, and it matches it equally well with the catalytic constants permuted, so the growth column is as expressive a readout as the head but its k_{cat} content is not what it reads. On this endpoint, at this width, the enzyme-constrained flux is a learned nonlinearity over the context rather than a physical prior that ranks genes.

4.5 Round 6: one medium against four, the layer on top of the solve, and two other embeddings ✗

Round 5 (Section 3) put FBA's four growth ratios beside the context head and reached the ladder's level. Round 6 (jobs 2251 to 2256, three cells per GPU) asks three questions of that result on the same cell, batch 32 with the class weight at 200 epochs, three seeds each, paired against the round-5 cell (Table 38, Figure 24): whether one medium's ratio carries what four do; whether the flux layer adds anything once the solve is present, with the forced growth column plus the affine (no context head) and the gated readout plus the affine, both at constraint scale 0; and whether the backbone on another node embedding does better, on codon frequency (64 dimensions, the encoding that reaches 0.82 with FBA on the ladder from a support vector classifier) and on the species-LM composite the expression line uses (the 5′ and 3′ species language model, ProtT5 and CaLM, 3,332 dimensions), with and without the affine.

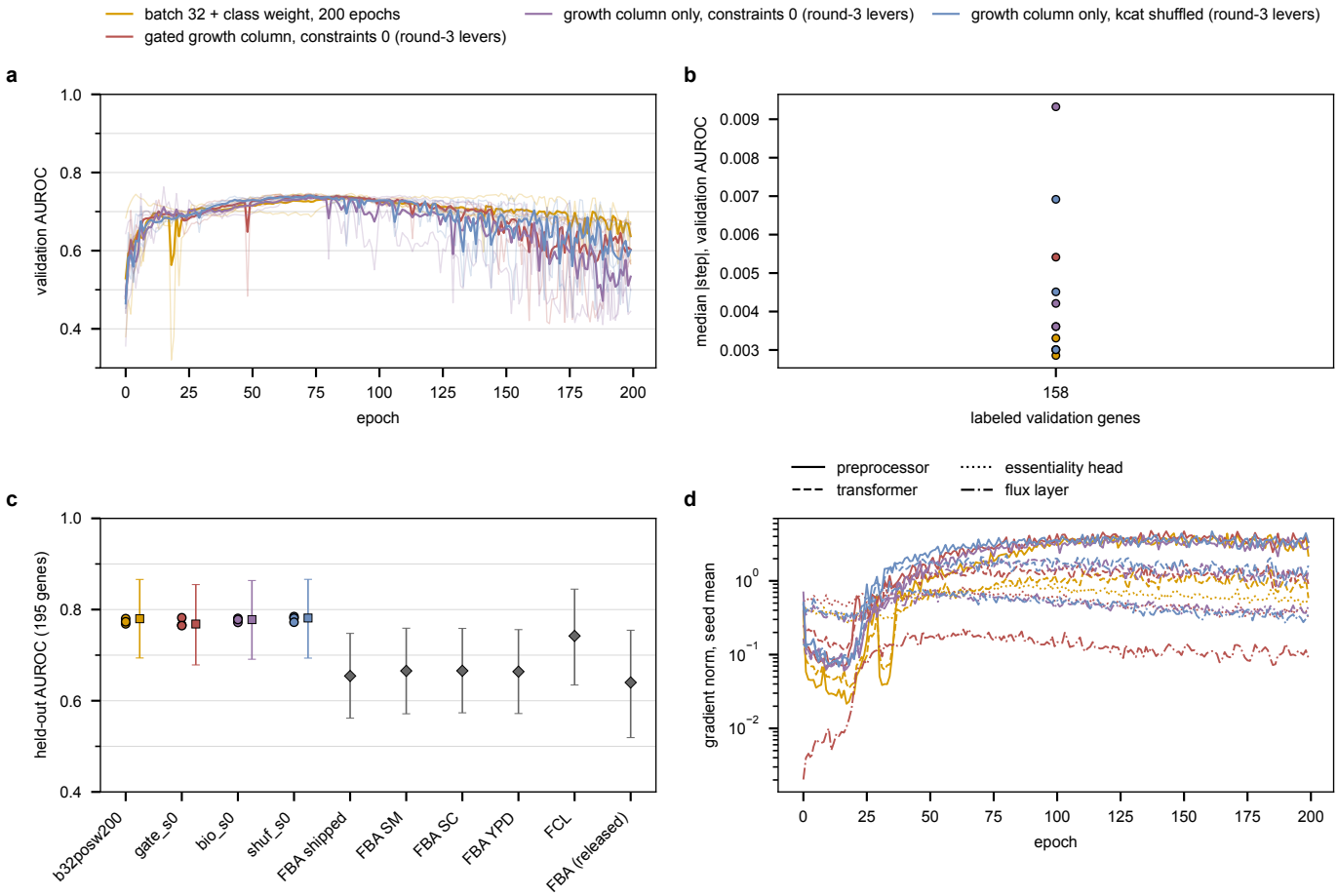


Figure 23: Round 4, the module on the round-3 cell. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 158 for every cell. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w15`.

Table 38: Round 6, one medium, the layer on top of the solve, and two other embeddings. The round-3 pooled cell and the round-5 FBA-readout cell, then: the FBA affine on the YPD ratio alone; the forced and the gated growth column of the flux layer with the FBA affine beside them (constraint scale 0); and the pooled backbone opened on codon frequency, and on the species-LM composite (5' and 3' species LM, ProtT5, CaLM), with and without the FBA affine. Batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
context head + FBA ratio, YPD only	0.795	0.022	0.819, 0.775, 0.789 ↑	0.808 [0.73, 0.89]	18, 98, 61	0.0034
growth column + FBA ratios, no context head	0.818	0.014	0.834, 0.811, 0.809 ↑	0.820 [0.74, 0.90]	74, 83, 57	0.0052
gated growth column + FBA ratios	**0.821**	0.011	0.810, 0.821, 0.832 ↑	0.827 [0.75, 0.90]	116, 43, 92	0.0041
codon-frequency backbone + FBA ratios	0.731	0.043	0.747, 0.682, 0.764 ↓	0.721 [0.62, 0.81]	16, 172, 14	0.0130
species-LM composite backbone + FBA ratios	0.816	0.013	0.802, 0.825, 0.822 ↑	0.819 [0.73, 0.90]	149, 65, 65	0.0031
species-LM composite backbone	0.778	0.005	0.783, 0.779, 0.773	0.776 [0.69, 0.85]	30, 46, 23	0.0035

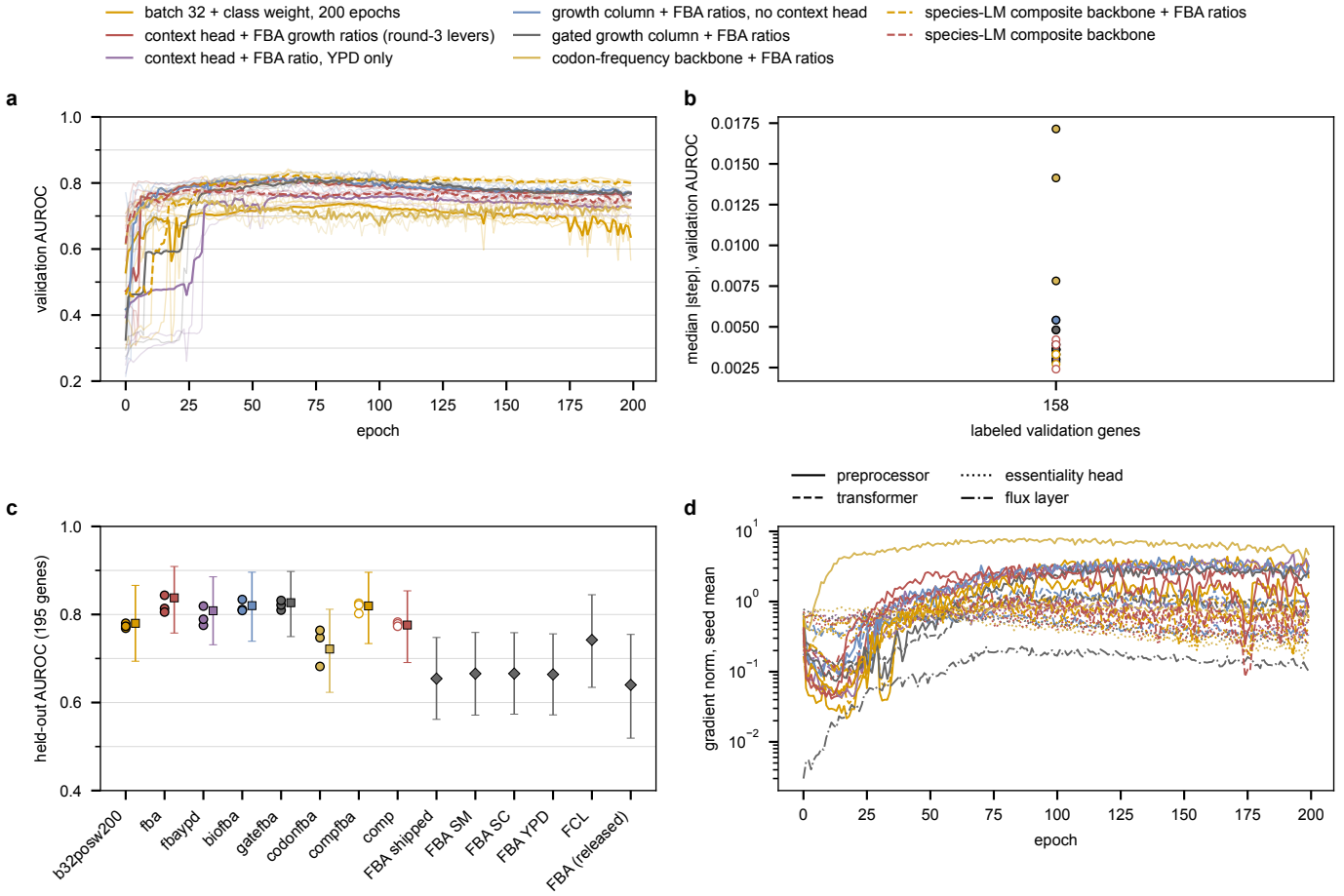


Figure 24: Round 6. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm; the second tier of arms is dashed. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points, hollow for the second tier), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w17`.

Four media carry more than one. The YPD ratio alone scores 0.819, 0.775 and 0.789 against 0.814, 0.805 and 0.844 with all four, two of three seeds below by 0.03 to 0.05, a seed mean of 0.795 against 0.821; the one column still lifts the pooled cell by 0.02.

The learned layer adds nothing on top of the solve. The forced growth column with the affine beside it and no context head scores 0.834, 0.811 and 0.809, a mean of 0.818, and the gated readout with the affine 0.810, 0.821 and 0.832, a mean of 0.821, each one of three seeds above the round-5 cell and each within its seed spread. The growth column is as good a companion to the solve as the MLP head, and no better; its gradient norm, 0.4 to 1.1 in the forced cells, says the

layer trains rather than idles. Put beside rounds 2, 4 and 5, the pattern is one statement: the module's ranking is what the backbone gives it, and the physics that ranks genes on this endpoint is the solve's optimum, which the projection layer does not compute.

The embeddings do not move the ceiling. The species-LM composite scores 0.783, 0.779 and 0.773 without the affine, against ProtT5's 0.780, 0.769 and 0.773 at 2.7 times the parameters, and 0.802, 0.825 and 0.822 with it, against 0.814, 0.805 and 0.844: within noise both times. The backbone on codon frequency with the affine scores 0.747, 0.682 and 0.764, a mean of 0.731, below the pooled ProtT5 cell and 0.09 below the ladder's codon rung with FBA (0.818), so the 64 codon numbers that a support vector classifier reads well are not read well through a 12-wide transformer; the selected epochs (16, 172, 14) and the step of 0.013 in panel b say the cell never settled.

4.6 The growth column is a signed free coordinate ✗

Four rounds put the layer's growth column level with the pooled head and blind to its own k_{cat} table while one linear program per deletion lifts every model by 0.05. The probe in Table 39 reads the column itself. It rebuilds a forced cell from its result file, deletes each of the 982 labeled genes through the same batches training used, and reads v_{growth} at initialization and under the trained weights, scoring the negative column against the label (a deletion that stops growth is essential) and the column against FBA's ratio.

Table 39: The flux layer's growth column, read for every labeled deletion at initialization and under the cell's trained weights (`flux_growth_probe.py`). Median v_{growth} over the 982 labeled deletions (mmol/gDW/h; the wild type's FVA range on YPD is 0.48 to 0.54); AUROC of the NEGATIVE growth column against the released label on the train, validation and held-out genes, so 0.5 is chance and a value below 0.5 is a ranking with the sign of growth inverted; Spearman of the column against FBA's knockout-to-wild-type ratio on the shipped bounds and on YPD.

cell	weights	median v	train	val	held-out	ρ shipped	ρ YPD
w13_fluxbio_e80_s2700	initialization	-0.27	0.618	0.555	0.547	+0.12	+0.11
w13_fluxbio_e80_s2700	trained (epoch 77)	-2.41	0.091	0.266	0.241	-0.26	-0.22
w15_bio_s0_e200_s2700	initialization	-0.27	0.618	0.555	0.547	+0.12	+0.11
w15_bio_s0_e200_s2700	trained (epoch 108)	-1.43	0.048	0.258	0.228	-0.27	-0.24
w18_biobox_e200_s2700	initialization	45.08	0.402	0.389	0.326	-0.09	-0.05
w18_biobox_e200_s2700	trained (epoch 180)	0.08	0.103	0.250	0.233	-0.25	-0.22
w18_biobox_e200_s2701	initialization	50.26	0.512	0.503	0.603	-0.14	-0.08
w18_biobox_e200_s2701	trained (epoch 95)	0.02	0.637	0.696	0.609	+0.10	+0.05
w18_biobox_e200_s2702	initialization	53.07	0.572	0.602	0.506	+0.17	+0.15
w18_biobox_e200_s2702	trained (epoch 140)	0.27	0.125	0.265	0.215	-0.28	-0.23
w18_bioprojbox_e200_s2700	initialization	0.00	0.500	0.500	0.500	–	–
w18_bioprojbox_e200_s2700	trained (epoch 2)	0.00	0.500	0.500	0.500	–	–
w18_bioprojbox_e200_s2701	initialization	0.00	0.500	0.500	0.500	–	–
w18_bioprojbox_e200_s2701	trained (epoch 2)	0.00	0.500	0.500	0.500	–	–
w18_bioprojbox_e200_s2702	initialization	0.00	0.500	0.500	0.500	–	–
w18_bioprojbox_e200_s2702	trained (epoch 2)	0.00	0.500	0.500	0.500	–	–
w21_warmbio_e200_s2700	initialization	-0.07	0.247	0.259	0.337	-0.35	-0.43
w21_warmbio_e200_s2700	trained (epoch 112)	-0.10	0.272	0.215	0.383	-0.38	-0.38
w21_warmbio_e200_s2702	initialization	-0.10	0.304	0.305	0.392	-0.40	-0.44
w21_warmbio_e200_s2702	trained (epoch 157)	-0.20	0.070	0.114	0.215	-0.37	-0.39
w21_warmgate_e200_s2700	initialization	-0.07	0.247	0.259	0.337	-0.35	-0.43
w21_warmgate_e200_s2700	trained (epoch 181)	-0.10	0.422	0.328	0.410	-0.39	-0.45
w24_warmoffbio_e200_s2700	initialization	0.38	0.798	0.698	0.683	+0.47	+0.57
w24_warmoffbio_e200_s2700	trained (epoch 53)	0.55	0.689	0.638	0.623	+0.54	+0.64
w24_warmoffgate_e200_s2700	initialization	0.38	0.798	0.698	0.683	+0.47	+0.57
w24_warmoffgate_e200_s2700	trained (epoch 85)	0.55	0.735	0.748	0.670	+0.56	+0.64
w26_lpbio_e200_s2700	initialization	0.00	0.500	0.500	0.500	–	–
w26_lpbio_e200_s2700	trained (epoch 2)	0.00	0.500	0.500	0.500	–	–
w26_lpgate_e200_s2700	initialization	0.00	0.500	0.500	0.500	–	–
w26_lpgate_e200_s2700	trained (epoch 78)	0.00	0.500	0.500	0.500	–	–
w29_lpoffbio_e200_s2700	initialization	0.19	0.744	0.767	0.734	+0.40	+0.50
w29_lpoffbio_e200_s2700	trained (epoch 11)	0.00	0.500	0.500	0.500	–	–
w29_lpoffgate_e200_s2700	initialization	0.19	0.744	0.767	0.734	+0.40	+0.50
w29_lpoffgate_e200_s2700	trained (epoch 2)	0.42	0.779	0.775	0.768	+0.43	+0.54
w30_lpcapbio100_e200_s2700	initialization	0.10	0.765	0.682	0.730	+0.49	+0.55
w30_lpcapbio100_e200_s2700	trained (epoch 2)	0.00	0.501	0.500	0.500	+0.02	+0.01
w30_lpcapgate100_e200_s2700	initialization	0.10	0.765	0.682	0.730	+0.49	+0.55
w30_lpcapgate100_e200_s2700	trained (epoch 6)	0.33	0.685	0.635	0.624	+0.41	+0.59

At initialization the column carries a weak signal in the right direction: 0.55 on the held-out genes and a Spearman of +0.11 with the solve, nothing like FBA's 0.65, and no deletion reaches zero growth. Training inverts it. Under the

round-4 forced cell's weights the negative column reads 0.23 on the held-out genes, which is that cell's 0.77 with the sign of growth reversed: the affine learned a positive slope, an essential deletion gets the *higher* growth, and the column is anticorrelated with the solve (−0.25). The round-2 forced cell does the same (0.24, −0.26). The median growth under the trained weights is −1.4 and −2.4 mmol/gDW/h against a wild-type FVA range of 0.48 to 0.54 on YPD: for most deletions the column is negative, which growth cannot be.

The column can do this because the arm's alternations end on the balance projection, so $Sv = 0$ holds to machine precision and the box does not: 58 percent of reactions sit outside their bounds at the selected epoch (0.64 on the round-2 cell, 0.59 on the round-6 cell), the growth reaction's floor of zero among them. v_{growth} is then a signed, unbounded coordinate the backbone writes through the abundance head, and the readout uses it as a free latent. That is the mechanism behind the shuffled-k_{cat} parity and behind the layer matching the head: the quantity the forced readout reads is not growth.

4.7 Round 7: the growth column with the bounds enforced ✗

Round 7 (jobs 2257 and 2258, three cells per GPU) puts the forced readout on two layers whose bounds hold exactly: the same alternations ending on the clip instead of the balance, and the pure box form, $v = \ell + (u - \ell)\,\sigma(z)$, with no projection at all. Same cell otherwise, batch 32 with the class weight at 200 epochs, constraint scale 0, three seeds, paired against the round-4 forced cell and the pooled cell (Table 40, Figure 25); the probe runs on every cell (Table 39).

Table 40: Round 7, the growth column with the bounds enforced. The round-3 pooled cell, the round-4 forced growth column (projection ending on the balance, box open), and the forced growth column on two layers whose bounds hold exactly: the projection ending on the box, and the pure box parameterization. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
growth column only, constraints 0 (round-3 levers)	**0.777**	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
growth column only, projection ending on the box	0.500	0.000	0.500, 0.500, 0.500 ↓	0.500 [0.50, 0.50]	2, 2, 2	0.0000
growth column only, box parameterization	0.720	0.097	0.767, 0.609, 0.785	**0.781** [0.69, 0.87]	180, 95, 140	0.0074

With the clip last, the growth column is zero for every one of the 982 deletions, at initialization and after training: the clip pins the growth reaction at its floor, the cells score 0.500 exactly on three of three seeds and select epoch 2, and the balance residual, the loose end now, is 0.85 to 1.0 of the layer's own turnover. With the box form the column lives in $[0, u]$ and starts near $u/2$ (median 45 to 53 mmol/gDW/h against a wild-type 0.5); $Sv = 0$ is then never approached, the balance residual reads 2.0 at every epoch, and the cells score 0.767, 0.609 and 0.785, a seed mean of 0.720 below the pooled cell's 0.774. The probe on the trained box cells inverts the sign on two of three seeds as before (0.23 and 0.22 on the held-out genes, Spearman −0.25 and −0.28 with the solve) and reads 0.61 with +0.10 on the third.

The rounds close the module question at the layer rather than at the readout. Whichever constraint the layer holds exactly, the other is violated by an amount comparable to the fluxes themselves: 58 percent of reactions out of bounds when the balance is exact, a balance residual of 0.85 to 2.0 when the bounds are. The alternations never reach the polytope that FBA optimizes over, so the growth column is either a free signed coordinate, a constant zero, or an unbalanced sigmoid, and none of them is the quantity whose optimum over that polytope ranks essential genes at 0.65 on its own and lifts the transformer by 0.05. On this endpoint the physics that helps is the solve, and the layer would have to reach feasibility before its growth column could carry it.

4.8 Round 10: the layer warm-started from the deletion's own solve ✗

If the layer cannot reach the polytope from a free proposal, it can be handed a point on it. One parsimonious FBA solve per labeled gene on YPD, the medium the essentiality cell's box is built on, gives a reference flux per deletion (`ess_reference_flux_table.py`, 1,121 genes, 104 of them with zero growth, about eight minutes of CPU), and under the warm-start parameterization the layer proposes that reference plus a null-space residual of relative size α before its alternations. The growth column then starts at the deletion's own pFBA growth rather than at a free coordinate, and the negative of that growth ranks the 195 held-out genes at 0.645 on its own. Round 10 (jobs 2284 to 2287) runs the forced growth column and the gated column with the context head on the anchored layer at constraint scale 0, with α frozen at 0.01, learned from 0.01, and frozen at 0.1, each at the three seeds, paired against the round-4 forced and gated cells and the round-5 affine (Table 41, Figure 26). Two reads were fixed first: the probe at initialization must find the

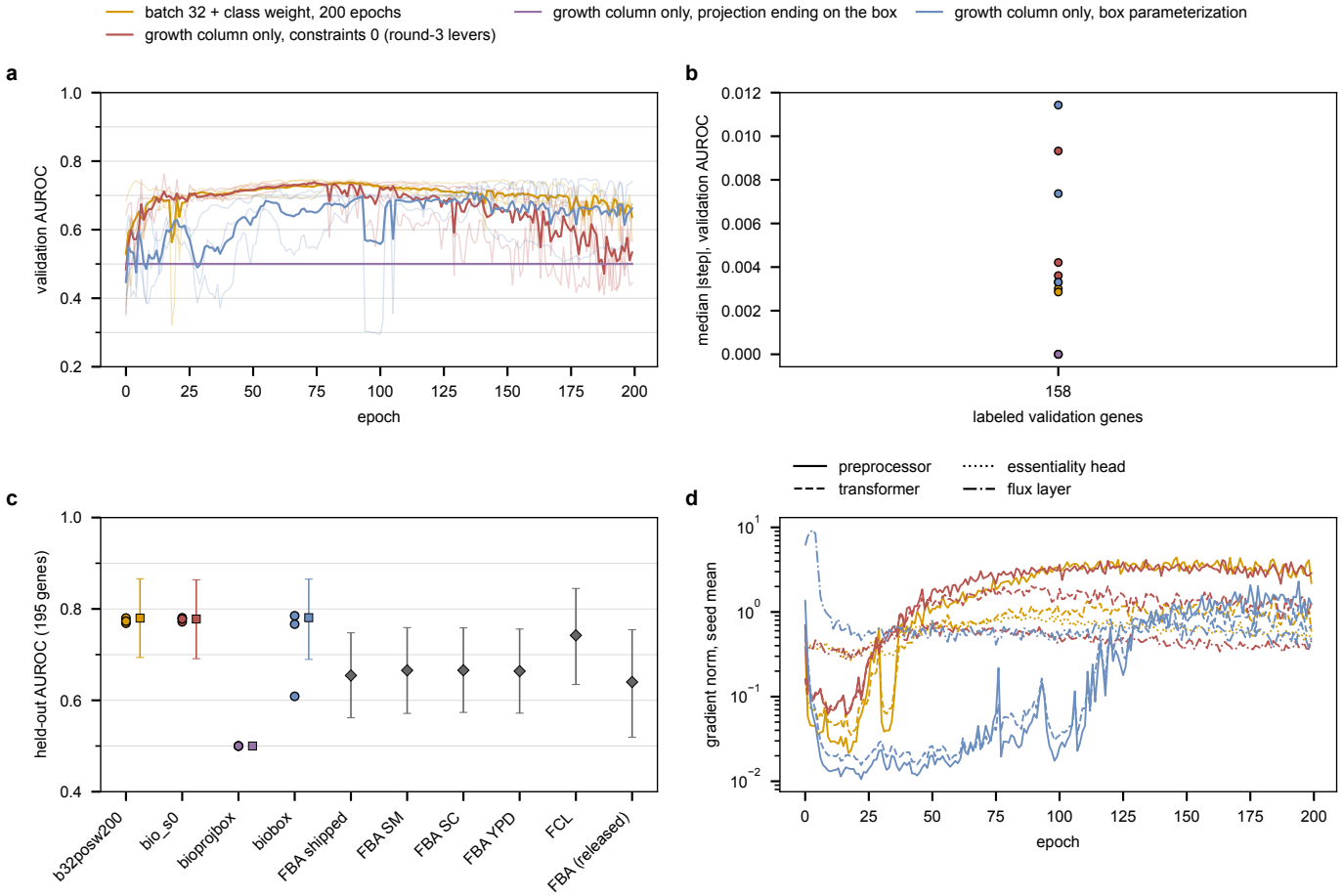


Figure 25: Round 7, the growth column with the bounds enforced. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w18`.

growth column at the solve's level, about 0.65 on the held-out genes, or the warm start did not reach the readout; and the gated cell passes if it sits at the round-5 affine's 0.82 with three of three seeds above the round-4 gated cell.

Table 41: Round 10, the flux layer warm-started from the deletion's own solve. The pooled cell, the round-4 forced and gated growth columns, and the round-5 FBA affine, against the forced and gated growth columns whose layer proposes the deletion's own parsimonious FBA flux on YPD plus a null-space residual of relative size alpha, with alpha frozen at 0.01, learned, or frozen at 0.1. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
growth column, pFBA warm start, alpha 0.01 frozen	0.704	0.073	0.629, 0.706, 0.776	0.801 [0.71, 0.90]	112, 144, 157	0.0222
gated column + context, pFBA warm start, alpha 0.01 frozen	0.776	0.034	0.810, 0.776, 0.742	0.784 [0.70, 0.87]	181, 70, 4	0.0037
gated column + context, pFBA warm start, alpha learned	0.775	0.035	0.812, 0.772, 0.741	0.796 [0.72, 0.87]	146, 71, 4	0.0036
gated column + context, pFBA warm start, alpha 0.1 frozen	0.751	0.044	0.784, 0.767, 0.701	0.777 [0.69, 0.86]	79, 83, 3	0.0035

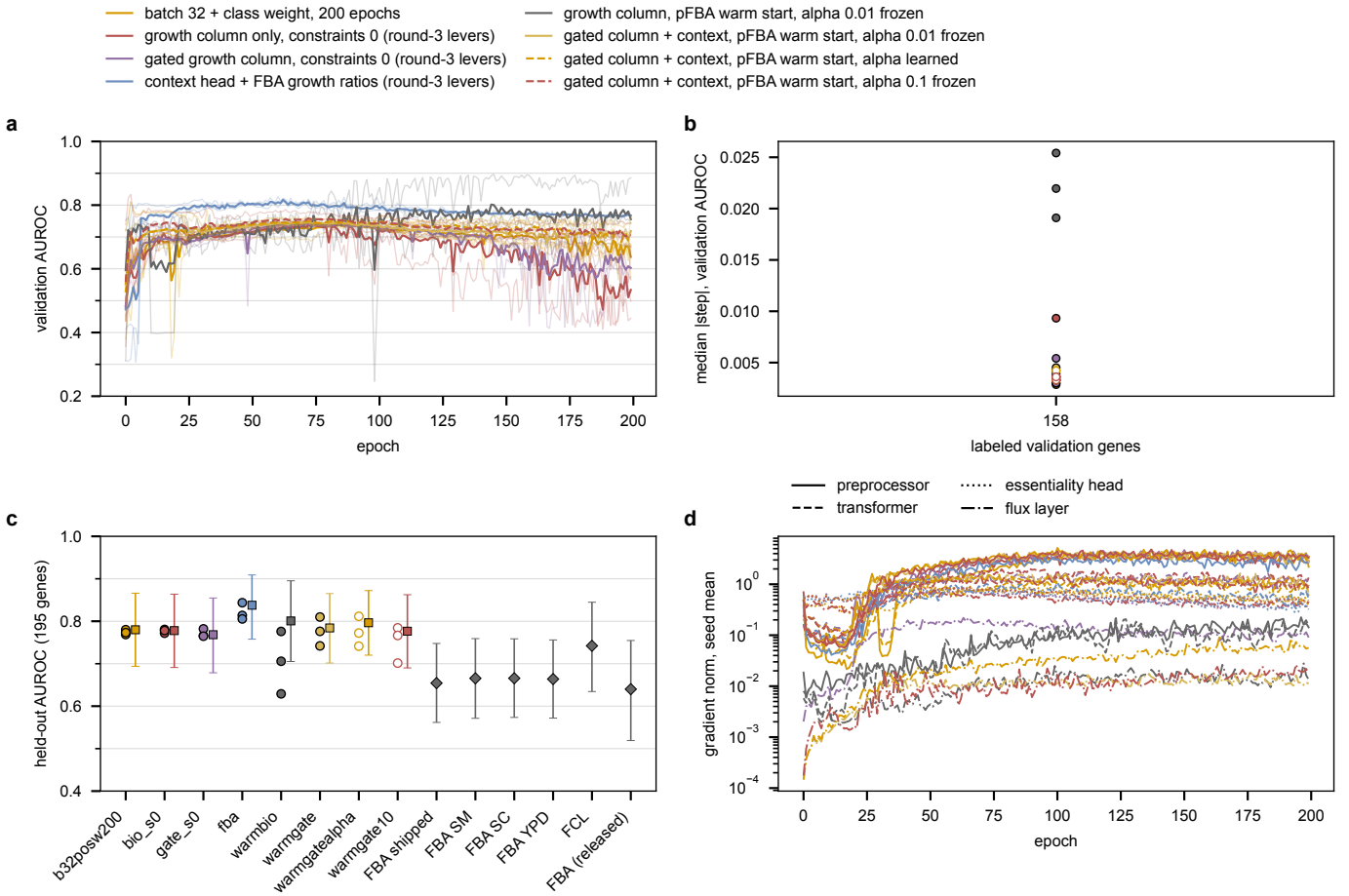


Figure 26: Round 10, the layer warm-started from the deletion's own solve. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w21`.

Neither read holds. The probe at initialization finds the growth column negative (median −0.07 to −0.10 mmol/gDW/h against references of 0 to 0.5) and anti-correlated with the solve it was started from: Spearman −0.43 to −0.44 against the YPD ratio, and the negative column ranks the held-out genes at 0.34 to 0.39, which is the solve's ranking with its sign reversed. The layer's own diagnostics say where the reference goes: the anchored layer's box is the GEM's bounds tightened by the enzyme-capacity ceilings and the second-law directions, the reference violates it on 28 to 39 percent

of reactions at initialization, and the proposal the alternations return sits 0.45 to 0.57 of the reference's norm away from it at $\alpha = 0.01$. The projections that make the flux balanced move it half its length off the point it was handed and reverse the growth column's correlation with the solve on the way. Trained, the forced cells read 0.629, 0.706 and 0.776 against the round-4 forced cell's 0.772, 0.781 and 0.778, down on three of three seeds, and the probe on their trained weights inverts further (0.22 to 0.41); the gated cell reads 0.810, 0.776 and 0.742 against 0.766, 0.782 and 0.765, one seed up and two down, with seed 2702 selecting epoch 4 and its gate ending at -2.0. Letting α learn changes nothing on the held-out genes (0.812, 0.772, 0.741) while α grows to 0.19, 0.02 and 0.11 and the box violation on one seed rises from 0.30 to 0.45, the job-1946 failure mode in miniature; α frozen at 0.1 reads 0.751. The warm start hands the layer the solve and the anchored layer's constraint set does not keep it. Round 13 repeats the warm start on the stoichiometry-only layer, whose box is the GEM's own bounds, so the reference is inside it by construction.

4.9 Round 13: the warm start on the stoichiometry-only layer ✗

Round 13 (jobs 2297 and 2298) repeats the warm start on the stoichiometry-only layer, whose box is the GEM's own bounds scaled by the availability gate, with no capacity ceiling and no second-law direction, forced and gated, α frozen at 0.01, three seeds each (Table 42, Figure 27). The reads are round 10's: the probe at initialization at the solve's level, and the gated cell at the round-5 affine's 0.82 with three of three seeds above the round-4 gated cell.

Table 42: Round 13, the warm start on the stoichiometry-only layer. The pooled cell, the round-4 forced and gated columns and the round-10 warm-started columns on the anchored layer, against the same warm start on the stoichiometry-only layer, whose box is the GEM's own bounds rather than the capacity- and second-law-tightened one, so the deletion's pFBA reference sits inside it. Alpha frozen at 0.01, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
growth column, pFBA warm start, alpha 0.01 frozen	0.704	0.073	0.629, 0.706, 0.776	0.801 [0.71, 0.90]	112, 144, 157	0.0222
gated column + context, pFBA warm start, alpha 0.01 frozen	0.776	0.034	0.810, 0.776, 0.742	0.784 [0.70, 0.87]	181, 70, 4	0.0037
growth column, GEM box only, pFBA warm start, alpha 0.01 frozen	0.664	0.058	0.654, 0.726, 0.612 ↓	0.676 [0.55, 0.79]	53, 2, 23	0.0236
gated column + context, GEM box only, pFBA warm start, alpha 0.01 frozen	**0.816**	0.019	0.794, 0.829, 0.825 ↑	**0.802** [0.72, 0.88]	85, 5, 118	0.0042

Both reads hold, and this is the first module cell in the document that carries the solve. At initialization the growth column is positive (median 0.38 mmol/gDW/h), its Spearman with the solve is +0.47 to +0.57 across the four media, and its negative ranks the held-out genes at 0.683, above FBA's own 0.654 to 0.666 on the same genes; the alternations leave the reference where it was, 0.3 percent of its norm away, with a box violation of 0.24 to 0.29 that is the availability gate's untrained scaling of the bounds rather than a displacement of the flux. The forced cells read 0.654, 0.726 and 0.612, a mean of 0.664, which is the solve's number: trained, the column keeps its sign (Spearman +0.54 to +0.64) and its ranking (0.62 to 0.67), and the layer's learned parts add nothing to it. The gated cells read 0.794, 0.829 and 0.825 against the round-4 gated cell's 0.766, 0.782 and 0.765, up by 0.028, 0.047 and 0.060 on three of three seeds, a seed mean of 0.816 against the affine's 0.821, and on the two seeds that train past epoch 5 the gate ends at +0.97 and +0.90, the first positive gates in the document: the head adds the growth column with its physical sign. The third seed selects epoch 5 with its gate at zero and still reads 0.829, the seed-to-seed spread the round-9 cells also show.

Put beside round 10, the comparison isolates the loss. The same reference on the anchored layer arrived sign-reversed and half its norm away; on the GEM's own box it arrives intact. The capacity ceilings and the second-law directions, tightened onto a flux that was solved without them, are what the projections could not reconcile, and the layer that solves (round 15) applies those constraints inside the solve instead of after it.

4.10 Round 15: the layer that solves ✗

The projection route carries the solve only when handed it. Round 15 replaces the alternations with the solve: under the `lp` parameterization the layer's flux is the growth optimum of the linear program $\max\{v_{\text{growth}} : Sv = 0,\ \ell \le v \le u\}$ over the layer's own bounds, one HiGHS solve per genotype in a forked pool, and the backbone reaches the prediction through the bounds alone, with the gradient the bound duals give (`torchcell.metabolism.lp_layer`; the envelope theorem, checked against finite differences on the GEM). On the anchored layer the bounds are the GEM's scaled by availability and capped by the enzyme-capacity ceilings from the abundance head, so this is the GECKO construction

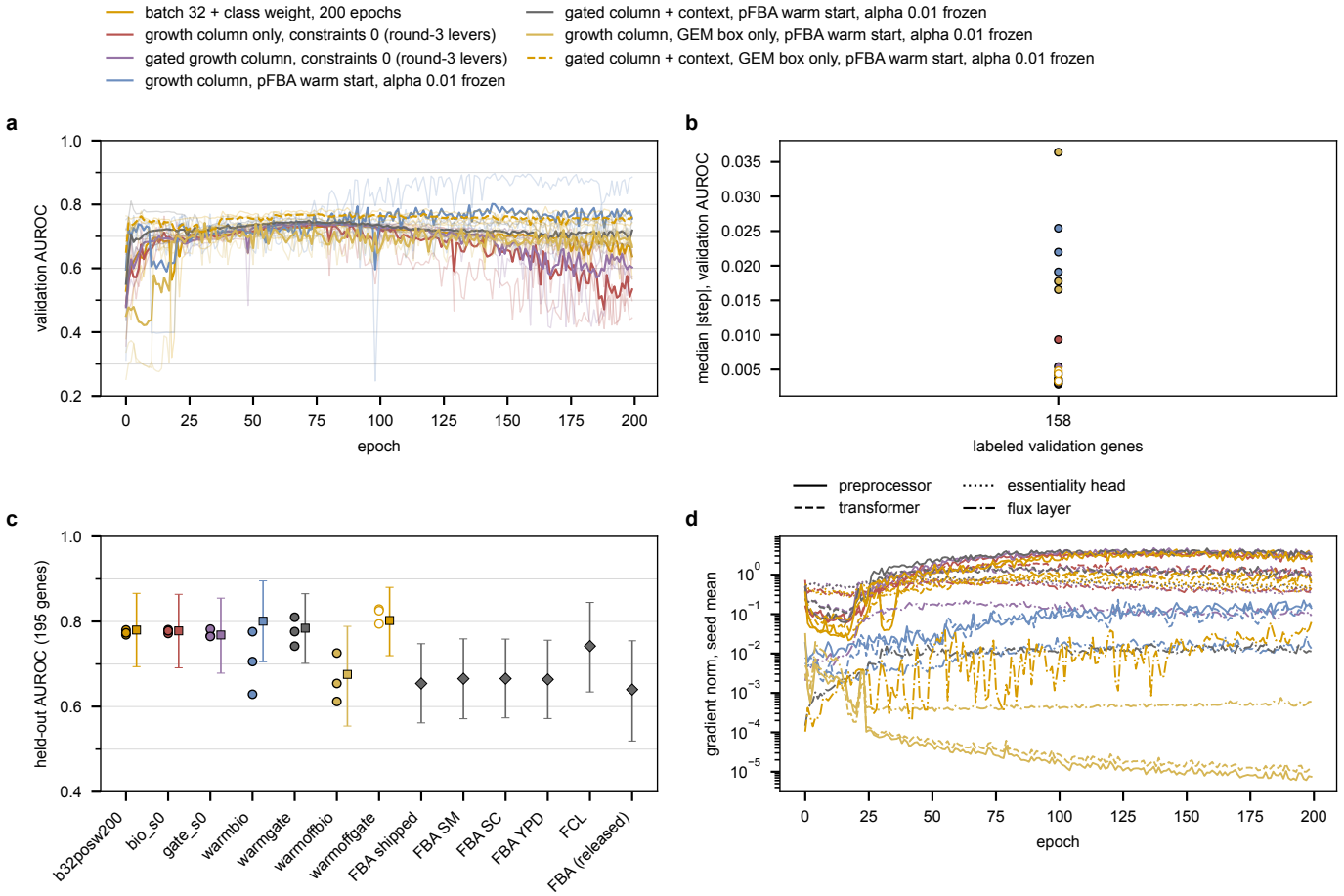


Figure 27: Round 13, the warm start on the stoichiometry-only layer. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w24`.

with a learned enzyme pool. Round 15 (jobs 2304 to 2306) runs it forced, forced with the second-law box, and gated, three seeds each (Table 43, Figure 29); Figure 28 draws the construction.

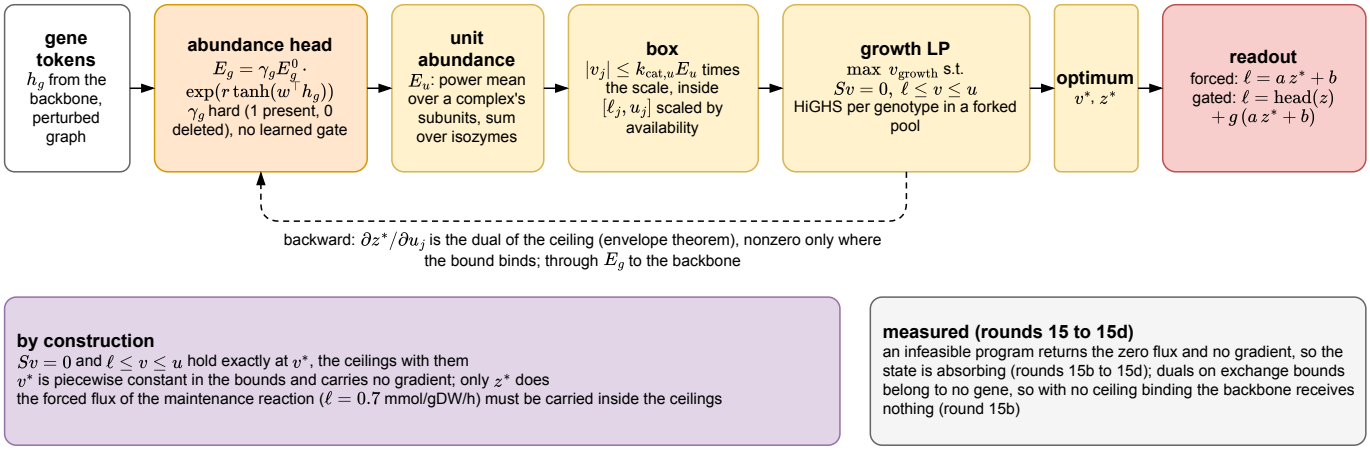


Figure 28: The layer that solves, rounds 15 to 15d. The abundance head sets each gene's enzyme abundance with availability a hard fact, the unit abundance and the k_{cat} ceilings form the box, and the flux is the growth optimum of the linear program over that box, one HiGHS solve per genotype; the program's value z^* enters the readout. Mass balance and the box hold exactly at the optimum. The dashed path is the backward pass: the derivative of z^* in a bound is that bound's dual, nonzero only where the bound binds, so the backbone is trained through the ceilings and only through them; an infeasible program returns no flux and no gradient. Colors as in Figure 17. Drawn in draw.io from `notes/assets/drawio/028-module.gen.py`.

Table 43: Round 15, the layer that solves. The pooled cell, the round-4 forced and gated growth columns and the round-5 FBA affine, against a flux layer whose flux is the growth optimum of a linear program under the layer's own bounds (availability and the enzyme-capacity ceiling from the abundance head), with the bound duals as its gradient: the forced growth column, the same with the box tightened to the second-law direction before the solve, and the gated column with the context head. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
growth column, the layer that solves	0.500	0.000	0.500, 0.500, 0.500 ↓	0.500 [0.50, 0.50]	2, 2, 2	0.0000
growth column, the layer that solves, second-law box	0.500	0.000	0.500, 0.500, 0.500 ↓	0.500 [0.50, 0.50]	2, 2, 2	0.0000
gated column + context, the layer that solves	0.773	0.012	0.771, 0.786, 0.763	0.770 [0.68, 0.86]	78, 81, 63	0.0028

Every forced cell scores 0.500 exactly and selects epoch 2, because the program is infeasible for every genotype at every epoch: the probe finds the growth column at zero for all 982 deletions at initialization and under the trained weights. The gated cells read 0.771, 0.786 and 0.763, the pooled cell's 0.774, with gates within 0.07 of zero, which is the head working alone beside a dead path. The cause is measured on one genotype's bounds taken from the layer: under the GEM's own bounds the program solves at the wild-type growth of 0.0858, and under the layer's bounds it has no feasible point, with or without the maintenance flux released and with the deleted gene's reactions reopened. What differs is the capacity ceiling: the layer's upper bounds have a median of 0.084 mmol/gDW/h and a tenth of the reactions below 0.007, against 100 on the GEM, because the predicted k_{cat} tier multiplied by the abundance head's nominal abundances gives ceilings two to four orders below the fluxes growth needs. The alternating layer never noticed, because it never satisfied the box; the solve cannot proceed without it. That is the same fact as round 10's, read exactly: the enzyme-capacity box as parameterized contains no flux that meets the model's maintenance requirement, let alone one that grows (Section 4.2 puts the shortfall at 0.54 of 0.70), and every arm built on it has been reading a flux off the polytope. Round 15b runs the solve on the stoichiometry-only box, where round 13 showed the polytope holds growth. The capacity ceiling that admits growth is then calibrated the way GECKO calibrates its pool, by one scale on every ceiling (`capacity_calibration.py`, through the layer's own forward pass on a batch of 32 held-out genotypes at initialization): the program is infeasible up to a scale of 10, feasible for 31 of the 32 from a scale of 30 (the thirty-second is lethal) with a median growth of 0.042, reaches 0.10 at 100 and 0.20 at 1,000, and at 10,000 returns the unconstrained wild type's 0.535, where the ceilings no longer bind. Round 15c runs the layer that solves at 100 and 1,000, where bounds the genes own bind at the optimum.

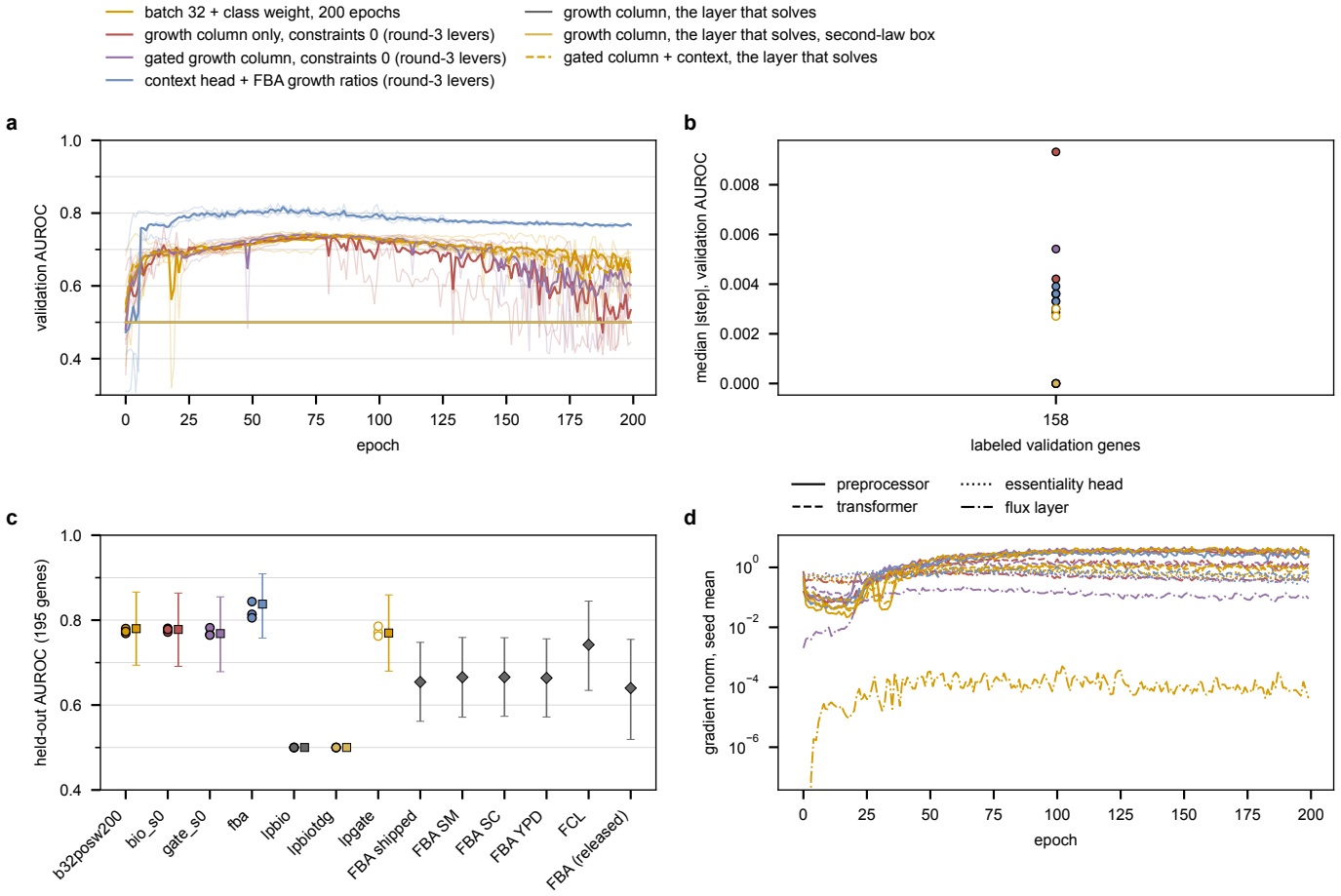


Figure 29: Round 15, the layer that solves on the anchored box. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w26`.

4.11 Round 15b: the layer that solves on the stoichiometry-only box ✗

Round 15b (jobs 2314 and 2315) runs the same solve on the stoichiometry-only layer, whose bounds are the GEM's scaled by the availability gate, forced and gated, three seeds each, beside the round-13 warm-started columns on the same box and the round-5 affine (Table 44, Figure 30); the read is round 10's.

Table 44: Round 15b, the layer that solves on the stoichiometry-only box. The pooled cell, the round-5 FBA affine and the round-13 warm-started columns on the stoichiometry-only layer, against the growth LP under that layer's box (the GEM's bounds scaled by availability, no capacity ceiling): the forced growth column and the gated column with the context head. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
growth column, GEM box only, pFBA warm start, alpha 0.01 frozen	0.664	0.058	0.654, 0.726, 0.612 ↓	0.676 [0.55, 0.79]	53, 2, 23	0.0236
gated column + context, GEM box only, pFBA warm start, alpha 0.01 frozen	0.816	0.019	0.794, 0.829, 0.825 ↑	0.802 [0.72, 0.88]	85, 5, 118	0.0042
growth column, the layer that solves, GEM box	0.613	0.098	0.500, 0.669, 0.669 ↓	0.669 [0.58, 0.76]	11, 51, 2	0.0000
gated column + context, the layer that solves, GEM box	**0.824**	0.021	0.800, 0.837, 0.835 ↑	0.837 [0.76, 0.91]	2, 103, 4	0.0046

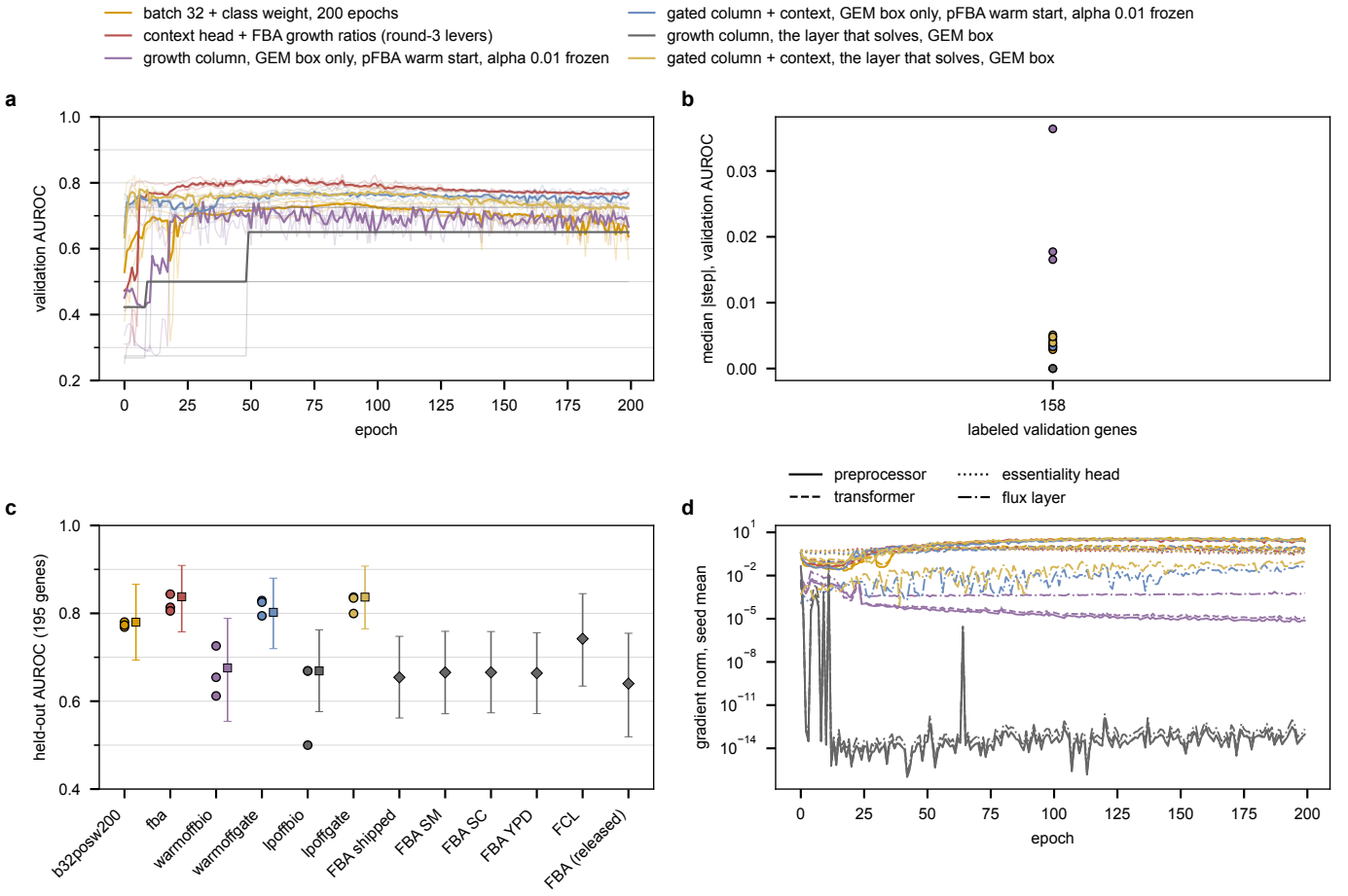


Figure 30: Round 15b, the layer that solves on the stoichiometry-only box. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w29`.

The program is feasible here and its growth column is growth from the first forward pass: the probe at initialization finds a positive column with 10 percent of the 982 deletions at exactly zero growth, the lethal ones, a Spearman of +0.40 to +0.50 with FBA's ratio across the four media, and a ranking of 0.734 on the held-out genes, above FBA's own 0.654 to 0.666 on the same genes. The layer that solves under the availability-scaled box is a better ranker than

the unscaled solve before any training, which is the uniform enzyme ceiling the gate imposes doing GECKO's work in miniature. The gated cells read 0.800, 0.837 and 0.835 against the round-4 gated cell's 0.766, 0.782 and 0.765, up by 0.034, 0.055 and 0.070 on three of three seeds, a seed mean of 0.824 against the affine's 0.821 and a pooled bootstrap of 0.837 [0.76, 0.91]: both halves of the read hold, and this is a module cell at the affine's level with the solve inside the forward pass and the gradient through it. The seed that trains longest (2701, epoch 103) ends with its gate at +1.42, the largest positive gate in the document, and its growth column ranks the held-out genes at 0.768 on its own after training, above its 0.734 at initialization; the other two seeds select epochs 2 and 4 with gates near zero, so on those the number is the head's early ranking and the column's contribution is not read.

The forced cells fail in two ways that are the linear program's own. On two seeds the gradient into the backbone is 10^{-14} for the whole run and the cell reads 0.669, the solve's number with a learned threshold: at the optimum the binding constraints are the exchange bounds, which no gene owns, so the duals on the learnable bounds are zero and the backbone has nothing to move. On the third seed the early gradient moves the availability gate until the program is infeasible for every genotype (epoch 11), the column is zero from then on, and the cell reads 0.500; an infeasible program returns no gradient, so the state is absorbing. The gated cell avoids both because the head carries the loss while the column contributes where its duals are live. A layer that solves therefore needs two more things the rounds name: bounds the genes own that bind at the optimum (the capacity ceilings, once calibrated to admit growth), and a barrier that keeps the program feasible under training.

4.12 Round 15c: the layer that solves with calibrated capacity ceilings ✗

Round 15c (jobs 2316 to 2318) runs the solve on the anchored box with the capacity ceilings scaled by 100 and by 1,000, the range the calibration put growth in with the ceilings still binding, gated at both scales and forced at 100, three seeds each (Table 45, Figure 31). Bounds the genes own now bind at the optimum: the layer reports 3 to 8 percent of reactions at their capacity ceiling at the selected epoch, against zero on the GEM box.

Table 45: Round 15c, the layer that solves with calibrated capacity ceilings. The pooled cell, the round-5 FBA affine and the round-15b cells on the GEM box, against the growth LP on the anchored box with the enzyme-capacity ceilings scaled by 100 and by 1,000 (capacity_calibration.py: the program admits growth from a scale of 30 and the ceilings stop binding near 10,000), forced and gated. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844 ↑	0.838 [0.76, 0.91]	28, 70, 61	0.0036
growth column, the layer that solves, GEM box	0.613	0.098	0.500, 0.669, 0.669 ↓	0.669 [0.58, 0.76]	11, 51, 2	0.0000
gated column + context, the layer that solves, GEM box	**0.824**	0.021	0.800, 0.837, 0.835 ↑	0.837 [0.76, 0.91]	2, 103, 4	0.0046
growth column, the layer that solves, capacity x100	0.614	0.198	0.500, 0.843, 0.500	**0.843** [0.76, 0.91]	2, 29, 2	0.0001
gated column + context, the layer that solves, capacity x100	0.800	0.017	0.820, 0.795, 0.786 ↑	0.790 [0.71, 0.87]	6, 88, 63	0.0037
gated column + context, the layer that solves, capacity x1000	0.795	0.012	0.809, 0.786, 0.791 ↑	0.793 [0.71, 0.87]	6, 51, 75	0.0039

The forced column carries the prediction alone on one seed. At scale 100 seed 2701 starts with 87 percent of its genotypes infeasible, is pulled back to none by epoch 50 by the gradient from the feasible rows, selects epoch 29 and reads 0.8425 on the held-out genes: the growth column of a linear program under learned enzyme ceilings, with no context head and no appended ratio, at the level of the affine (0.821) and of the round-9 backbone's seeds, 0.18 above FBA's own ranking on the same genes. The other two seeds enter the absorbing state within five epochs, 97 percent of genotypes infeasible from then on, and read 0.500. The probe at initialization reads the column at 0.730 with a Spearman of +0.49 to +0.55 against the solve, the same start as the GEM box's. The gated cells stay feasible for every genotype throughout and read 0.820, 0.795 and 0.786 at scale 100 and 0.809, 0.786 and 0.791 at 1,000, three of three seeds above the round-4 gated cell at both scales, with gates ending between +0.05 and +1.04, but below the GEM box's 0.824 and the affine's 0.821, so the calibrated ceilings cost the gated cell 0.02 to 0.03 against the plain box while binding.

What the one seed shows is the module's upper bound so far: the solve with learned capacity ceilings is a 0.84 ranker on its own, and the physics is doing it, since the column is growth (10 percent of deletions at exactly zero, the lethal ones) and the ceilings bind. What the other two show is that the linear program's infeasible region is absorbing under gradient descent on its bounds, which round 15d tests with more slack and a smaller step.

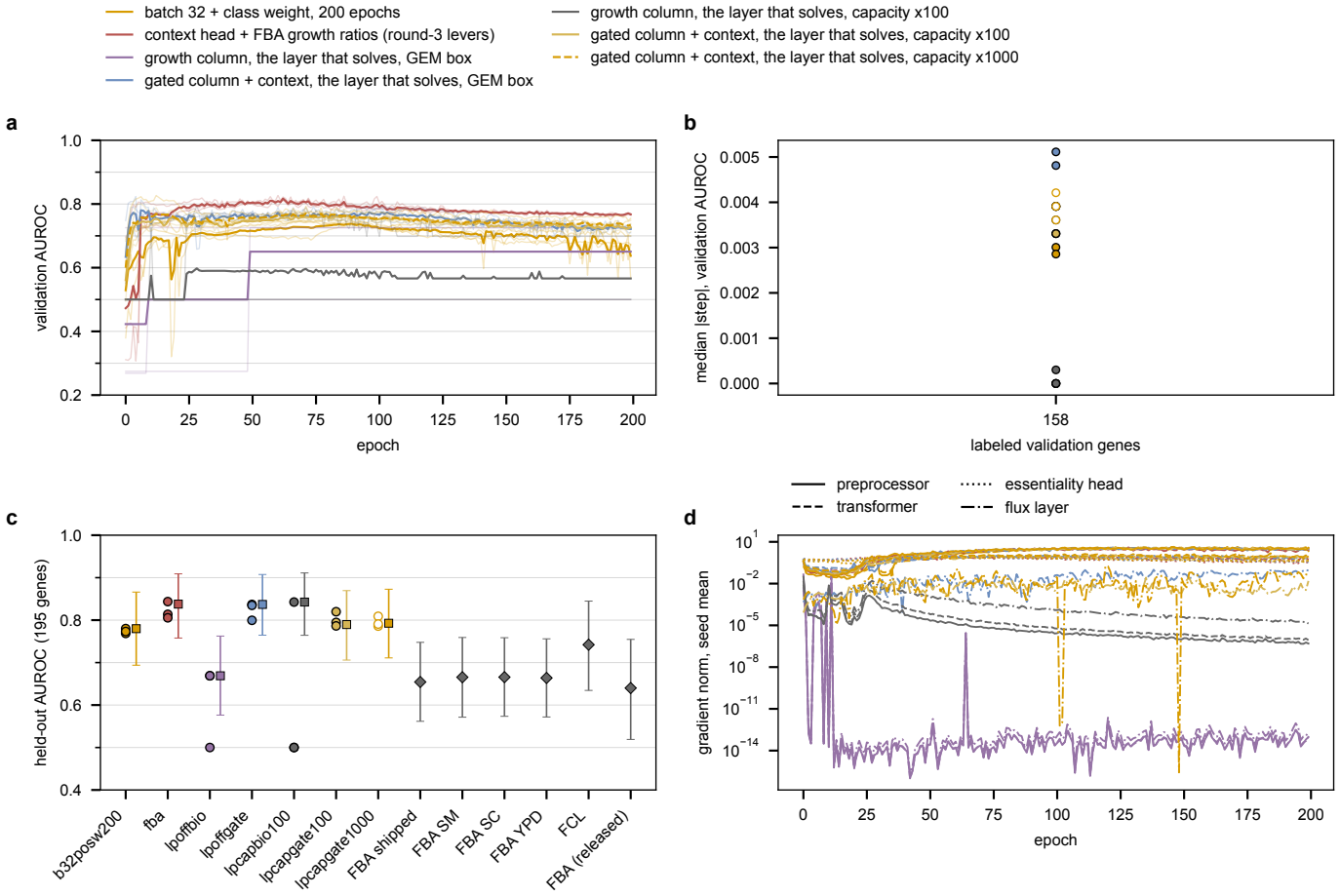


Figure 31: Round 15c, the layer that solves with calibrated capacity ceilings. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w30`.

4.13 Round 15d: the forced layer that solves against the absorbing state ✗

Round 15d (jobs 2319 and 2320) reruns the forced cell at capacity scale 1,000, where the calibration leaves ten times the slack, and at scale 100 with the learning rate at 3×10^{-4}, three seeds each (Table 46, Figure 32).

Table 46: Round 15d, the forced layer that solves against the absorbing state. The pooled cell, the round-5 FBA affine and the round-15c forced cell at capacity scale 100, against the same cell at scale 1,000 and at scale 100 with the learning rate at 3×10^{-4}. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	0.838 [0.76, 0.91]	28, 70, 61	0.0036
growth column, the layer that solves, capacity x100	0.614	0.198	0.500, 0.843, 0.500	**0.843** [0.76, 0.91]	2, 29, 2	0.0001
growth column, the layer that solves, capacity x1000	0.714	0.106	0.615, 0.826, 0.702	0.819 [0.73, 0.90]	73, 30, 45	0.0000
growth column, the layer that solves, capacity x100, lr 3e-4	0.582	0.142	0.500, 0.745, 0.500 ↓	0.745 [0.64, 0.84]	2, 119, 2	0.0000

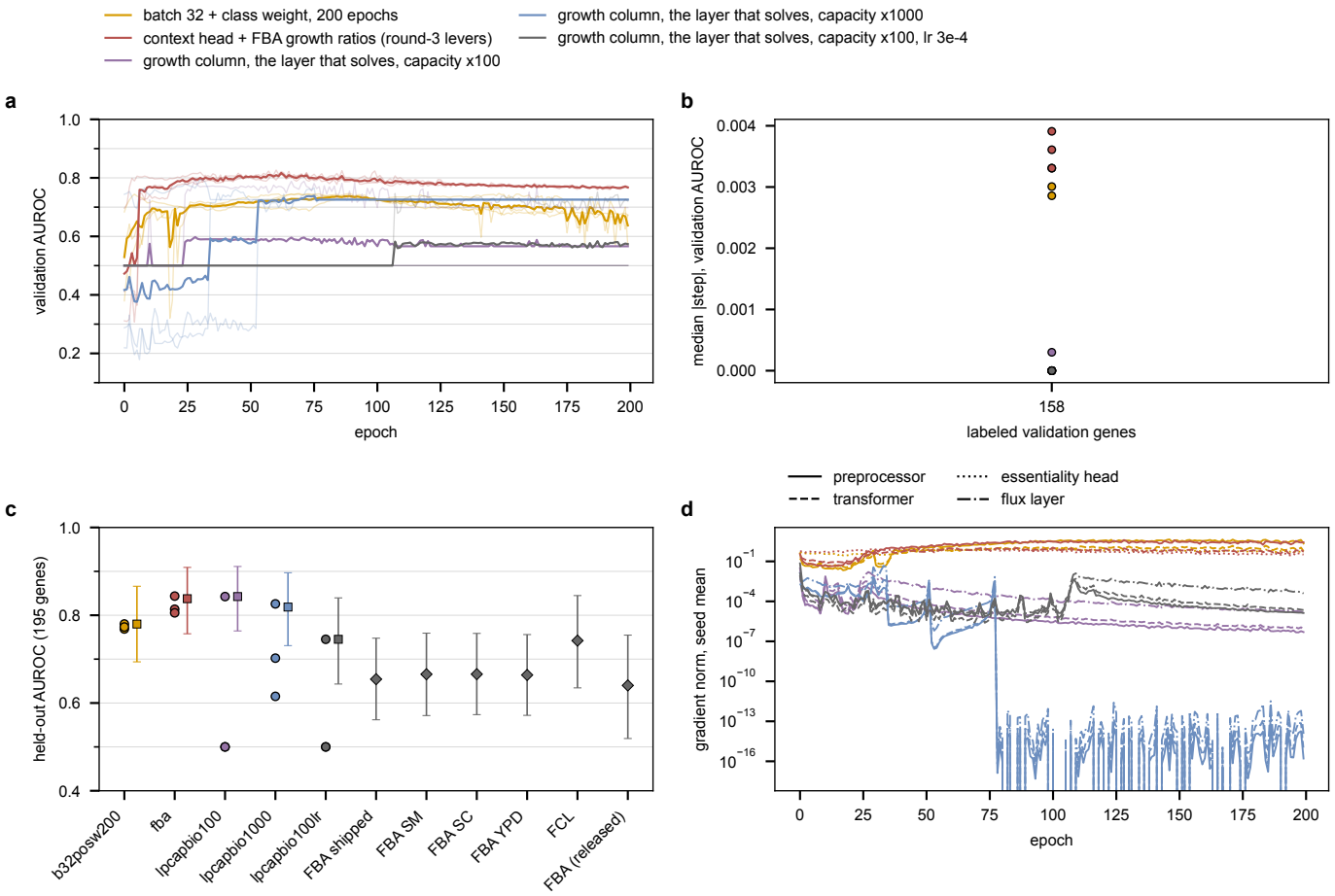


Figure 32: Round 15d, the forced layer that solves against the absorbing state. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w31`.

Slack removes the absorbing state and the step size does not. At scale 1,000 no genotype's program is infeasible at any epoch on any seed, and the forced column reads 0.615, 0.826 and 0.702, a mean of 0.714 with a spread of 0.106: one seed near the affine's level and two near the solve's, from a column that is growth throughout. At scale 100 with the smaller step two seeds are absorbed by epoch 5 as before and the third is infeasible on 73 to 97 percent of genotypes until epoch 100, recovers, and selects epoch 119 at 0.745. The absorbing state is a property of how much room the ceilings leave, and with room the forced column is a 0.62 to 0.83 ranker whose spread across seeds is the largest in the document. Across rounds 15c and 15d the forced layer that solves has reached 0.84 and 0.83 on two of nine seeds, so

the module can carry the whole prediction at the affine's level, and does so unreliably; the gated form is the reliable one, and it reads the affine's level on the GEM box.

4.14 Round 18: the elastic layer that solves, feasible by construction ✗

Rounds 15b to 15d leave the forced layer that solves with one failure mode, the absorbing infeasible state, and one unmeasured question, whether the growth column alone is a seed-robust 0.83 ranker once that state cannot occur. Round 18 (jobs 2321 to 2324) removes the state by construction rather than by slack in the ceilings: the elastic program of Section 4.2, every forced flux priced at $M = 1$ per mmol/gDW/h of shortfall, so that the value the readout sees is growth where maintenance is met and a negative number where it is not, and the gradient on the ceilings that block maintenance is the dual of the elastic row, of magnitude M, in place of no gradient at all. The program is solved in micromole units, because HiGHS's presolve at its absolute tolerances reads ceilings of 10^{-6} as an empty box and reports the feasible program infeasible; in those units it solves at the hard program's cost. Four cells, three seeds each, all on the `nodiss` arm: the forced column at ceiling scale 100 and 1,000 (the two scales of rounds 15c and 15d), the forced column at 100 with the protein pool held by construction, and the gated column at 100. The reads were fixed before launch: the forced cells pass if all three seeds are above the round-4 forced column and none is below 0.80, the level the hard program reached on two of nine seeds; the pool cell is read paired against the elastic cell without it; the gated cell passes if all three seeds are above the round-4 gated cell with a seed mean at or above the GEM box's 0.824. The infeasible fraction reading zero at every epoch is a check on the construction, not a result. Beside the score, every cell logs what the program did: the fraction of genotypes with unmet maintenance, the fraction on which a gene-owned bound carries a dual (the fraction on which the backbone receives any gradient through the solve), and the gradient norm of each part of the layer (Figure 34). Figure 33 draws the construction.

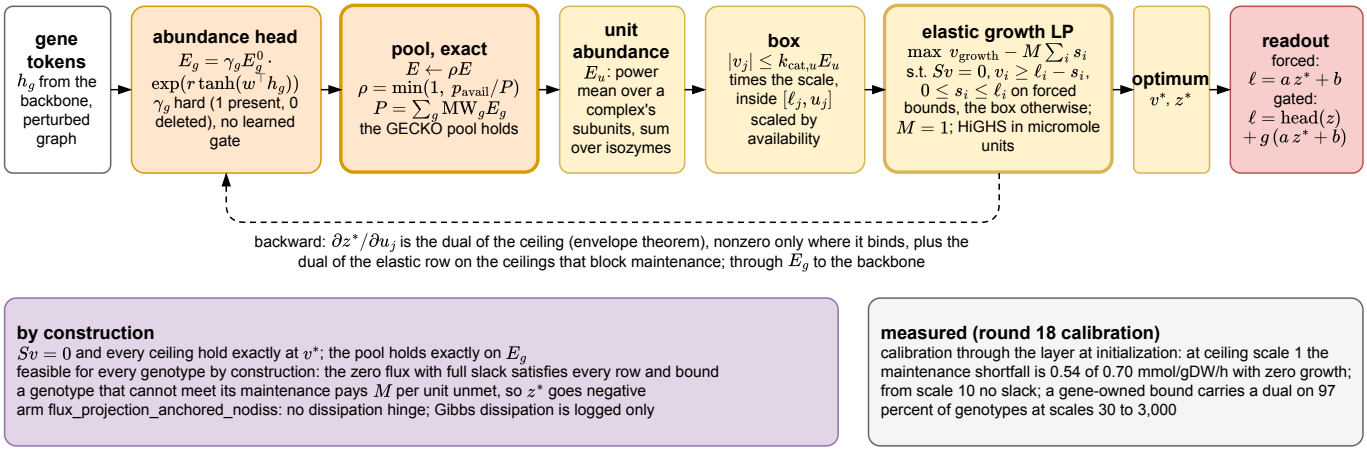


Figure 33: The elastic layer that solves with the pool by construction, round 18. Two blocks change against Figure 28, drawn with a heavier border: the abundance vector is projected onto the protein budget before it sets the ceilings, so the GECKO pool constraint holds exactly, and the growth program is elastic, every forced flux (the ATP maintenance reaction, lower bound 0.7 mmol/gDW/h) carrying a priced slack so that the zero flux with full slack is feasible for every genotype and the value z^* falls by M per unit of maintenance the ceilings cannot carry. The backward pass gains the dual of the elastic row on the ceilings that block maintenance. The arm carries no dissipation hinge; the dissipation is logged only. Colors as in Figure 17. Drawn in draw.io from `notes/assets/drawio/028-module.gen.py`.

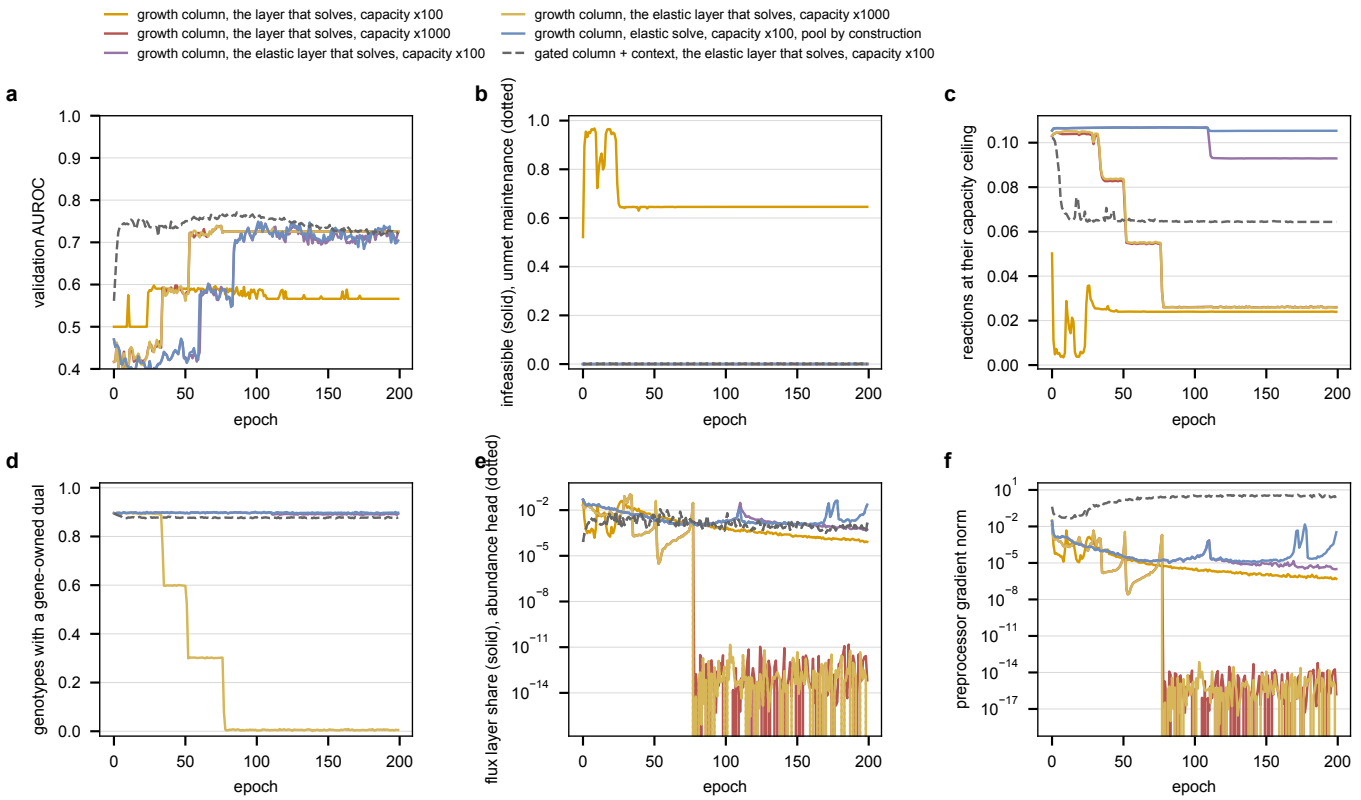


Figure 34: The layer that solves, probed per epoch. Seed mean per arm on the cells whose flux is a linear program. **a**, Validation AUROC. **b**, The fraction of genotypes whose program was infeasible (solid) and, on the elastic program, the fraction with unmet maintenance (dotted). **c**, The fraction of gene-owned reactions at their capacity ceiling. **d**, The fraction of genotypes on which a gene-owned bound carries a dual, logged from round 18. **e**, The flux layer's share of the summed per-module gradient norm (solid) and the abundance head's share (dotted, round 18). **f**, The preprocessor's gradient norm. Written by `experiments/028-gene-essentiality/scripts/lp_probe_readout.py`.

Table 47: Round 18, the forced column of the elastic layer that solves. The round-4 forced column of the alternating layer (the paired comparator), the round-5 FBA affine and the hard program's forced cells at capacity scale 100 and 1,000 (rounds 15c and 15d), against the elastic program (every forced flux priced at 1 per mmol/gDW/h of shortfall, feasible for every genotype by construction) at scale 100 and 1,000, and at 100 with the protein pool held by construction. Arm `flux_projection_anchored_nodiss`, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	0.838 [0.76, 0.91]	28, 70, 61	0.0036
growth column, the layer that solves, capacity x100	0.614	0.198	0.500, 0.843, 0.500	**0.843** [0.76, 0.91]	2, 29, 2	0.0001
growth column, the layer that solves, capacity x1000	0.714	0.106	0.615, 0.826, 0.702	0.819 [0.73, 0.90]	73, 30, 45	0.0000
growth column, the elastic layer that solves, capacity x100	0.714	0.024	0.704, 0.741, 0.697 ↓	0.697 [0.56, 0.82]	106, 132, 180	0.0103
growth column, the elastic layer that solves, capacity x1000	0.713	0.110	0.607, 0.826, 0.707	0.819 [0.73, 0.90]	73, 30, 45	0.0000
growth column, elastic solve, capacity x100, pool by construction	0.706	0.037	0.704, 0.743, 0.670 ↓	0.697 [0.57, 0.82]	106, 126, 187	0.0097

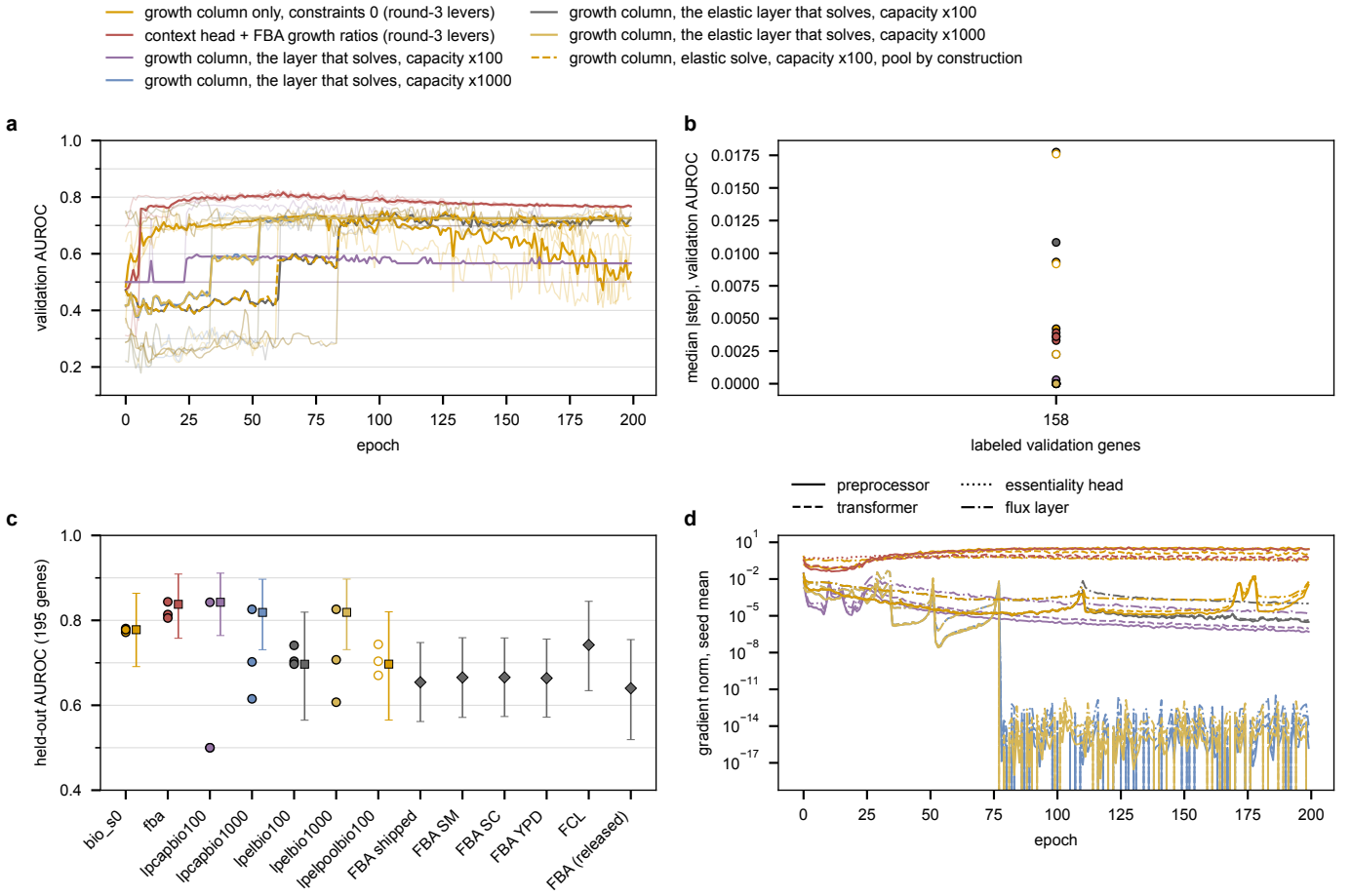


Figure 35: Round 18, the forced column of the elastic layer that solves. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w32`.

Table 48: Round 18, the gated column of the elastic layer that solves. The round-4 gated cell (the paired comparator), the round-5 FBA affine, the gated hard program on the GEM box (round 15b) and at capacity scale 100 (round 15c), against the gated elastic program at scale 100. Arm `flux_projection_anchored_nodiss`, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
gated column + context, the layer that solves, GEM box	**0.824**	0.021	0.800, 0.837, 0.835 ↑	0.837 [0.76, 0.91]	2, 103, 4	0.0046
gated column + context, the layer that solves, capacity x100	0.800	0.017	0.820, 0.795, 0.786 ↑	0.790 [0.71, 0.87]	6, 88, 63	0.0037
gated column + context, the elastic layer that solves, capacity x100	0.803	0.018	0.816, 0.810, 0.783 ↑	0.800 [0.72, 0.88]	6, 95, 63	0.0038

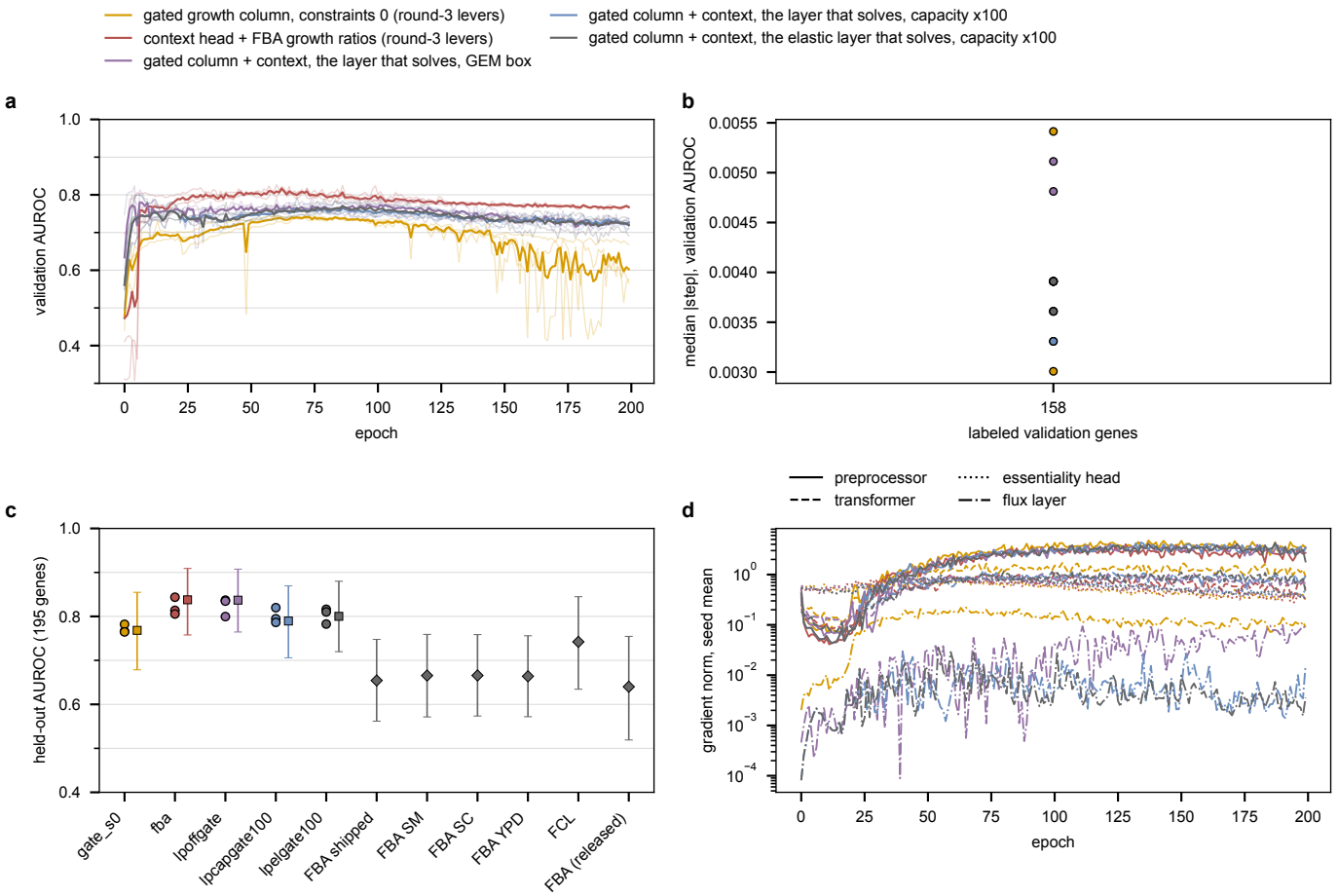


Figure 36: Round 18, the gated column of the elastic layer that solves. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w33`.

The construction did what it was built to do and the score did not follow. On every cell no genotype's program is infeasible at any epoch, the maintenance flux is met on every genotype at the selected epoch, and a gene-owned bound carries a dual on 88 to 90 percent of genotypes, so the two seeds that the hard program absorbed at scale 100 now train from the first step (Figure 34b, d). The forced column reads 0.704, 0.741 and 0.697 at scale 100, a seed mean of 0.714 with a spread of 0.02 where the hard program's was 0.20, three of three seeds below the round-4 forced column, and none at 0.80: the read fails on both clauses. At scale 1,000 the elastic cell reads 0.607, 0.826 and 0.707, the hard program's 0.615, 0.826 and 0.702 to within a seed's noise, which is what the construction promises when no program is infeasible, since the two programs are then the same program. With the pool held by construction the column reads 0.704, 0.743 and 0.670: the projection never engaged (the factor stayed at 1 on every batch), so the abundance head never asked for more protein than the budget and the cell is the one without it. The gated cell reads 0.816, 0.810 and 0.783, three of three above the round-4 gated cell and level with the hard gated cell at the same scale (0.800), below

the GEM box's 0.824, so its read fails on the second clause.

What the histories add is the diagnosis the score alone would not give. The forced column is not overfitting through the abundance head: its training loss barely moves, from 1.41 to 1.25 on the seed that trains best and from 1.26 to 1.25 on another, against 1.33 to 0.31 for the alternating layer's forced coordinate (round 4) and 1.26 to 0.17 for the gated cell beside it, and its validation curve sits at 0.72 to 0.79, the solve's own ranking at initialization (0.73) with a learned sign: two seeds spend their first sixty epochs at 0.23 to 0.30, the inverted affine, before the sign flips (Figure 35a). The gradient is present, since the duals reach the abundance head on nine genotypes in ten and it carries 10^{-3} to 10^{-2} of the summed norm (Figure 34e), but the path it travels is stiff: the head is one linear map of the width-12 token, twelve weights shared by every gene, trained at the backbone's rate, and each step moves ceilings whose duals are of order 0.01 per mmol/gDW/h. Under the hard program at scale 1,000 the same probe shows the other failure, the backbone's gradient collapsing to 10^{-13} after epoch 78 when no gene-owned bound binds any longer (Figure 34d, e). Round 19 tests the stiffness directly, with the layer's parameters at ten times the learning rate and with a two-layer abundance head, and reads the training loss at the selected epoch against round 18's 1.25 beside the score.

4.15 Round 19: the step and the head's capacity ✗

Round 19 (jobs 2326 to 2329) moves the two things that make the path through the solve stiff: the flux layer's parameters at ten times the learning rate (a second optimizer group), a two-layer abundance head in place of the twelve-weight linear map, and both, on the forced elastic column at ceiling scale 100, with both changes behind the gate as well, three seeds each (Tables 49 and 50, Figures 37 and 38); the read is round 18's, paired against the round-18 elastic cells.

Table 49: Round 19, the forced elastic column with a faster layer and a deeper head. The round-18 forced elastic cell at capacity scale 100 (the paired comparator), the round-4 forced column and the round-5 FBA affine, against the same elastic cell with the flux layer's parameters at ten times the learning rate, with a two-layer abundance head, and with both. Arm `flux_projection_anchored_nodiss`, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
growth column, the elastic layer that solves, capacity x100	0.714	0.024	0.704, 0.741, 0.697	0.697 [0.56, 0.82]	106, 132, 180	0.0103
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778 ↑	0.778 [0.69, 0.86]	108, 74, 84	0.0057
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
growth column, elastic solve, x100, flux layer lr x10	0.677	0.042	0.704, 0.629, 0.697	0.699 [0.57, 0.82]	115, 64, 182	0.0044
growth column, elastic solve, x100, two-layer abundance head	0.752	0.072	0.823, 0.754, 0.678	0.818 [0.74, 0.89]	2, 10, 85	0.0048
growth column, elastic solve, x100, two-layer head + lr x10	0.720	0.070	0.799, 0.691, 0.669	0.787 [0.69, 0.87]	9, 16, 191	0.0000

Table 50: Round 19, the gated elastic column with a faster layer and a deeper head. The round-18 gated elastic cell at capacity scale 100 (the paired comparator), the round-4 gated cell and the round-5 FBA affine, against the gated elastic cell with the two-layer abundance head and the flux layer at ten times the learning rate. Arm `flux_projection_anchored_nodiss`, constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
gated column + context, the elastic layer that solves, capacity x100	0.803	0.018	0.816, 0.810, 0.783	0.800 [0.72, 0.88]	6, 95, 63	0.0038
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765 ↓	0.768 [0.68, 0.85]	67, 71, 77	0.0040
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
gated column + context, elastic solve, x100, two-layer head + lr x10	0.774	0.025	0.786, 0.790, 0.745 ↓	0.791 [0.71, 0.87]	85, 66, 10	0.0021

Neither lever is the limit. The faster layer reads 0.705, 0.629 and 0.697, below round 18 on two seeds; the deeper head 0.823, 0.754 and 0.678, above on two, with its best seed selected at epoch 2; both together 0.799, 0.691 and 0.669. The training loss at the selected epoch is 1.24 to 1.26 in every cell, round 18's number, and where the deeper head does fit further later in the run (to 1.07 on two seeds) the validation curve has already peaked and decays, so the extra capacity memorizes and does not rank. Behind the gate the two changes cost the cell: 0.786, 0.790 and 0.745 against the round-18 gated cell's 0.816, 0.810 and 0.783, with the gate ending negative on every seed (-0.51, -0.45, -0.09), the head learning to subtract a column the faster layer has bent toward the training set. No arm passes either clause. Rounds 18 and 19 together say that the forced column cannot be fit through the ceilings by any of feasibility, slack, step size or head capacity, and the next subsection measures why.

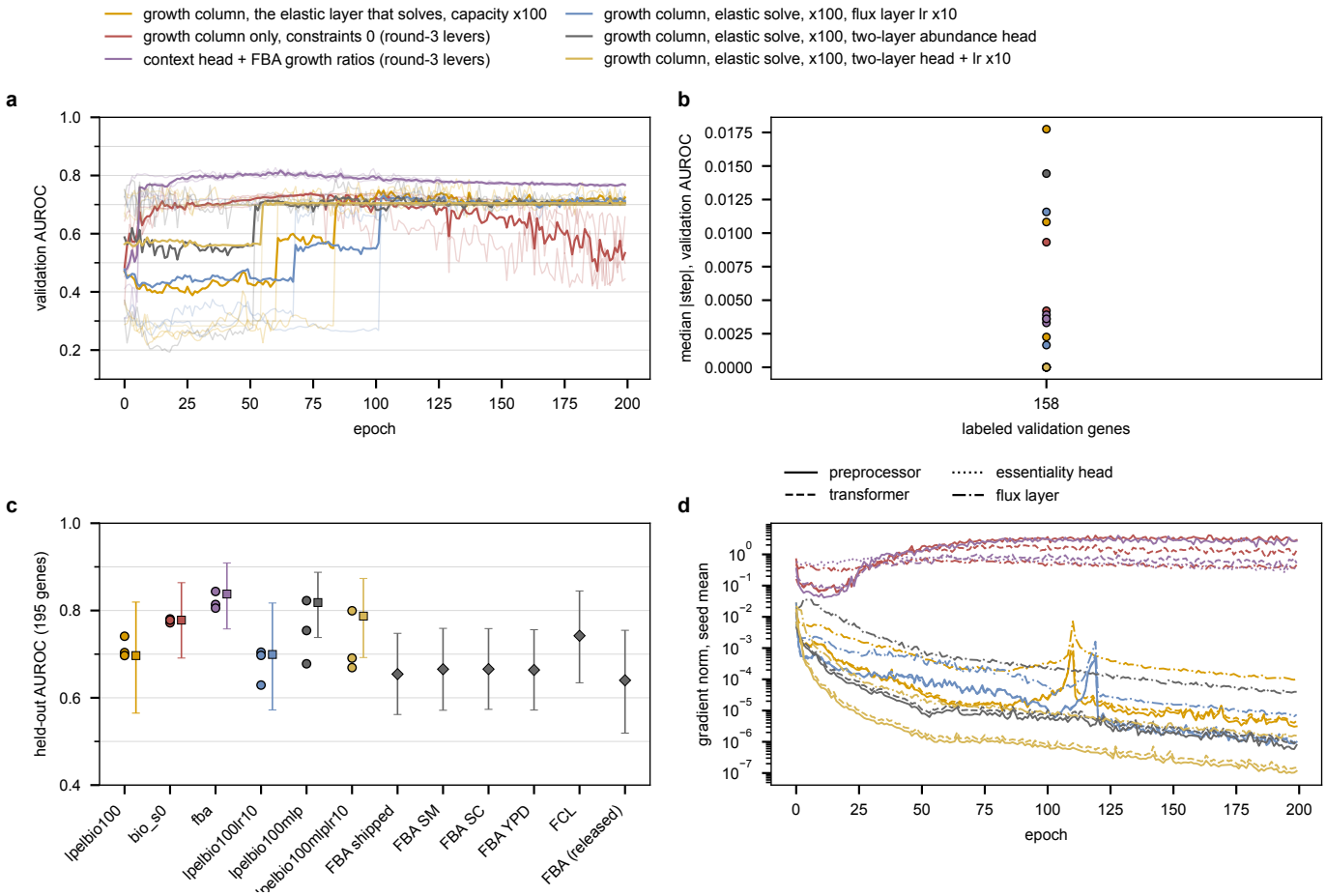


Figure 37: Round 19, the forced elastic column with a faster layer and a deeper head. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w34`.

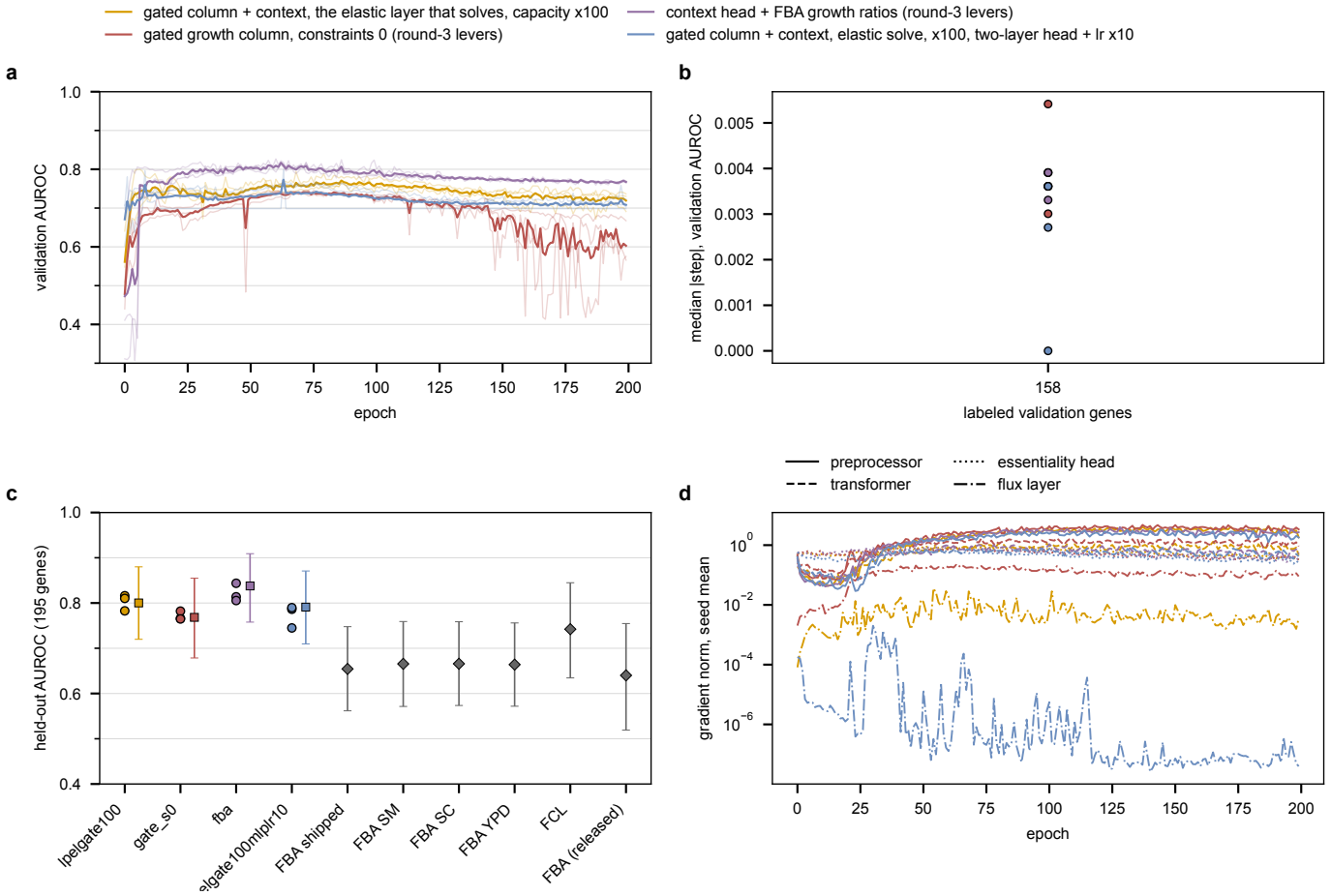


Figure 38: Round 19, the gated elastic column with a faster layer and a deeper head. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w35`.

4.16 Why the ceilings cannot carry the ranking ✗

The gradient the layer that solves hands the backbone is the dual of the bounds that bind at the optimum, so what the backbone can learn through the solve is bounded by which bounds those are and whether they differ between genotypes. `lp_dual_overlap.py` reads them off the elastic cell at initialization on 64 held-out genotypes at three ceiling scales (`results/lp_dual_overlap.json`). At scale 100 the median genotype has six gene-owned reactions at their ceiling, of 2,709 that carry a gene, and the median Jaccard overlap between two genotypes' binding sets is 1.0: the same six reactions bind whichever gene is deleted, with a union of 96 over the 64 genotypes carried by one outlier genotype with 90 and by the lethal deletions, whose programs bind nothing. At scale 30 the median is eight reactions with a union of 22, at 1,000 two with a union of four, and the median dual is 0.005 to 0.009 per mmol/gDW/h throughout. The deleted gene's own reactions never bind, because the deletion has already closed them: the gene's identity reaches the value through the shape of the polytope, which the head cannot move, and never through a bound the head owns. The abundance head can therefore raise or lower one growth level for every genotype at once, which the readout's intercept absorbs, and it cannot move one genotype's value relative to another's, which is what the labels ask for. That is the mechanism behind rounds 18 and 19: the elastic program removes the absorbing state and the gradient arrives on nine genotypes in ten, and it arrives on the same six ceilings for all of them. What the growth column carries is the solve's own ranking of the deletions, 0.73 at initialization, with a learned sign and threshold; the constructions that let the module reach the affine's level are the ones that put a learned head beside the column (the gated cells, 0.80 to 0.82) rather than behind it.

4.17 Round 20: the gated forms on six seeds, and the last forced probes ✗

Round 20 (jobs 2332 to 2335) does two things the dual-overlap probe asks for. It puts the two gated forms of the layer that solves, the GEM-box cell of round 15b and the elastic cell of round 18, on three more seeds (2703 to 2705) so the reliable form is read on six, and it runs the two probes of the forced column the probe leaves open: the elastic column at ceiling scale 30, where the median genotype has eight binding reactions rather than six, with the flux layer at ten times the learning rate, and the scale-100 column with the abundance head's reach raised from e^3 to e^6, so a head that wants to move a ceiling by a decade can (Table 51, Figure 39). The gated read is round 18's, extended to six seeds: every seed above its paired gated cell without the solve (round 4, 0.771), and a seed mean at or above the GEM-box gated cell's 0.824. The forced read is unchanged.

Table 51: Round 20, the gated forms on six seeds and the last forced probes. The round-4 gated cell (the paired comparator) and the round-5 affine, against the gated layer that solves on the GEM box (round 15b, seeds 2700 to 2702) and on seeds 2703 to 2705, the gated elastic cell at capacity scale 100 (round 18) and on seeds 2703 to 2705, and two last probes of the forced elastic column: at ceiling scale 30 with the flux layer at ten times the learning rate, and at 100 with the abundance head's reach doubled to e^6. Constraint scale 0, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
context head + FBA growth ratios (round-3 levers)	**0.821**	0.020	0.814, 0.805, 0.844 ↑	**0.838** [0.76, 0.91]	28, 70, 61	0.0036
gated column + context, the layer that solves, GEM box	0.814	0.020	0.787 to 0.837 (6)	0.816 [0.74, 0.89]	2 to 103	0.0045
gated column + context, the elastic layer that solves, capacity x100	0.804	0.013	0.783 to 0.816 (6)	0.803 [0.72, 0.88]	6 to 95	0.0038
growth column, the elastic layer that solves, capacity x100	0.714	0.024	0.704, 0.741, 0.697 ↓	0.697 [0.56, 0.82]	106, 132, 180	0.0103
growth column, elastic solve, x30, flux layer lr x10	0.630	0.074	0.653, 0.548, 0.690 ↓	0.722 [0.61, 0.83]	110, 2, 187	0.0092
growth column, elastic solve, x100, reach e^6, flux layer lr x10	0.652	0.029	0.669, 0.619, 0.669 ↓	0.741 [0.63, 0.84]	154, 177, 159	0.0032

The gated forms are reliable and stop short of the affine. On seeds 2703 to 2705 the GEM-box gated cell reads 0.818, 0.807 and 0.787 and the elastic gated cell 0.795, 0.808 and 0.814; over six seeds the two arms read 0.814 (sd 0.020) and 0.804 (sd 0.013), pooled 0.816 and 0.803, twelve seeds of twelve above the gated cell without the solve (0.771). Neither reaches the second clause: the GEM-box form's three-seed 0.824 was the top of its spread, and six seeds put both forms 0.01 to 0.02 under the affine's 0.821 (0.814, 0.805 and 0.844 by seed), a gap inside one seed standard deviation of either arm. The gate ends at −0.95, 0.68 and 0.80 on the GEM box and −1.01, −1.41 and −0.79 on the elastic program: the head uses the column with either sign, and the sign is not what the seeds agree on. The forced probes fail as the probe of Section 4.16 predicts. At scale 30 with the faster layer the column reads 0.653, 0.548 and 0.690, the middle seed selected at epoch 2 with the maintenance slack active on every genotype; at scale 100 with the head's reach at e^6 it reads 0.669, 0.619 and 0.669, and the two better seeds end with no gene-owned dual at all, the head having pushed every ceiling off the optimum, while the third sits in full slack. A wider reach hands the head both exits from the ceilings, above the optimum and below the maintenance, and neither one ranks the deletions.

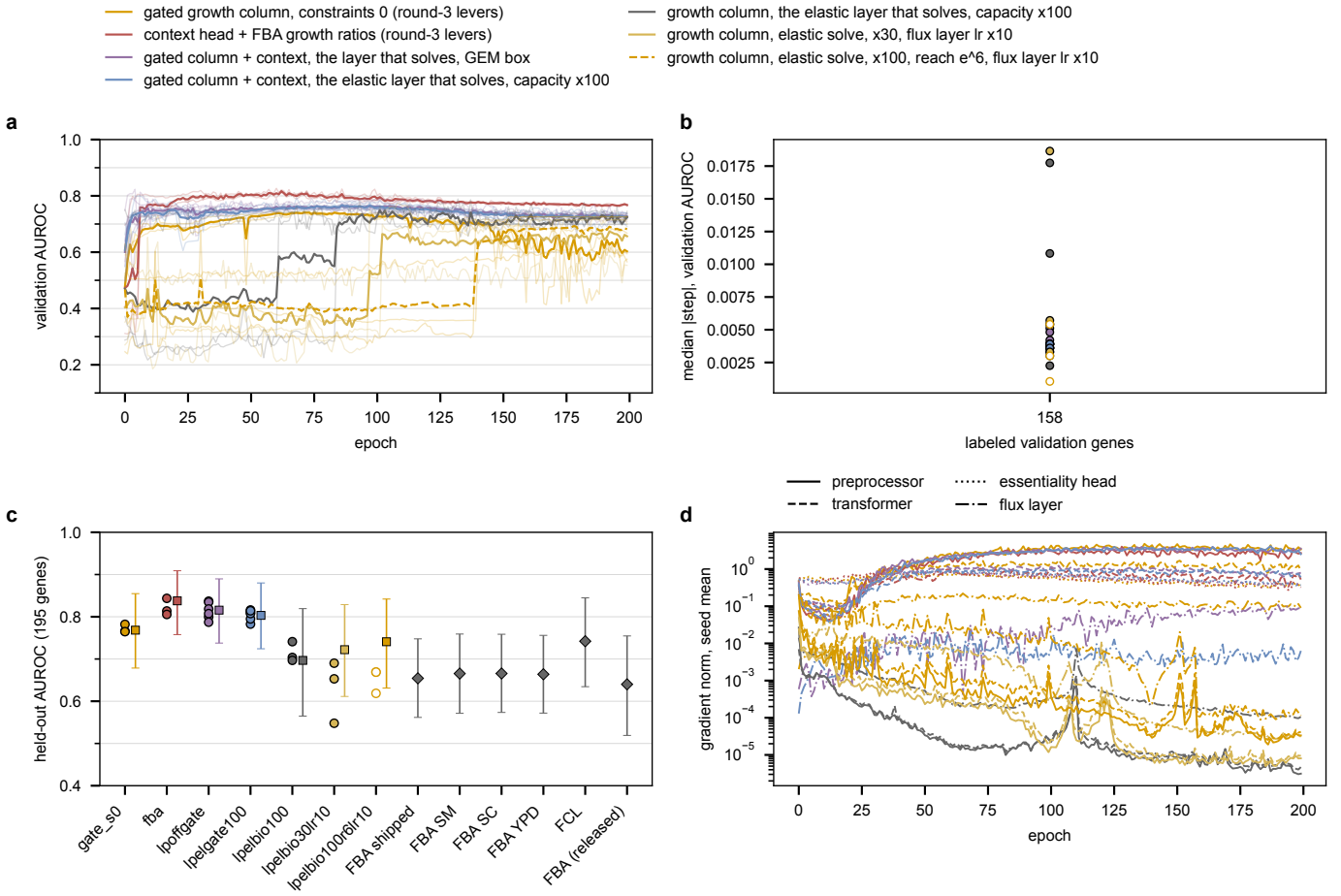


Figure 39: Round 20, the gated forms on six seeds and the last forced probes. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w36`.

4.18 Round 21: the gated cell with a head over the solved flux ✗

If the ceilings cannot move one genotype's growth against another's, the thing the solve does hold per genotype is the flux vector it returns: round 11 measured the deletion's own pFBA vector at 0.77 as an encoding, above its growth ratio at 0.65. Round 21 (jobs 2336 and 2337) reads that vector from inside the cell. The `gated_flux` readout (Figure 40) is the round-18 gated elastic cell plus a zero-initialized linear head over $\log(1 + |v^*_{\mathcal{R}}|)$, v^* the optimum the elastic program returns, detached, over all 4,131 reactions or over the 274 exchange reactions (the uptake and secretion profile), constraint scale 0, three seeds (Table 52, Figure 41). The head's weight norm is logged each epoch. The read is paired against the round-18 gated elastic cell by seed, the pass being three of three above it and a seed mean at or above the affine's 0.821.

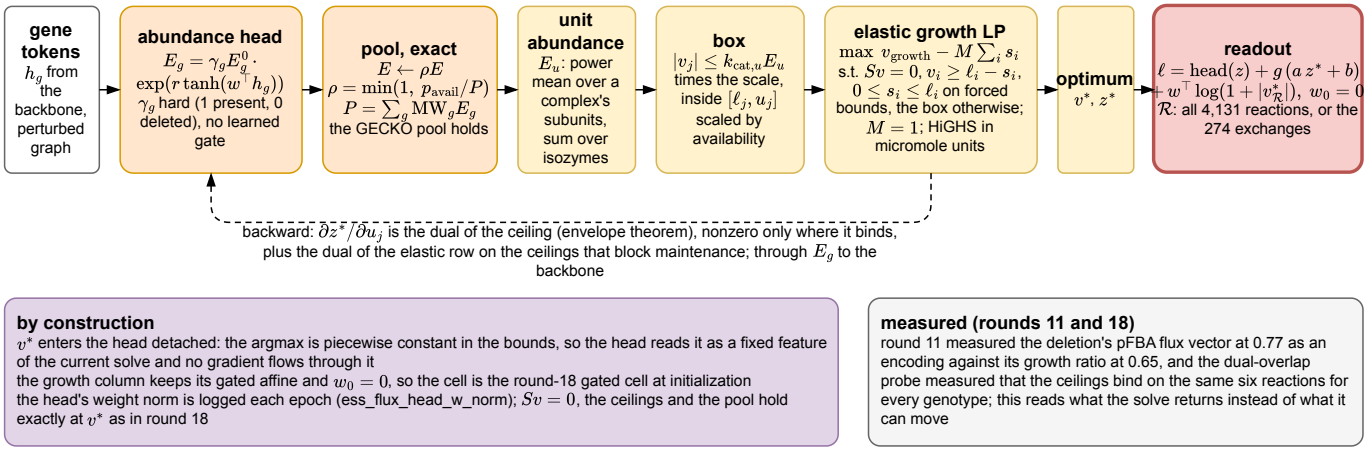


Figure 40: The gated elastic cell with a head over the solved flux, round 21. One block changes against Figure 33, drawn with a heavier border: the readout adds a zero-initialized linear head over $\log(1 + |v^*_{\mathcal{R}}|)$, the flux vector the solve returns restricted to a reaction set $\mathcal{R}$ (all 4,131 reactions, or the 274 exchange reactions). The vector enters detached, since the argmax is piecewise constant in the bounds, so the head reads the solve's answer as a feature and the backbone keeps receiving the duals through the growth column alone. The growth column keeps its gated affine, so the cell is the round-18 gated cell at initialization. Colors as in Figure 17. Drawn in draw.io from `notes/assets/drawio/028-module.gen.py`.

Table 52: Round 21, the gated elastic cell with a head over the solved flux. The round-18 gated elastic cell (the paired comparator) and the round-5 affine, against the same cell with a zero-initialized linear head over $\log(1 + |v^*|)$ of the solved flux, over all 4,131 reactions and over the 274 exchange reactions. Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
gated column + context, the elastic layer that solves, capacity x100	0.803	0.018	0.816, 0.810, 0.783	0.800 [0.72, 0.88]	6, 95, 63	0.0038
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
gated column + context + a head over the solved flux (all reactions)	**0.868**	0.015	0.883, 0.853, 0.868 ↑	**0.868** [0.79, 0.93]	66, 83, 91	0.0158
gated column + context + a head over the solved exchange fluxes	0.835	0.020	0.831, 0.856, 0.817 ↑	0.843 [0.76, 0.92]	83, 197, 89	0.0045

Both heads pass, and the head over every reaction passes by more. Over all 4,131 reactions the cell reads 0.883, 0.853 and 0.868 (mean 0.868, sd 0.015; pooled 0.868 with a gene bootstrap of 0.792 to 0.934), above the round-18 gated elastic cell on every seed (0.816, 0.810, 0.783) and 0.047 above the affine's 0.821, which is the first module cell to pass a pre-registered read in this document and the first to sit above the affine on the ProtT5 backbone. Over the 274 exchange reactions it reads 0.831, 0.856 and 0.817 (mean 0.835), also above its comparator on every seed and above the affine's mean, with one seed selected at epoch 197. Two logged quantities say what changed. The training loss at the selected epoch is 0.61 to 0.79, where every forced cell of rounds 15 to 20 sat at 1.24 to 1.26: the head fits, and the growth column never did. The gate on the growth column ends at 0.01, 0.05 and 0.09 under the full head and 0.24 to 0.42 under the exchange head, so the growth column is used little or not at all once the head can read the vector behind it, and the head's weight norm grows to 3.3 to 5.6 from zero. What is measured is that the flux vector the elastic program returns for a deletion, read linearly, ranks the held-out genes at 0.87 on a backbone whose own pooled cell ranks them at 0.77. Which part of the cell makes that vector informative, the deletion's closure of its reactions through the GPR rules, the learned ceilings that reshape the optimum per genotype, or the head's reach over the exchange and transport columns, is what Section 4.20 separates.

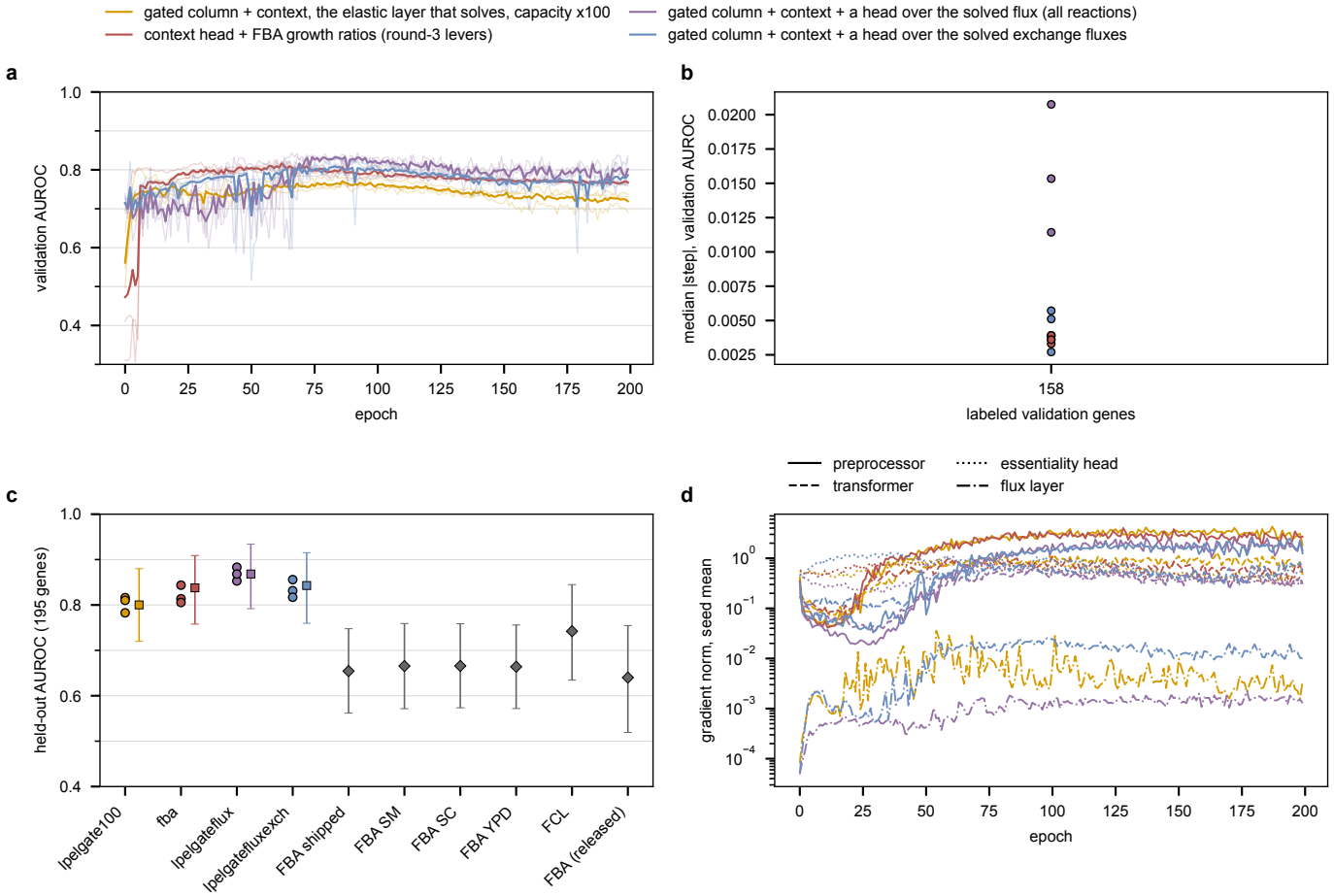


Figure 41: Round 21, the gated elastic cell with a head over the solved flux. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w37`.

4.19 Round 23: the module on the ladder's top input ✗

Every module cell so far sits on the ProtT5 backbone, where the affine adds 0.05 to the pooled cell (0.771 to 0.821). Section 2.6 found the input that carries the rest of the ladder, the 173-column pathway multi-hot beside ProtT5, and round 22 (Section 3.13) runs the affine on it. Round 23 (jobs 2341 to 2344) asks whether the module delivers the same bump there: the three gated forms of the layer that solves, the GEM-box solve of round 15b, the elastic solve of round 18 and the elastic solve with the flux head of round 21, each on ProtT5 with the pathway multi-hot, beside the pooled cell on that input with no readout as the module-free control, three seeds (Table 53, Figure 42). The read pairs each form by seed against the control and against the round-22 affine on the same input: a form passes if it is above the control on three seeds of three and at or above the affine on three of three.

Table 53: Round 23, the gated module forms on the ladder's top input. The pooled cell on ProtT5 with the pathway multi-hot and no readout (the module-free control) and the round-22 FBA affine on that input (the paired comparator), against the three gated forms of the layer that solves on the same input: the GEM-box solve (round 15b), the elastic solve at capacity scale 100 (round 18) and the elastic solve with the head over the solved flux (round 21). Constraint scale 0, batch 32, class weight, 200 epochs, seeds 2700 to 2702. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
pathway multi-hot + ProtT5 backbone, no readout	0.860	0.002	0.859, 0.859, 0.862	0.861 [0.79, 0.92]	36, 56, 46	0.0040
pathway multi-hot + ProtT5 backbone + FBA ratios	**0.890**	0.005	0.889, 0.895, 0.885 ↑	0.890 [0.83, 0.94]	31, 45, 55	0.0054
pathway multi-hot + ProtT5; gated column + context, the layer that solves, GEM box	0.868	0.005	0.862, 0.871, 0.869 ↑	0.867 [0.80, 0.93]	47, 61, 40	0.0052
pathway multi-hot + ProtT5; gated column + context, elastic solve, x100	0.867	0.007	0.859, 0.872, 0.869	0.872 [0.81, 0.93]	43, 58, 34	0.0048
pathway multi-hot + ProtT5; gated + a head over the solved flux (all reactions)	0.845	0.087	0.716 to 0.911 (6)	**0.910** [0.86, 0.95]	11 to 150	0.0203

The input does most of the work, the affine the rest, and the flux head is the one form that reaches past it. The pooled cell on ProtT5 with the pathway multi-hot and no readout reads 0.859, 0.859 and 0.862 (mean 0.860), 0.09 above the same cell on ProtT5 alone, and the affine on that input reads 0.889, 0.895 and 0.885 (mean 0.890, Section 3.13). The two gated forms without the head sit between: the GEM-box solve at 0.862, 0.871 and 0.869 and the elastic solve at 0.859, 0.872 and 0.869, above the control on three seeds and on two, and below the affine on three of three. On seed 2700 the elastic cell's gate ended at exactly zero and its held-out score is the control's to four decimals, which is what a closed gate means: the cell is the pooled model. The flux head on this input reads 0.897, 0.716 and 0.903. The two seeds that fit are the two highest single cells in this document, 0.008 and 0.018 above the affine on their seeds, with the pooled per-gene logits of all three at 0.907; the third seed never fit, its training loss 1.02 at the selected epoch against 0.55 and 0.74, its validation curve flat at 0.69 to 0.72 for 200 epochs where the other two pass through a late transition near epoch 80 to a loss of 0.2 to 0.3. So the form fails the read as written, above the control on two seeds of three, and its seed mean of 0.839 is set by the seed that did not fit. Three more seeds (round 25) read 0.890, 0.752 and 0.911, the middle one unfit again, so over six seeds the form reads 0.897, 0.716, 0.903, 0.890, 0.752 and 0.911: four cells at 0.890 to 0.911, the highest single transformer cells in the document, and two at 0.72 to 0.75 that never left a training loss of 1.0, a seed mean of 0.845 with a standard deviation of 0.087. The pooled per-gene logits of the six read 0.910 [0.86, 0.95], the highest bootstrap point in the document, the two unfit seeds diluted rather than decisive in the mean logit. Hypothesis, untested: the transition is delayed rather than absent on the unfit seeds, since the round-21 cells on ProtT5 alone fit by epoch 66 to 91 on every seed and the pathway input adds 173 columns the head has to learn around; a longer budget on an unfit seed is the direct test.

4.20 Round 24: what carries the head over the solved flux ✗

Round 21's first pack passed its read (Section 4.18), and three things could be carrying it: the flux vector the program returns under the deletion, which the ladder's pFBA-vector encoding put at 0.77 on its own; the learned ceilings, which shape that vector per genotype through the abundance head; and the linear head's choice of reactions. Round 24 (jobs 2347 to 2349) separates them. The same head over the flux of the GEM-box solve (round 15b's program, no ceilings and no abundance head) says whether the enzyme-constrained part is needed: at or above round 21 on three seeds of three and it is not, below on three of three and it is, anything else undecided at this size. The head over the 2,709 gene-owned reactions only says whether the exchange and transport columns matter. And the passing cell runs on seeds 2703 to 2705, so its read stands on six (Table 54, Figure 43).

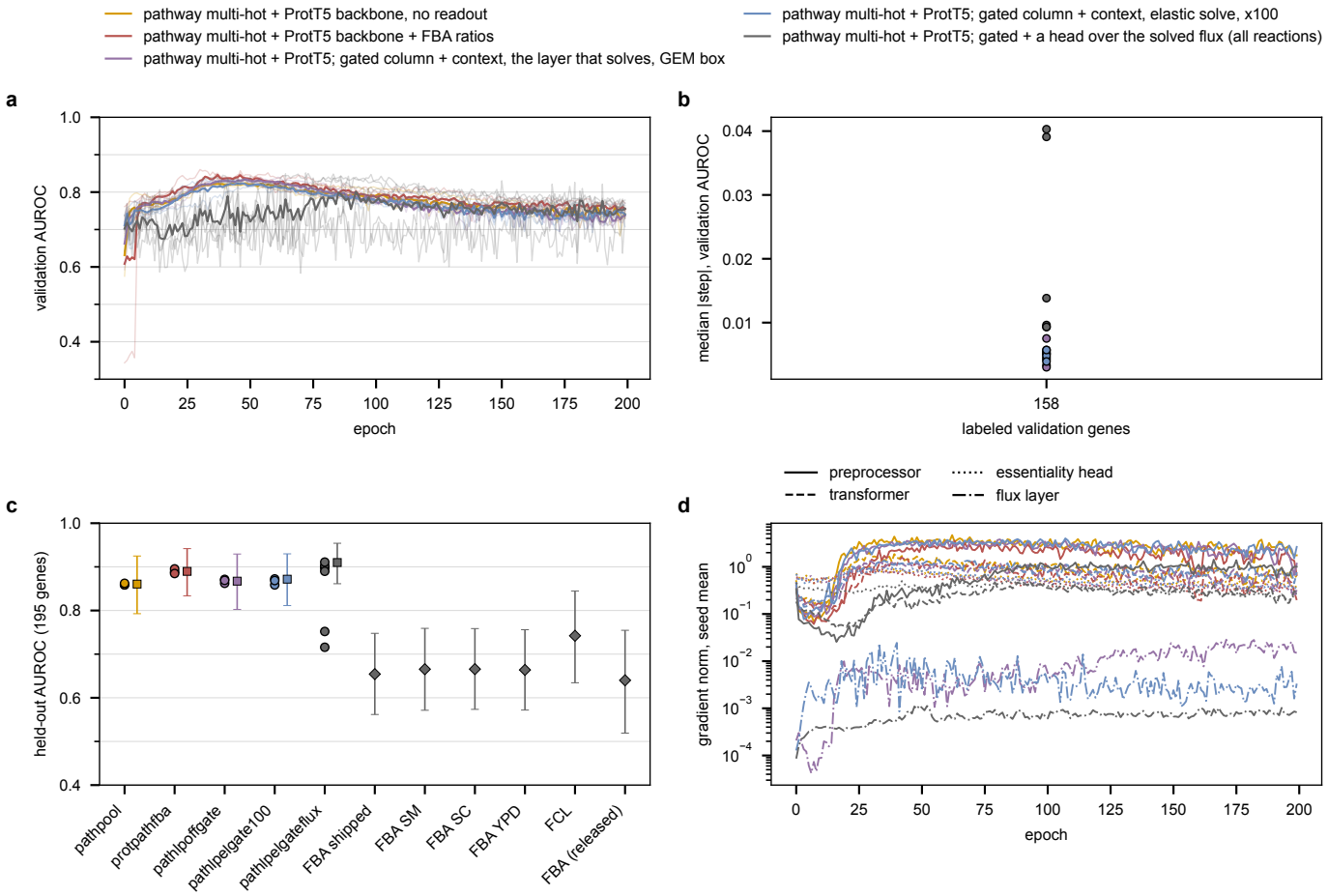


Figure 42: Round 23, the gated module forms on ProtT5 with the pathway multi-hot. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w39`.

Table 54: Round 24, what carries the head over the solved flux. The round-18 gated elastic cell, the round-5 affine on ProtT5 and the round-9 affine on the chromosome-pathway indicator with ProtT5 (the comparators), the round-21 cells (the head over all solved fluxes, on six seeds once seeds 2703 to 2705 land; the head over the 274 exchange fluxes), against the head over the 2,709 gene-owned fluxes, and the head over the flux of the GEM-box solve with no learned ceilings. Constraint scale 0, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
gated column + context, the elastic layer that solves, capacity x100	0.803	0.018	0.816, 0.810, 0.783	0.800 [0.72, 0.88]	6, 95, 63	0.0038
context head + FBA growth ratios (round-3 levers)	0.821	0.020	0.814, 0.805, 0.844	0.838 [0.76, 0.91]	28, 70, 61	0.0036
chrom-pathway + ProtT5 backbone + FBA ratios	**0.871**	0.015	0.860, 0.866, 0.888 ↑	**0.868** [0.80, 0.93]	53, 51, 38	0.0030
gated column + context + a head over the solved flux (all reactions)	0.862	0.013	0.847 to 0.883 (6)	0.866 [0.79, 0.93]	66 to 135	0.0179
gated column + context + a head over the solved exchange fluxes	0.835	0.020	0.831, 0.856, 0.817 ↑	0.843 [0.76, 0.92]	83, 197, 89	0.0045
gated column + context + a head over the solved gene-owned fluxes	0.861	0.009	0.871, 0.860, 0.853 ↑	0.861 [0.78, 0.93]	93, 86, 75	0.0197
gated column + context + a head over the GEM-box solve's flux (all reactions)	0.852	0.019	0.873, 0.836, 0.847 ↑	0.860 [0.78, 0.93]	68, 164, 49	0.0184

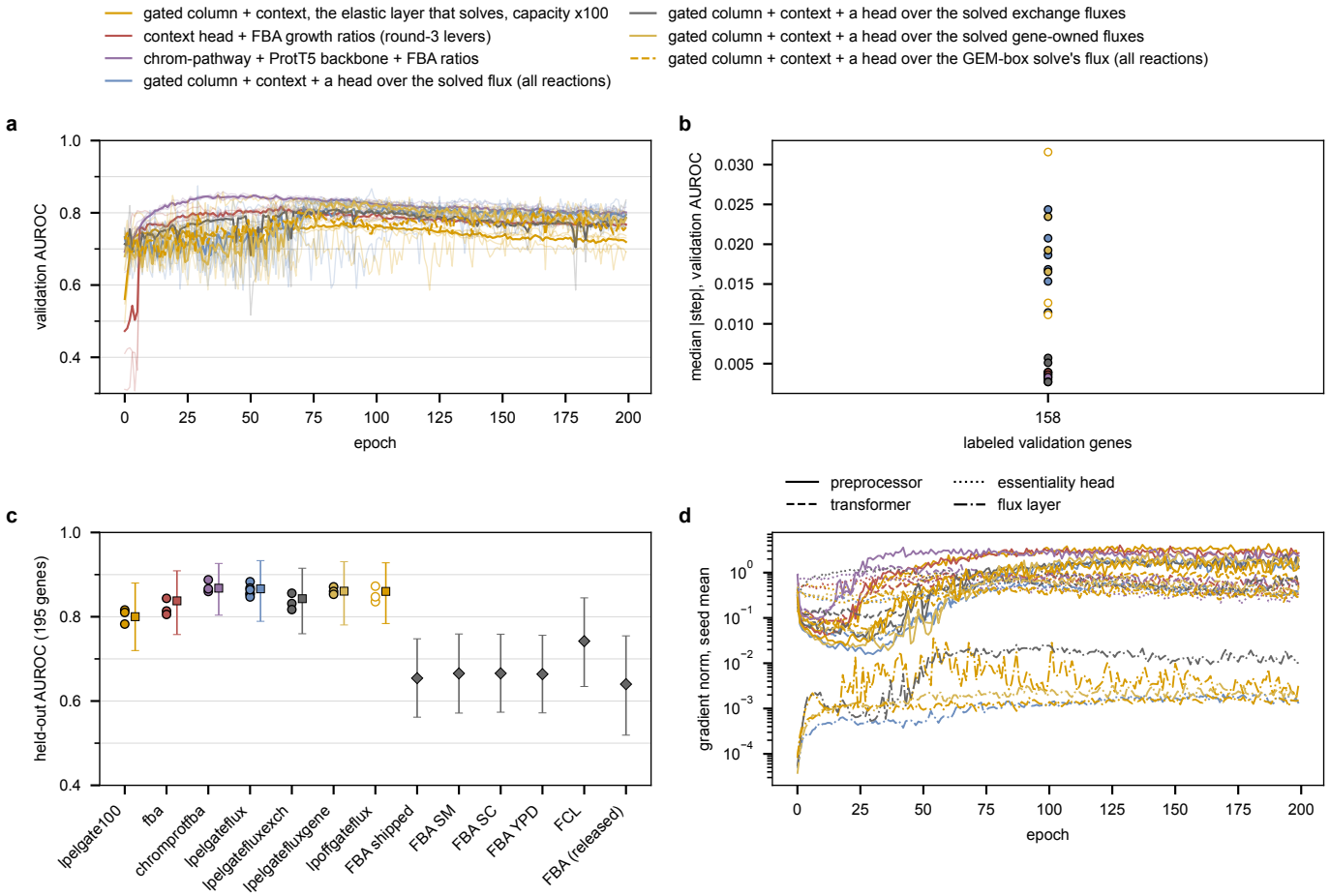


Figure 43: Round 24, what carries the head over the solved flux. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w40`.

The vector carries it, the ceilings sharpen it a little, and the gene-owned columns are where it lives. On six seeds the head over every solved flux reads 0.862 (sd 0.013, pooled 0.866 with a gene bootstrap of 0.789 to 0.933), every seed above its paired gated elastic cell, so the round-21 read holds on six. The same head over the GEM-box solve, with nothing learned in the layer, reads 0.873, 0.836 and 0.847 (mean 0.852), below the elastic cell on three seeds of three by 0.010, 0.017 and 0.021: by the read as written the learned ceilings are needed, and what they add is inside one seed standard deviation of either arm. With nothing learned the flux is a function of the genotype and the medium alone, and a linear head over it still sits 0.03 above the affine and 0.08 above the ladder's linear rung on the deletion pFBA vector (0.771); which of the transformer's context head, the class-weighted training, or the difference between

the program's argmax and the parsimonious point accounts for that gap is not measured. The head over the 2,709 gene-owned reactions reads 0.871, 0.860 and 0.853 (mean 0.861), level with the full head (below on two seeds, above on one), where the 274 exchange fluxes alone read 0.835: the uptake and secretion profile is the part of the vector that carries least.

4.21 Round 25: the flux head under the reported protocol ✗

The document's number is a fold ensemble (Section 3): five models trained on the five cross-validation files, their logits averaged on the held-out genes. Round 25 (jobs 2352 to 2354) runs the round-23 flux-head cell, the gated elastic solve with the head over every solved flux on ProtT5 with the pathway multi-hot, under that protocol, and on seeds 2703 to 2705 of the split file so the single-split read stands on six seeds and the number of seeds that miss the fitting transition is counted out of six rather than three. The ensemble is read beside the round-12 ensemble's 0.893 with its gene bootstrap, with no pass rule (Table 55); the six-seed read is round 23's.

Table 55: Every fold ensemble in the document on the 195 held-out genes: five models trained on the five cross-validation files (each a different fifth of the 787 labeled non-test genes held for validation), their per-gene logits averaged and scored once, Flux Cone Learning's own test protocol. Fold AUROCs are the single models on the held-out genes; the ensemble carries a 2,000-draw gene bootstrap. Written by `cv_ensemble_readout.py`. Bold marks the highest value in a column.

cell	folds	fold AUROCs	fold mean	ensemble [95% CI]
round 3: pooled transformer, ProtT5, no readout	5	0.780, 0.806, 0.797, 0.751, 0.789	0.785	0.795 [0.71, 0.88]
round 5: ProtT5 + FBA affine	5	0.814, 0.824, 0.839, 0.775, 0.853	0.821	0.822 [0.74, 0.90]
round 12: chromosome-pathway indicator + ProtT5 + FBA affine (the reported number)	5	0.860, 0.899, 0.889, 0.875, 0.884	0.881	0.893 [0.84, 0.94]
round 17: chromosome-pathway indicator + ProtT5 + the widened affine (ratios, statistics, pFBA vector)	5	0.863, 0.860, 0.887, 0.840, 0.882	0.867	0.901 [0.85, 0.95]
round 25: pathway multi-hot + ProtT5, gated elastic solve + the head over the solved flux	5	0.901, 0.702, 0.859, 0.904, 0.687	0.811	0.897 [0.84, 0.95]
MLP on the transformer's three inputs (no graph)	5	0.870, 0.892, 0.880, 0.866, 0.884	0.878	0.886 [0.82, 0.94]
graph convolution on the three inputs, 400 epochs	5	0.874, 0.807, 0.897, 0.888, 0.882	0.870	0.887 [0.83, 0.93]
graph convolution on the GO multi-hot alone, 400 epochs	5	0.914, 0.893, 0.918, 0.916, 0.906	0.909	**0.918** [0.87, 0.95]
graph convolution on the GO multi-hot alone, 800 epochs	5	0.905, 0.900, 0.915, 0.911, 0.909	0.908	0.915 [0.87, 0.95]
graph convolution on GO alone, 400 epochs, three seeds per fold	5	0.907, 0.899, 0.922, 0.916, 0.907	0.910	0.917 [0.87, 0.95]
graph convolution on GO alone, 800 epochs, three seeds per fold	5	0.906, 0.901, 0.919, 0.912, 0.909	0.909	0.915 [0.87, 0.95]

The ensemble is 0.897 [0.84, 0.95], one number with the reported 0.893, and it is made of three fold models at 0.90 and two that never fit. The folds read 0.901, 0.702, 0.859, 0.904 and 0.687; folds 1 and 4 stay at a training loss of 1.0 through their budget, the trace seed 2701 showed in round 23, so three of the eight cells of this form on the pathway input miss the fitting transition. The fold mean is 0.811 against round 12's 0.881, and by the rule round 17 fixed (fold models above round 12's on at least four folds of five, here two) the reported number stands. The fit folds are the highest fold models in the document; the form's problem is the third of its cells that do not fit, and the six-seed read of Table 53 says how often.

4.22 Round 27: the scalable form ✗

What the foundation-model setting can carry is a transform, not a layer. Round 24 measured that the head over the GEM-box solve, with nothing learned in the layer, reads 0.852 on ProtT5, 0.01 to 0.02 under the elastic cell with its ceilings and 0.03 above the affine; and with nothing learned the flux is a function of the genotype and the medium alone, so the program need only be solved once per genotype. Round 27 adds that cache to the solver (a solved program is reused whenever its bounds carry no gradient) and runs the cached GEM-box head on ProtT5 with the pathway multi-hot on six seeds (job 2368, then 2388 to 2390) and on the five folds (2369, then 2391), the cells of rounds 23 and 25 with the learned ceilings removed (Table 56, Figure 44; Table 55). The read is round 23's, and the wall time per cell is reported beside the number, since that is what the form is for.

Table 56: Round 27, the scalable form on the pathway input. The pooled control on ProtT5 with the pathway multi-hot, the round-22 affine on that input, and the round-23 flux head on the elastic solve (the comparators), against the same head over the GEM-box solve with nothing learned in the layer and the program cached per genotype, seeds 2700 to 2705. Constraint scale 0, batch 32, class weight, 200 epochs. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
pathway multi-hot + ProtT5 backbone, no readout	0.860	0.002	0.859, 0.859, 0.862	0.861 [0.79, 0.92]	36, 56, 46	0.0040
pathway multi-hot + ProtT5 backbone + FBA ratios	0.890	0.005	0.889, 0.895, 0.885 ↑	0.890 [0.83, 0.94]	31, 45, 55	0.0054
pathway multi-hot + ProtT5; gated + a head over the solved flux (all reactions)	0.845	0.087	0.716 to 0.911 (6)	0.910 [0.86, 0.95]	11 to 150	0.0203
the same cell, first attempt (three survivors of the six-cell pack; per-gene dumps overwritten)	**0.905**	0.005	0.911, 0.901, 0.903 ↑	–	55, 76, 78	0.0174
pathway multi-hot + ProtT5; gated + a head over the GEM-box solve's flux (cached, nothing learned in the layer)	0.882	0.055	0.772 to 0.917 (6)	**0.917** [0.87, 0.96]	42 to 61	0.0176

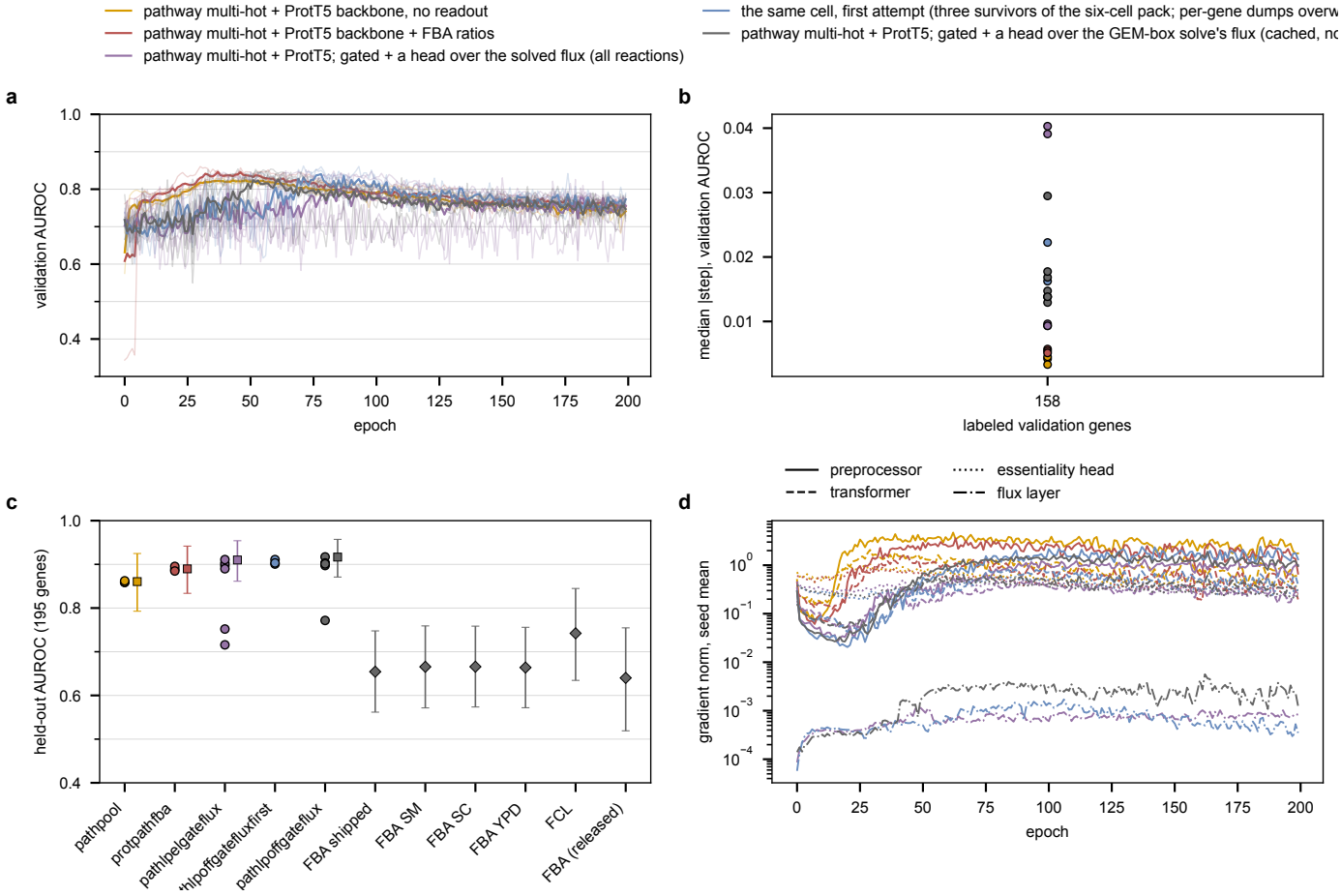


Figure 44: Round 27, the scalable form on the pathway input. a, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w43`.

The scalable form matches the best cells and, on nine runs, misses the fitting transition once. The six-cell pack (job 2368) lost three cells to the launcher's memory ceiling; its three survivors, seeds 2701 to 2703, read 0.911, 0.901 and 0.903 (mean 0.905, sd 0.005) with training losses of 0.53 to 0.77 and selected epochs 55 to 78, and their per-gene dumps were overwritten by the rerun, so the table carries their pinned numbers without a bootstrap. The rerun of all six seeds in pairs (jobs 2388 to 2390, 40 GB each) reads 0.902, 0.772, 0.897, 0.917, 0.903 and 0.901 on seeds 2700 to 2705 (mean 0.882, sd 0.055; pooled 0.917 with a gene bootstrap of 0.871 to 0.957, above the elastic form's 0.910 on the same six seeds), selected epochs 42 to 61, training losses 0.05 to 0.18 on the five that fit and 1.04 on seed 2701, which never fit. Seed 2701 is the seed the elastic form never fit at 200 epochs (0.716); the scalable form fit it in the first attempt (0.911) and not in the second (0.772), which is round 29's finding again, on a form with nothing learned in the layer and the solve cached: the miss is a chance event of the run and not of the seed or of the layer's ceilings. Over the nine runs of

this form, eight fit and read 0.897 to 0.917; over the twelve runs of the elastic form on the same input (six seeds, five folds and the 400-epoch rerun), nine fit (Section 4.23). The wall time was 139 minutes per cell at six cells on one card and 83 minutes at two, against 86 for the uncached elastic form at three: at either packing the layer's own forward on the card, not the solve, is the cost, and the cache's hit rate was not logged. The fold read of this form was lost to the same ceiling and reruns at 120 GB (job 2391) at the time of writing.

4.23 Round 29: the unfit seed under a longer budget ✗

A third of the elastic flux-head cells never fit within 200 epochs (rounds 23 to 25), and the constant learning rate leaves two readings open: the fitting transition is delayed on those seeds, or it is absent. Round 29 reruns the round-23 cell on seed 2701, the seed whose 200-epoch run never fit, with the same program and a 400-epoch budget (job 2387; Table 57, Figure 45). Because the learning rate is constant, the first 200 epochs of the two runs are the same program on the same seed, so the pair also reads whether a run is reproducible.

Table 57: Round 29, the unfit seed under a 400-epoch budget. The round-23 flux-head cell on ProtT5 with the pathway multi-hot, seed 2701, whose 200-epoch run never fit, beside the same program and seed run for 400 epochs. Batch 32, class weight, one seed; the two runs diverge from the first epoch, so the pair reads a chance event, not a delay. Bold marks the highest value in a column. ↑ (↓) marks an arm whose held-out AUROC is above (below) the first arm's on every one of the same three seeds.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
pathway multi-hot + ProtT5; gated + a head over the solved flux (all reactions)	0.716	0.000	0.716	0.716 [0.60, 0.82]	150	0.0391
the same cell, 400 epochs	**0.889**	0.000	0.889	**0.889** [0.84, 0.94]	82	0.0165

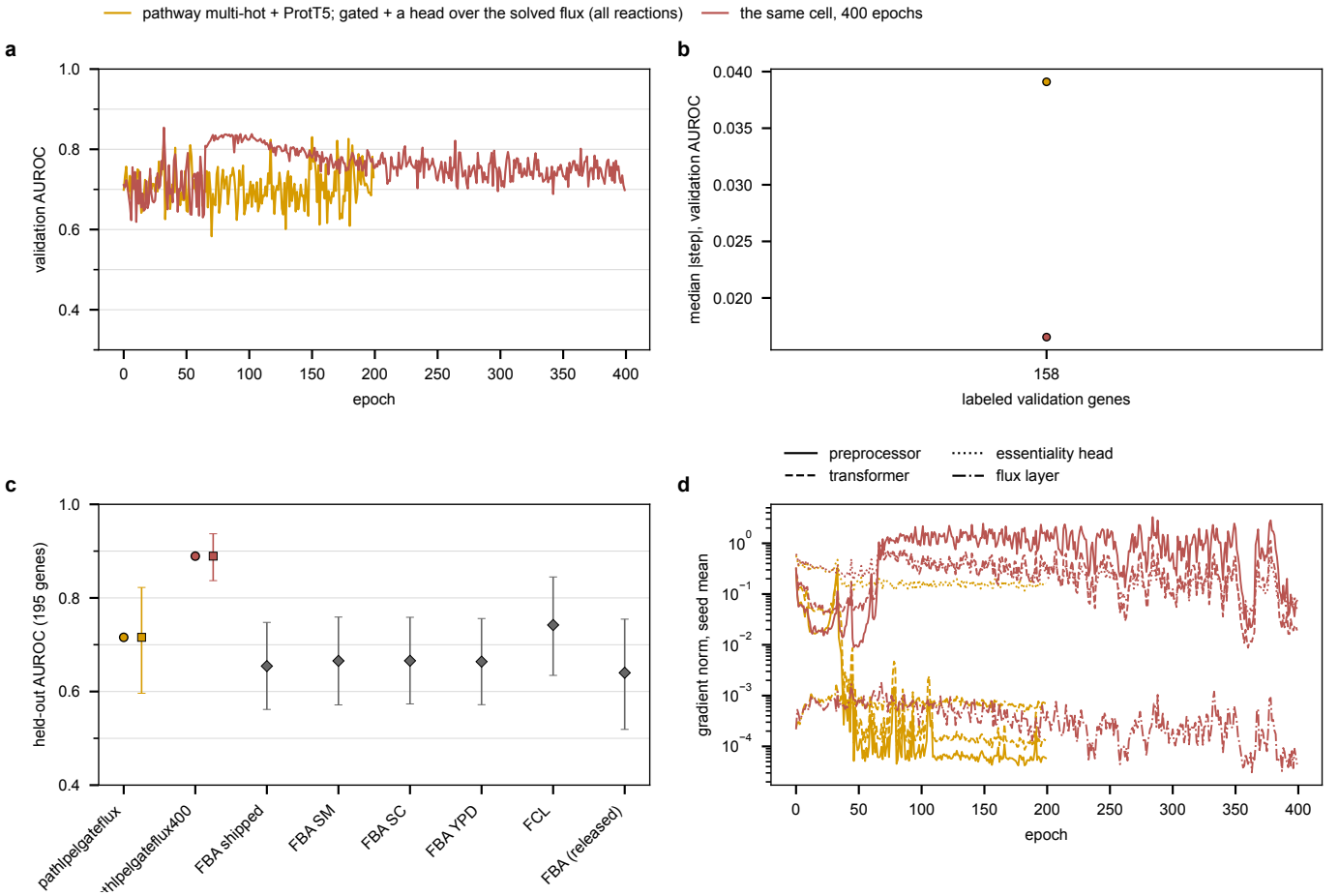


Figure 45: Round 29, the unfit seed under a 400-epoch budget. a, Validation AUROC per epoch for the two runs of seed 2701. **b**, The median absolute epoch-to-epoch step per cell against the validation size. **c**, Held-out AUROC per cell (points) and the gene bootstrap (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w45`.

The transition is neither delayed nor absent on the seed; it is a chance event of the run. The 400-epoch run fits between epochs 60 and 80 (training loss 0.98 at epoch 60, 0.50 at 80, 0.07 at 399), selects epoch 82 and reads 0.889 [0.84, 0.94]

on the held-out genes, against 0.716 for the 200-epoch run of the same seed, which never left a training loss of 0.95. The two runs are not the same trajectory: their training losses differ from the first epoch (1.128 against 1.125) and are unrelated by epoch 50 (0.987 against 0.951), so the seed fixes the initialization and not the path, and the solve inside the forward, the sampler and the card's kernels leave the path to chance. What the 200-epoch budget could not tell apart, a seed that fits late from a seed that does not fit, is therefore the wrong question. Across the seventeen elastic flux-head cells of rounds 21, 23 and 25 (six on ProtT5, six on the pathway input, five folds), the training loss first falls under 0.6 between epochs 39 and 137 in fourteen and never in three, and the two latest transitions (epochs 136 and 137) read 0.864 and 0.702 held out, so a late transition is no guarantee either. Past the transition the 400-epoch run overfits (validation AUROC 0.836 at epoch 82, 0.70 at 399, with the prediction spread growing twelvefold), so the longer budget buys nothing after the selected epoch. The scalable form of round 27 is not free of the chance: its rerun of the same seed fit in one attempt and not in the other (0.911 against 0.772), with the solve cached and nothing learned in the layer, so the layer's ceilings are not where the chance enters.

4.24 Where the gradient goes ✗

Every cell logs a gradient norm per module each epoch, and read across the module arms (Figure 46) they say which of the three failure modes each arm is in. On the forced arms the whole gradient passes through the layer and the layer passes it on: the flux layer carries 10 to 40 percent of the summed module norm and the preprocessor's norm climbs to 1 to 3, the same level as under the pooled head, so the backbone is trained through the layer and the growth column is a trained, free coordinate that ranks no better than the head. On the gated arms the layer carries 0.3 to 3 percent, the head carries the rest, and the gate ends between -0.2 and $+0.2$ across seeds under constraint scale 0 (seed mean $+0.05$) and at $+0.2$ on every seed with the FBA affine beside it, so the backbone is trained through the MLP and the flux path is a small correction. On the warm-started forced arm with α frozen the backbone receives no gradient at all: the preprocessor's norm sits at 0.02 to 0.1 for the whole run, two orders below every other arm, because the flux is 99 percent a fixed reference and only the two parameters of the growth affine train; the cell is the affine on the projected reference, and the projection reversed its sign. The warm-started gated arms leave the flux path at 0.3 percent of the gradient and push the gate to -0.8, which is the head learning to subtract a sign-reversed growth column. None of the arms is gradient-starved except the one that was meant to be a fixed function, so the alternating module's null on this endpoint is not a gradient-flow problem; the gradient reaches the layer, and what that layer computes is not growth. The layer that solves has the opposite gradient problem, which rounds 15b and 15c measure: its gradient is exactly the dual of the binding bounds, which is zero when no gene-owned bound binds and absent when the program is infeasible.

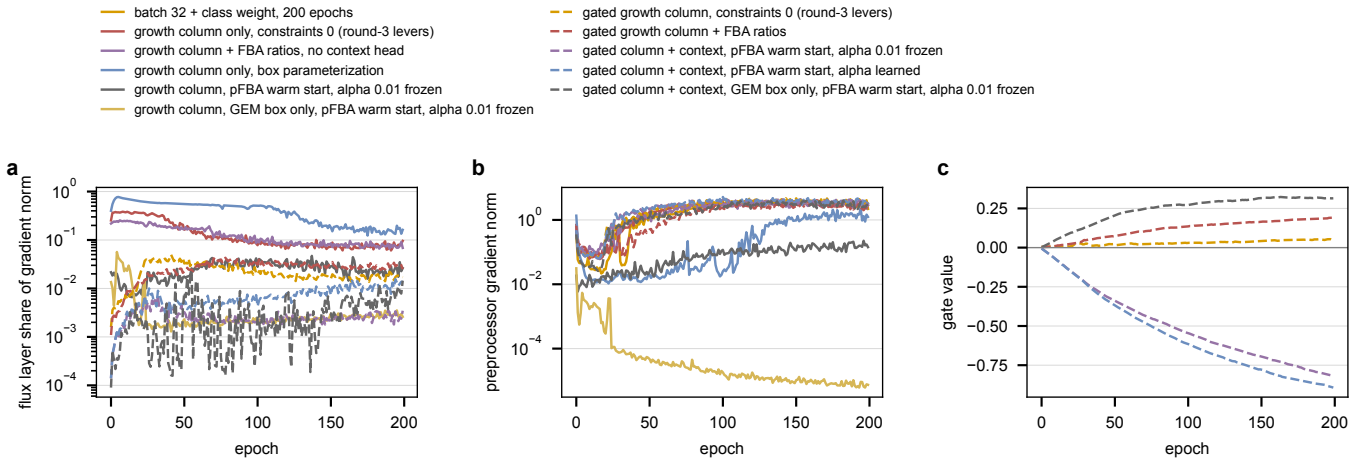


Figure 46: Where the gradient goes on the module arms. **a**, The flux layer's share of the summed per-module gradient norm per epoch, seed mean, on every module arm of rounds 4 to 13; solid lines are the forced arms and the pooled reference, dashed the gated arms. **b**, The preprocessor's gradient norm per epoch, seed mean, on the same arms. **c**, The gate's value per epoch on the gated arms. Written by `experiments/028-gene-essentiality/scripts/grad_flow_readout.py`.

4.25 Why the solve helps and the layer does not ✗

What FBA contributes to the prediction is one number per deletion: the maximum of growth over the polytope $\{Sv = 0,\ \ell \le v \le u\}$ with the deleted gene's reactions closed. That number ranks the held-out genes at 0.65 alone, its four-medium version lifts a logistic regression, a random forest and the transformer by 0.05 each, and the solve has more to give than the number: eight statistics of the gene's reactions under the wild-type solve reach 0.80 with no sequence, and the whole deletion flux vector 0.77. The value is in the optimum over the constraint set and in the flux that attains it.

The module holds the same constraint set and computes something else. Its flux is K alternating projections of a proposal the backbone writes, and the rounds measure, in order, that the projections stop short of the polytope (58 percent of reactions out of bounds when the balance is exact, a balance residual of 0.85 to 2.0 when the bounds are), that the growth column of a flux off the polytope is a free coordinate the readout learns to use with either sign (the probe: 0.23 on the held-out genes under trained weights, Spearman -0.25 with the solve), that the learned parts of the layer are not what the column carries (the shuffled-k_{cat} control matches the real table), and that the layer's constraint set rejects the solve when the solve is supplied as the proposal (round 10: the reference violates the anchored box on a third of its reactions and the returned flux sits half the reference's norm away, sign reversed). The gradient is not the cause; it reaches the layer on every arm that lets it.

The comparison also bounds what a feasible layer could add. A layer that reached the polytope would return a feasible flux, not the optimal one, unless the backbone learned to write proposals whose projection is the optimum; the solve finds the optimum by construction. And a growth column is one number, where the solve's flux vector scores 0.11 to 0.13 more AUROC on the ladder than its growth ratio does. A module that helped on this endpoint would therefore have to solve rather than project, with the backbone setting the bounds it solves under (the capacity ceilings from the abundance head are that interface already), and read more of the resulting flux than its growth column. The first of those is a linear program per genotype per forward pass with the gradient taken through its dual, about 50 milliseconds each. Rounds 13, 15 and 15b test both routes on both boxes. On the GEM's own box the solve survives the projections (round 13) and the program that solves is feasible and ranks at 0.73 before training (round 15b), and in both cases the gated module cell reaches the affine's level, 0.816 and 0.824, three of three seeds above the round-4 gated cell, with the growth column entering through a positive gate. On the anchored box the projections reverse the solve (round 10) and the program has no feasible point (round 15), because the enzyme-capacity ceilings as parameterized exclude growth; with those ceilings scaled to admit growth the program solves, the ceilings bind, and on one of three seeds the growth column alone ranks the held-out genes at 0.84 with no head and no appended ratio (round 15c), the module's upper bound so far. The module helps, then, exactly when its flux is on the polytope the solve optimizes over, and the constraint set that was meant to make it more than the solve is what kept it off until it was calibrated.

5 Findings ✗

The two questions the document opened with have the following answers on the released Merzbacher 2025 split, every number at a validation-selected epoch on the 195 held-out genes with a gene bootstrap, and every gain attributed by a paired, pre-registered read.

The held-out number is 0.893 [0.84, 0.94]. A width-12 cell graph transformer reading the chromosome-pathway indicator and ProtT5, with FBA's four growth ratios on its logit, under Flux Cone Learning's own five-fold ensemble protocol; 0.15 above Flux Cone Learning's 0.742 on the same genes and 0.06 above the classical ladder's best rung. Its per-cell mean over seven seeds is 0.869.

The inputs are the number. The gain over the backbone alone (0.774) is the four solver ratios (to 0.821) and the pathway multi-hot beside the protein embedding (to 0.871); the five gene properties in the same encoding, expression included, add nothing, and neither do width, depth, the preprocessor, ESM2, codon frequency or the species-LM composite. A two-layer MLP on the same three inputs reads 0.870 per cell in six seconds and 0.886 as a fold ensemble; graph convolution over the nine gene graphs reads 0.888. Three architectures are one number at this resolution.

The pathway multi-hot is the indicator. Split into its blocks, the 197-column encoding's gain is the 173 pathway columns: with ProtT5 and the ratios they read 0.880 on the MLP and 0.890 on the transformer, above the full encoding (0.870 and 0.871), while the chromosome one-hot (0.809, 0.822) and the seven properties (0.798, 0.824) sit at the level of ProtT5 with the ratios. On the transformer the pathway columns need the sequence embedding beside them (0.766 alone); on the MLP they do not (0.828). The pooled transformer on ProtT5 with the pathway multi-hot and no readout reads 0.860.

Keeping the edge type is the one architectural choice that registers. Of the message-passing forms of MATH-490 and Gene_Graph on the transformer's three inputs, relational attention over the nine graphs as nine edge types reads 0.889, GraphSAGE 0.887, relational convolution 0.860, graph convolution over the untyped union 0.847 and the graph isomorphism network 0.705, against the MLP's 0.870. The 1-WL-expressive aggregator is the weakest, and the typed-edge attention is level with the transformer's best single cell.

GO through the graph is the highest number in the document. Two layers of graph convolution over the union of the nine gene graphs on the propagated GO multi-hot alone, no sequence and no solve, read 0.912, 0.910 and 0.904 at 400 epochs, 43 seconds per seed, and under the reported five-fold protocol 0.917 [0.87, 0.96] against the transformer's 0.893 [0.84, 0.94], the folds above on four of five and the same 0.917 with three seeds per fold;

the curve is flat past 400 epochs, the sequence embedding and the ratios add nothing to it, and on six seeds it reads 0.898 to 0.912 (mean 0.905, sd 0.005) with no run that misses. On the ladder GO is the strongest single encoding (0.808 linear, 0.844 with a random forest and the ratios). The transformer is the one model that does not take it: 0.833 beside ProtT5 and a cost of 0.04 beside the pathway block.

The genes every model misses are the ones the solve misses. Thirteen of the fifteen hardest essential held-out genes have a knockout growth ratio of 1.0 (CDC19 covered by PYK2, PET9 by AAC1 and AAC3), and the hardest nonessential genes are metabolic paralogs (TDH1, TDH2, TDH3, PFK1, PFK2) the annotation vectors make look essential; the six transformer arms agree on their failures at a Spearman of 0.4 to 0.9, and the GO convolution fails elsewhere (0.15 to 0.4 with them), halving the misrank on FRQ1, AUR1, PGA3, LCB1 and LCB2 and repairing twelve of the misranked paralogs. Isozyme redundancy, read the wrong way in both directions, is where the remaining AUROC sits, and the two model families miss different halves of it (Section 3.16). Averaging their existing predictions with equal weight and nothing fitted scores 0.946 [0.91, 0.98] on the split-file cells (the six-seed GO convolution with the six-seed scalable form, from 0.905 and 0.918 alone) and 0.949 [0.91, 0.98] under the fold protocol, from families at 0.917 and 0.897 alone: the maximum of the eleven and six post hoc averages tried, reported as such (Section 3.17).

Off the metabolic genes, only the graph transfers. On the 4,674 genome-wide genes the released split never labeled (974 essential by SGD, 3,700 nonessential by a fitness record, none with a GEM reaction, so FBA reads exactly 0.5 and label origin coincides with class), models trained on the 629 released genes read: the MLP on the transformer's inputs 0.509, the transformer's reported recipe 0.574, graph convolution on ProtT5 alone 0.749, on the three inputs 0.744, on GO alone 0.702, degree and clustering alone through the physical graph 0.753, and the raw degree in the physical graph, nothing trained, 0.740 (0.594 on the released 195). The GO convolution's margin is a property of the metabolic genes; off them the number is the gene's degree in the protein interaction graph, a genome-wide signal the metabolic genes barely vary in, which any input through the graph reaches and no input without it does. The other way, the genome-wide labels as training data leave the GO convolution at 0.913 on the 195 and cost the three-input models 0.03 to 0.05 (Section 3.18).

Structure is not the signal. The 1-WL test separates 1,118 of 1,121 single-deletion graphs on the physical graph and all of them on the union, so expressivity does not bound a message-passing model here, and the WL kernel, all-ones and degree inputs all rank at chance. What is written on the nodes carries the number.

Accuracy is the wrong statistic. yeast-GEM's own benchmark reports 0.902 on a gene set whose majority class alone gives 0.856, with recall 0.41; FBA on 9.0.2 reproduces it at 0.907, the released labels agree with the Stanford list on 1,102 of 1,104 shared genes, and the transformer's balanced accuracy at the loss's decision boundary is 0.78 against 0.63 to 0.72 for the solvers.

The solve carries the module's value, and the alternating layer cannot compute it. Routed through the module's growth column the prediction never beats the backbone alone. The layer's alternating projections never reach the polytope the solve optimizes over: with the balance exact the box is violated on 58 percent of reactions, the second law on half, the protein pool by up to seven times, the dissipation limit by three; the growth column is a free signed coordinate the readout learns to use with either sign; the k_{cat} content is unread (the shuffled table matches); and handed the solve's own point, the anchored constraint set discards it.

The enzyme-capacity box as parameterized contains no growing flux. A linear program under the layer's bounds is infeasible for every genotype, because the predicted k_{cat} tier times the nominal abundances caps the median reaction at 0.084 mmol/gDW/h. Every anchored arm in this document read a flux off the polytope for that reason. Scaled by 30 or more the box admits growth; the ceilings still bind up to a scale near 3,000.

On the polytope, the module helps. On the GEM's own box the solve survives the projections and the program that solves is feasible and ranks the held-out genes at 0.73 before training, above FBA's 0.66; the gated module cell reaches the level of the solve appended as an input, 0.816 by warm start and 0.824 by the layer that solves, three of three seeds above the round-4 gated cell, with the growth column entering through a positive gate. On six seeds the two gated forms read 0.814 (GEM box) and 0.804 (elastic program, feasible by construction), twelve seeds of twelve above the gated cell without the solve and 0.01 to 0.02 under the affine.

The ceilings cannot carry the ranking, so the forced growth column cannot be trained. With the program elastic, feasible for every genotype and its gradient arriving on nine genotypes in ten, the forced column still reads 0.70 to 0.74 and its training loss never leaves 1.25, under ten times the learning rate, a deeper head, a wider reach and two ceiling scales. The dual-overlap probe measures why: the same six gene-owned ceilings bind whichever gene is deleted (median Jaccard 1.0), and the deleted gene's own reactions never bind because the deletion has closed them, so the head can move one growth level for every genotype and cannot move one genotype against another.

The solve's flux vector, read by a linear head, is the module's number. The gated elastic cell with a zero-initialized linear head over $\log(1+|v^*|)$ of the solved flux reads 0.883, 0.853 and 0.868 on the ProtT5 backbone, the first module cell to pass a pre-registered read, 0.047 above the FBA affine on that backbone and with the growth column's gate ending near zero. Over the 274 exchange fluxes alone it reads 0.835. On ProtT5 with the pathway multi-hot the same cell reads 0.897 and 0.903 on two seeds, the two highest single cells in the document and above the affine on that input (0.890), and 0.716 on a third seed whose training loss never fell. The vector carries it: with nothing learned in the layer the same head reads 0.852, the ceilings add 0.01 to 0.02, and the gene-owned columns hold it (0.861 against 0.835 for the exchanges). Under the fold-ensemble protocol on the pathway input it reads 0.897 [0.84, 0.95], one number with the reported 0.893, from three fold models at 0.90 and two that never fit; three of this form's seventeen cells miss the fitting transition within 200 epochs, and the miss is a chance event of the run and not of the seed: the unfit seed rerun under the same program fits at epoch 70 and reads 0.889, the two runs diverging from their first epoch (Sections 4.20, 4.21 and 4.23). The form that scales, the program solved once per genotype with nothing learned in the layer and its flux read through a zero-initialized head, reads 0.897 to 0.917 on eight of nine runs over six seeds (pooled 0.917 [0.87, 0.96] on the six-seed rerun, above the elastic form's 0.910), and misses the transition on the ninth, a seed it had fit in the previous attempt, so the chance is not in the ceilings (Section 4.22). Metabolism enters the cell as a perturbation-conditional transform that holds mass balance and the bounds exactly, costs nothing on the card, and cannot hurt at initialization.

Where the gradient goes is not the problem. On the alternating layer the gradient reaches the flux layer on every arm that lets it; the backbone's preprocessor, three quarters of the parameters, is where fitting continues after the validation curve peaks.

What carries forward to a model trained over every phenotype in the database is the part of this that did not depend on the 195 genes. Metabolism enters as a perturbation-conditional data transform: the genome-scale program solved once per genotype and medium on the GEM's own box, cached, and read through a zero-initialized head over $\log(1+|v^*|)$, which cost nothing on the card, held mass balance and the bounds exactly, read 0.90 to 0.92 on eight runs of nine where the affine read 0.89, and cannot hurt at initialization; the learned enzyme-constrained layer, whose ceilings added 0.01 to 0.02 at five times the wall time and whose forced growth column could not be trained, is a fine-tuning module and not a backbone component. Annotation vectors enter through message passing over the gene graphs and not through the transformer's token: the GO multi-hot read 0.917 through two layers of convolution and 0.833 through the transformer, and the pathway block's gain on the transformer was the one annotation it took. The gene graphs enter as a message-passing input and not as a regularizer, since off the metabolic genes the models without the graph as an input fell to chance and the ones with it held 0.75. What the 195 genes could not decide, which of three architectures on the same inputs is better, the genome-wide set decided for the graph; what neither set can decide is whether that transfer is essentiality or the deletion collection's coverage.

6 Open ✗

The held-out number is close to saturated: the round-12 ensemble's interval, 0.893 [0.84, 0.94], spans what 195 genes with 31 essential can resolve, and every lever on the transformer side has been moved (width, depth, the preprocessor, the budget, the class weight, six encodings). What remains is attribution and the module.

- The pathway multi-hot carries the round-9 gain (rounds 14 and 22, and the split on the cheap models); which pathways, and whether a pathway-level flux statistic from the solve would substitute for the annotation, is not measured.
- The round-16 cell (the round-11 affine on the round-9 backbone) and the round-9 cell are one cell to the resolution of 195 genes (rounds 16 and 17); a larger held-out set, such as the genome-wide labels scored on genes the released split never touched, is what would separate them.
- The MLP, the graph convolution and the transformer on the same three inputs are one number as fold ensembles (0.886, 0.888, 0.893), and graph convolution on GO alone is 0.917 above them with overlapping intervals. On the 4,674 genome-wide genes the released split never touched (round 30) the architectures separate: the graph convolution transfers at 0.75 and the GO margin does not survive (0.70), while the MLP and the transformer's recipe read 0.51 and 0.57. Open on that set: whether a model trained on the metabolic genes transfers because it reads essentiality or because it reads what the graph encodes about SGD-annotated against deletion-collection genes (label origin coincides with class there, and for a viable deletion the two are close to one definition). Degree and clustering alone reach the 0.75 on that set, so the question is what degree in the physical graph measures there; a genome-wide set whose nonessential labels come from the same source as the essential ones would separate origin from class, and the physical graph's own coverage of the deletion collection against the SGD inviable set is the first thing to count.

- The GO multi-hot is the ladder's strongest encoding and the cheap models take it (graph convolution 0.887), and the transformer does not (0.833 beside ProtT5, a cost of 0.04 beside the pathway block, round 26), and a cell 350 times wider reads it the same (round 28), so capacity is not the reason. What separates 173 sparse pathway columns from 2,362 dense GO columns in this backbone is not measured; a per-block projection, the GO vector entering as its own token, and the pooled readout itself are the untested constructions, and the answer decides whether annotation vectors scale as inputs to this backbone.
- The forced growth column is closed by the dual-overlap probe (Section 4.16): the ceilings that bind are the same for every genotype, so no step size, head or reach trains it (rounds 18 to 20). A construction that changes which bounds bind per genotype, such as ceilings the deleted gene's neighbors own, or an objective other than biomass, since the layer that solves accepts any objective column, is untested.
- The head over the solved flux is the module's number, and the form that scales (round 27: the program solved once per genotype, nothing learned in the layer, a zero-initialized head) reads 0.90 to 0.92 on eight runs of nine and misses the fitting transition on one, where the learned-ceiling form misses it on three cells in seventeen. Open on it: its fold read, lost to the launcher's memory ceiling and in flight at 120 GB; what in the forward makes the transition a chance event of the run rather than of the seed (round 29: the same seed and program fit at epoch 70 in one run and never in another, the two diverging from the first epoch, and the scalable form repeats it with the solve cached and nothing learned in the layer), and whether pinning the sampler and the card's kernels makes the seed reproducible; the cache's hit rate and where the wall time goes when the solve is cached, since six cached cells on one card ran no faster per cell than three uncached; and the flux entering the cell representation as a token rather than the logit as a head, which is the multi-phenotype form.
- The two model families, annotation through the graph and the solve through the transformer, miss different genes, and their post hoc average reads 0.949 under the fold protocol. A model trained to hold both, or a learned weighting, is untested, and the average's number carries the selection of the best of six.
- The layer that solves has the wrong shape for a training loop: the program runs on the CPU inside the forward pass, the GPU waits on it (0 to 13 percent utilization sampled during round 19, against a 1.6 hour cell where the alternating layer takes 19 minutes), and the bytes moved are not the cost, the synchronization is. Three constructions would remove it, in order of how much they change: a per-genotype basis warm start of the simplex through HiGHS's own interface, which keeps the exact solve and the exact duals; a batched first-order solve on the GPU; a smooth interior-point solve at a fixed barrier, which changes what the layer computes and belongs to its own rounds.
- The module's own hyperparameters have not been moved on this endpoint: the k_{cat} tier (measured against predicted) and the thermodynamic source. Given round 15, the k_{cat} tier is the one that sets whether the capacity box holds a growing flux at all.